

Studies on Dry Root Rot of Chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler

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CERTIFICATE-I

*This is to certify that the thesis entitled “Studies on dry root rot of Chickpea caused by **Rhizoctonia bataticola** (Taub.) Butler.” submitted in partial fulfilment of the requirements for the degree of **DOCTOR OF PHILOSOPHY** in **PLANT PATHOLOGY** of **Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior** is a record of the bonafied research work carried out by **JAGDISH KUMAR PATIDAR (ID RA/GW/103/2006)** under my guidance and supervision. The subject of the thesis has been approved by the student’s Advisory Committee and the Director of Instruction.*

No part of the thesis has been submitted for any other degree or diploma or has been published. All the assistance and help received during the course of this investigation has been acknowledged by the scholar.

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*This is to certify that thesis the entitled “**Studies on dry root rot of Chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler**” submitted by **JAGDISH KUMAR PATIDAR** to the Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior in partial fulfilment of the requirements for the degree of **Ph. D. in AGRICULTURE** the department of **PLANT PATHOLOGY** has been accepted after the evaluation by the External Examiner and by the Student’s Advisory Committee after an oral examination on the same.*

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LIST OF SYMBOL

Symbol	Abbreviation	Stand for
@		At the rate of
%		Per cent
±		Plus or minus
°C		Degree centigrade
μ		Micron
	Avg.	Average
	CD	Critical difference
	CFU	colony forming units
	cm	Centimeter
	Conc.	Concentration
	cv.	Cultivar
	<i>et al.</i>	Co- workers
	Fig.	Figure
	g	Gram
	ha	Hectare
	hrs	Hours
	i.e.	That is
	kg	Kilogram
	L	Litre
	M	Meter
	mg	Milligram
	mm	Millimeter
	NS	Non significant
	PDA	Potato dextrose agar
	ppm	Parts per million
	psi	Pound square inch
	RH	relative humidity
	SEm ±	Standard error of mean
	spp.	Species
	DRR	Dry root rot
	Temp.	Temperature
	<i>viz.,</i>	Namely
	W / W	Weight by weight
	WP	Wettable powder

Chapter-I

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is the premier pulse crop grown in more than 50 countries, originated in south west Asia and cultivated from ancient times both in Asia and European countries. It is the world's second most important food legume next to common bean. Asia accounts 89.20% of the chickpea area and 84.47% of production. The major chickpea producing countries, which contribute about 88% of the global chickpea production, include India (64.58%), Australia (7.22%), Myanmar (4.62%), Pakistan (4.27%), Turkey (3.76%), and Ethiopia (3.67%) (Anon., 2015-16a). In India, chickpea is grown in an area of 8.35 million hectares with a production of 7.82 million tonnes and productivity of 936 kg ha⁻¹. Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Andhra Pradesh, Tamil Nadu, Odisha and Karnataka together contribute 89.47 % of production and 90.39 % of area in the country (Anon., 2015-16a). In Madhya Pradesh it cultivated in about 30.17 lakh ha. with an average annual production of 33.64 lakh metric tonnes having the productivity of 1115 kg/ha (Anon. 2015-16b).

Chickpea belongs to Leguminaceae (Fabaceae) family usually grown after rainy season on conserved soil moisture during winter in the tropics; in spring in the temperate and mediterranean regions. It is the important grain legume grown for protein rich seeds for human consumption and to maintain the soil fertility by its nitrogen fixing capability. Chickpea seeds contain 23% protein, 64% carbohydrates, 47% starch, 5% fat, 6% crude fibre, 6% soluble sugar and 3% ash, minerals such as calcium (202 mg), phosphorus (312mg), iron (10.2mg), vitamin C(3.0 mg),calorific value (360 cal), small amounts of B complex, fibre (3.9g) and moisture (9.8g) [Awasthi *et al.*, 1991; Dhawan *et al.*, 1991; Dutta *et al.*, 1985].Predominantly chickpea is being consumed as dal or variety of snack foods, sweets and condiments. Husk and split grains are useful as livestock feed (Malik and Tufail, 1981). Acidic liquid from glandular hairs of the plant contain 94% malic acid and 6% oxalic acid, has medicinal value and used in preparation of vinegar (Alam, 1989).

Chickpea cultivation is often subjected to significant yield losses due to insects and diseases ranging from 5-10% in temperate and 50-100% in tropical regions (Van Emden *et al.*, 1988). Currently chickpea is attacked by 172 pathogens viz., 67 fungi, 3 bacteria, 22 viruses and mycoplasma, and 80 nematodes reported

from 55 countries. Maximum number of pathogens infecting chickpea (89) had been reported from India while in other countries, it varied from 1 to 40. The pathogens that are most widely distributed are *Ascochyta rabiei* (35 countries), *Fusarium oxysporum* f. sp. *ciceri* (32 countries), *Uromyces ciceris arietini* (25 countries), bean leaf roll virus (23 countries), *Rhizoctonia bataticola* (21 countries), *Botrytis cinerea* (15 countries), *Sclerotinia sclerotiorum* (15 countries) and in 9 countries cucumber mosaic virus (Nene et al., 1996).

Dry root rot of chickpea caused by necrotrophic fungus, *Rhizoctonia bataticola* (Taub.) Butler is emerging as a serious threat to the chickpea production worldwide (Pande and Sharma, 2010). Dry root rot generally appears during late flowering and podding stages and the infected plants appear completely dried. Among the several constraints affecting the productivity of chickpea, 10-35 per cent loss in yields are due to wilt and dry root rot diseases (Mahendra Pal, 1998). Among them, dry root rot caused by *Rhizoctonia bataticola* is becoming severe in most of the chickpea growing regions of Madhya Pradesh. *R. bataticola* is a polyphagous soil borne pathogen infecting over 500 plant species worldwide causing huge losses. Though, the fungus is seed and soil borne (Dhingra and Sinclair, 1994), however, soil borne inoculum is more important in causing infection and disease development.

In chickpea field, the onset of the disease appears as scattered drying of the plants. Affected plants are usually straw coloured, but in some cases the lower leaves and stems show brown discolouration. The tap root appears black, rotten and devoid of most of the lateral and fine roots. The dead root become quite brittle and shows shredding of bark. Dark minute sclerotial bodies can be seen on the roots exposed or inside the wood. When the dry stem of the collar region is split vertically, sparse mycelium or minute sclerotia can be seen in the pith (Nene et al., 1991).

Only data on few specific diseases in chickpea are available from different states in India (Singh and Sirohi, 2003; Nikam et al., 2011). However, no current information on occurrence and distribution of chickpea diseases in India is available. Chickpea diseases may cause yield losses up to 100% depending on time of infection [Pande et al., 2010]. The recommended method of managing the diseases is use of resistant varieties. A number of improved chickpea disease

resistant varieties have been multiplied and disseminated to farmers in many districts of Madhya Pradesh. However, their current prevalence in farmers' fields has not been documented.

Culture media is of fundamental importance for most pathological tests: to obtain pure cultures, to grow, count microbial cells, cultivate and select microorganisms. Without high quality media, the possibility of achieving accurate, reproducible and repeatable microbiological test results is reduced (Sandle, 2010).

The disease is soil borne due to prolonged saprophytic survival of the pathogen through sclerotia. Therefore, crop rotation as well as chemical methods for disease management become ineffective. Host plant resistance is an economical approach, but only a few sources of moderate resistance to dry root rot are available. Therefore, commercial cultivars with genetic resistance to the disease have not yet been developed (Pande *et al.*, 2004). The information is not available in respect of virulence/pathotypes of *R. bataticola* prevalent across the country and its correlation with morphological variability. Pathogenic variability is essentially required to develop area specific resistant cultivars.

Chemical control of dry root rot is not effective as *R. bataticola* has a broad host range and survives in soil for longer periods in the form of sclerotia. The sclerotia can survive up to 10 months even in the absence of the host plants and under prevailing dry soil conditions. Use of host plant resistance is the most economical approach for management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot have been identified, but high levels of resistance are scarce in cultivated genotypes (Anon., 2010).

Botanical extracts are biodegradable (Devlin and Zettel, 1999) and their use in crop protection is a practical sustainable alternative. It reduces environmental contamination and health hazards (Grange and Ahmed, 1988). Botanical fungicides are unique because they can be produced easily by the farmers and small industries (Roy *et al.*, 2005). Different plant extracts also used separately or in combination to control some other fungi by the farmers. Very few studies have conducted using plant extracts and fungicides in the country to control *R. bataticola* causing dry root rot of chickpea.

Disease management strategies primarily depend on sanitary practices and fungicide applications. However, development of fungicide resistance within

population of *M. phaseolina* became a problem, and alternative approaches that can be incorporated into integrated pest management of chickpea root rot disease are needed. Biological control is coming up as an alternative strategy for disease management, which is also ecofriendly.

Keeping this in view, the present investigation was conducted with the following objectives :

1. Survey of dry root rot of chickpea in Gwalior, Malwa and Nimar region of Madhya Pradesh.
2. Evaluation of culture media against growth of *R. bataticola*.
3. Evaluation of inoculation techniques to find out the most effective technique for artificial inoculation.
4. Studies on morphological, cultural and pathogenic variability in *R. bataticola*.
5. Screening of cultivar of chickpea against dry root rot pathogen.
6. Evaluation of botanicals against *R. bataticola*.
7. Evaluation of fungicides against *R. bataticola*.
8. Evaluation of *Trichoderma* strains against *R. bataticola*.
9. Evaluation of selected botanicals, bio control agents and chemicals for the management of dry root rot disease.

Chapter- II

REVIEW OF LITERATURE

Review of literature is a necessary step for any scientific study. It provides a theoretical framework, previous work and the basic interpretation of findings of study. An attempt has been made to review the literature, which is meaningful and had direct relevance to the study. The available researches have been presented under different heads.

Chickpea is the world's third most important grain legume globally grown in over 40 countries (Anwar *et al.*, 2009). Bagri *et al.* (2004) observed that chickpea suffers from seed borne fungal diseases viz., black root rot, dry root rot, wet root rot, seed rotting, root rot, stem rot, crown rot, foot rot, Sclerotinia wilt and gray mould. Amongst these diseases, dry rot such as dry root rots [*Macrophomina phaseolina* (Tassi.) Goid.] has been reported to cause severe losses right from seedling to maturity of the crop. *M. phaseolina* (Tassi.) Goid. has a wide host range and is responsible for causing losses on more than 500 cultivated and wild plant species (Khan, 2007).

2.1 : Association of fungal pathogens with chickpea seeds

Fakir (1983) found five seed borne fungi in chickpea seeds. *Fusarium oxysporum* caused wilt, seed rot by *M. phaseolina* (*Rhizoctonia*), collar rot by *Sclerotium rolfsii* and blight of chickpea by *Ascochyta rabiei*.

Ahmed (1985) reported a total of 50 pathogens of chickpea from different parts of the world, where 14 have been reported from Bangladesh.

Rahman and Ahmed (1985) detected seed-borne fungi of chickpea, where *Aspergillus* sp. was most predominant and pathogenic fungi *Colletotrichum dematium* and *Fusarium* sp. were also prevalent. *Macrophomina*, *Fusarium*, and *Colletotrichum* were responsible for causing seed rot and seedling infection. *Fusarium* wilt caused by *F. oxysporum* f. sp. *ciceri* is one of the important diseases of chickpea. The disease is prevalent almost all the chickpea-growing countries of the world.

Maden (1987) recorded few pathogenic fungi from chickpea seeds. Among them, *Ascochyta rabiei*, *Fusarium oxysporum*, *M. phaseolina*, *Rhizoctonia solani* etc. are predominant and were isolated from chickpea seeds.

Demirci *et al.* (1999) reported that fungal pathogens were isolated from plants and seeds of the chickpea cultivar Azizyw- 94 and their pathogenicity was tested individually or in combinations, in pot trials under controlled environmental conditions. The isolates *F. solani* f sp. *pisi* and *F. oxysporum* f sp. *ciceris* were more widespread than *F. acuminatum*, *F. avenaceum*, *F. equiseti*, *F. proliferatum*, *M. phaseolina*, and *Rhizoctonia solani*, *F. oxysporum* f.sp. *ciceris* was isolated from both plants and seeds while *F. oxysporum* f. sp. *pisi* was mainly isolated from plants. *F. oxysporum* f.sp. *pisi* and *R. solani* were the most pathogenic species when inoculated alone or in combination with other species.

Khan *et al.* (1999) recorded, nine fungal isolates from diseased chickpea and tested for pathogenicity in the green house.

Rauf (2000) recorded twenty-four seed borne fungi belonging to different genera using blotter method, from 145 seed samples of major legume crops in Pakistan. Of these fungi, *Alternaria alternata*, *Ascochyta* spp. *Colletotrichum* spp. *Fusarium* spp. and *M. phaseolina* were the most frequent and common pathogenic fungi in those crops.

Singh and Sirohi (2003) observed collar rot (*Sclerotium rolfsii* [*Corticium rolfsii*]) incidence decreased with crop age of chickpea. Wilt caused by *Fusarium solani*, *F. oxysporum* f.sp. *ciceris*, *Rhizoctonia solani* and *R. bataticola* [*M. phaseolina*] was observed at all growth stages.

Singh *et al.* (2004) observed nine fungal species namely, *A. alternata*, *Aspergillus flavus*, *A. niger*, *Curvularia lunata* [*Cochliobolus lunatus*], *F. moniliforme* [*Gibberella moniliformis*], *Helminthosporium sativum* [*Cochliobolus sativum*], *Mucor* sp., *Penicillium notatum* and *Rhizopus nigricans* [*R. stolonifer*] on seven seed samples of chickpea.

Toorray *et al.* (2005) isolated eight fungi, i.e. *Aspergillus niger*, *A. flavus*, *Rhizoctonia* sp., *Rhizopus* sp., *Fusarium* sp., *Alternaria* sp., *Curvularia* sp. and *Ascochyta* sp. from the seeds.

Delwar (2006) from Bangladesh reported 14 fungi comprising 12 genera namely *A. flavus*, *A. niger*, *Aspergillus* sp, *Botryodiplodia theobromae*, *Botrytis cinerea*, *C. lunata*, *Chaetomium globosum*, *Cladosporium* sp., *Colletotrichum*

dematium, *F. oxysporum*, *M. phaseolina*, *R. solani*, *Rhizopus stolonifer* and *Stemphylium sarciniforme* that were isolated from chickpea seeds collected.

Dawar et al. (2007) isolated a total number of 21 species belonging to 13 genera of fungi were from chickpea. Four species namely *Absidia glauca*, *R. solani*, *Syncephalastrum* sp., and *T. harzianum* were new report. Pathogenic fungi, viz. *F. moniliforme* [*Gibberella moniliformis*], *F. oxysporum*, *M. phaseolina* and *R. solani*, and saprophytic fungi like *A. niger* and *A. flavus* were predominant of them.

2.2 : Taxonomy

The taxonomy of the fungus *R. bataticola* (*M. phaseolina*) has been studied by number of workers all over the world to establish the exact systemic position with a valid name. The genus *Rhizoctonia* (means “root killer”) was introduced in 1815 by French mycologist Augustin De Candolle. *Rhizoctonia bataticola* (Taub.) Butler as a plant pathogen was recognized by Halsted (1890). Taubenhaus (1913) gave the name of the genus as *Sclerotium* because of absence of spores and the species name as *bataticola* because it was pathogenic to *Ipomea batatus* (L.) Lam. Briton Jones (1925) transferred the fungus to the genus *Rhizoctonia* based on the identification of cultures by Butler (1918). Ashby (1927) accepted *Macrophomina* and rejected the binomial *Macrophomina phaseoli* and proposed a new binomial *M. phaseolina* as the pycnidial stage of *R. bataticola* on the basis that *M. phaseoli* was the earliest applicable binomial. He was not aware that Tassi had earlier described *Macrophomina phaseolina*. Haigh (1930) suggested that *R. bataticola* be used for sclerotia isolate and pycnidial strains should be called *M. phaseolina*. Young (1944) who concluded that *R. bataticola* is a synonym of *Sclerotium bataticola* and that *Macrophomina phaseoli* is the pycnidial form of the same fungus. Reichert and Hellinger (1947) suggested that the *R. bataticola* should be discarded and *M. phaseoli* should be applied solely to strains forming pycnidia only, *S. bataticola* to those sclerotia only while those formatting both sclerotia and pycnidia should be reported to as *M. (Sclerotium) phaseoli*. Goidanich (1947) examined the original material of Tassi and compared with *M. phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinensis*, *Dothiorella cajani* and *D. phaseoli*. He confirmed that all of them were identical. He corrected the mistake made by Ashby.

According to the International Code of Botanical Nomenclature, the binomial *M. phaseolina* was the valid name for the pycnidial stage of *R. bataticola*.

However, now it is established that causal organism of dry root rot of chickpea is *R. bataticola* (Taub.) Butler and its systemic position is as follows (Alexopoulos and Mims, 1983).

Division	-	Mycotina
Sub division	-	Deuteromycotina
Form class	-	Deuteromycetes
Form sub class	-	Hyphomycetidae
Form order	-	Agonomycetales
Form family	-	Stilbellaceae
Genus	-	<i>Rhizoctonia</i>
Species	-	<i>bataticola</i>

The perfect stage of *R. bataticola* has been reported in India (Anon., 1963 and Ghosh *et al.*, 1964). They could obtain typical cultures of strain of saprophytic fungus from jute field. The fungus was described as *Orilia obscura* (Ghosh *et al.*, 1964) and was considered as perfect stage of *R. bataticola*.

2.3 : Occurrence and distribution of disease

An association of *Fusarium* sp. and *Rhizoctonia* sp. with wilted chickpea was first reported by Narasimhan (1929). Subsequently, in India, it was reported from Punjab by Luthra (1938), Madhya Pradesh (Sharma and Khare, 1969; Kotasthane *et al.*, 1979), West Bengal (Biswas and Gupta, 1981) and Haryana (Tripathi and Sharma, 1983). It has been reported from Egypt, Iran, Kenya, Lebanon, Mexico, Myanmar, Pakistan, Spain, Sri Lanka, Sudan, Syria, Tanzania, Turkey, Uganda, USA and Zambia. It causes considerable yield losses that vary from 5-50% and may cause 100% losses in susceptible cultivars under favourable conditions (Nene *et al.*, 1996).

In soils where soybean or susceptible hosts are planted successively year after year, *M. phaseolina* populations tend to increase. Cook *et al.* (1973) reported

an increase in microsclerotia in a field planted with soybeans for three years with 80, 120 to 149 microsclerotia per gram of soil.

The incidence of *R. bataticola* (*M. phaseolina*) on *Cicer arietinum* was higher in sandy than clay soil. Low soil moisture was conducive for disease development (Taya *et al.* 1988).

Pandey and Singh (1990) conducted the survey in Allahabad region of Uttar Pradesh and reported that the chickpea was infected by *R. bataticola* in 83% of villages with less than 25% disease intensity.

R. bataticola isolates of chickpea from several locations in Myanmar collected by Than *et al.* (1991) reported the possible occurrence of more than one type of isolate in a same field and the existence of closely related isolates at different places.

The incidence of dry root rot in chickpea caused by *R. bataticola* was studied in 20 locations in Uttar Pradesh and Uttaranchal, India during February to March of 1998-2000 and reported that disease was widespread in Uttar Pradesh. It was severe in bold seeded cultivars (Prajapati *et al.*, 2003).

Survey was conducted by Shubha and Gurha (2006) in different blocks of Bhundelkhand region of Uttar Pradesh and found infection of *R. bataticola* on chickpea which was ranged from 5-22%.

Chickpea fields in Shimoga, Raichur and Bangalore districts in Karnataka were surveyed by Gurha and Trivedi (2008) for the prevalence of dry root rot. *R. bataticola* was the predominant pathogen infecting 60 to 70% of the plants in the fields of Gulbarga, Manchalapur, Eklaapur, Raichur and Tengri.

Intensive roving survey was conducted by Manjunatha *et al.* (2011) during November to December, 2008 in the farmers' fields at Raichur, Gulbarga and Bidar districts in Karnataka. The maximum dry root rot incidence in chickpea was noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18 %) in Bidar. The overall incidence of dry root rot ranged from 1-19% across the districts. This was mainly attributed due to variation in soil type (black and red soil) and cultivars grown (Annegiri and JG 11) under rain fed condition.

The survey for the occurrence and severity of chickpea dry root rot was made during crop season 2010-11 in Jammu and Kashmir. The highest incidence

was up to 40% in Shangus and lowest up to 4.11% in Naina of Jammu and Kashmir. The incidence of dry root rot in chickpea caused by *R. bataticola* was observed in late October to mid November. The intensity of the disease was high in the month of February and March during late flowering and podding stage (Khan *et al.*, 2012).

A survey was conducted during 2010-2011 to obtain information on the distribution and incidence of chickpea diseases with respect to soil type, cultivar used, seed treatment in Central and Southern parts of India (Andhra Pradesh, Karnataka, Madhya Pradesh and Chhattisgarh). Local cultivars Annigeri (19.23%) followed by Harbora (12.82%) were most commonly grown by the farmers. However, among the improved varieties JG 11, JG 130, JG 16, JG 74 and Jawahar were most commonly grown covering 34.61% area. Dry root rot disease was found in all the sites surveyed where the incidence ranged from 8.9-10.3% irrespective of cultivar type and location. The local variety occupied 48% area in Karnataka while in Andhra Pradesh it was 39%. Disease incidence was lower on improved varieties (9%) as compared to local varieties (14%). Sixty three percent of the farmers practiced seed treatment with fungicides. The disease incidence in seed treated fields ranged from 6.63-12.40% as compared with untreated fields 9.22-20.02% (Ghosh *et al.*, 2013).

High levels of *M. phaseolina* colonization are positively correlated with poor seed quality and lower yields. After harvesting, roots loaded with microsclerotia start decaying and release the microsclerotia into the soil (Olaya *et al.*, 1996). Inoculum survival in the soil depends on soil moisture and the soil microbial community (Collins *et al.*, 1991)

Rajkrishan *et al.* (1999) studied the effect of edaphic factors and moisture regime on the incidence of *Macrophomina* root rot in sesame. Among the various soil types, sandy soil supported maximum disease (78.33%) as compared with clay soil (51.56%). Similarly, more disease incidence of 48.33% was observed when pots were irrigated after the interval of seven days as compared to pots receiving irrigation every day.

Sharma and Pande (2013) conducted study to quantify the role of temperature and soil moisture associated with infection, colonization and development of dry root rot under controlled environment. The dry root rot incidence

was significantly affected by high temperature and soil moisture deficit. Out of five temperature regimes (15°C, 20°C, 25°C, 30°C and 35°C) and four moisture levels (40%, 60%, 80% and 100%), a combination of high temperature (35°C) and soil moisture content (60%) predisposes chickpea to dry root rot. The study clearly demonstrates that high temperature coupled with soil moisture deficit may play an important predisposing factor for *R. bataticola* infection, colonization and development in chickpea.

2.4 Disease symptomatology

According to Dastur (1935) only mature plants were affected, showing bronzing of the leaves on one or more of the lower branches. The colour of leaves later changed to yellow and then brown. The affected branches and leaf stalks of diseased plants were stiff and turned upwards and leaflets stand more or less vertically and shed prematurely. The terminal part of the tap root was black or brown and shrivelled.

Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

According to Mehrotra and Aneja (1990) the first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant may become die. The leaves and stem of the affected plants are usually straw colored. The tap root is dark and quite brittle in dry soil.

Singh *et al.* (1990a) reported that roots inoculated with *R. bataticola* (*M. phaseolina*) caused disintegration of cortical tissues with mycelial and sclerotial bodies plugging the xylem vessels.

The dried plants due to disease generally appeared during flowering and podding time. These were scattered in the field. The seedlings can also get infected. Drooping of petioles and leaflets was confined to those at top of the plant. Leaves and stem of affected plant usually straw coloured but in some cases, the lower portion of the stem was brown. When the plant was uprooted, lower portion of the tap root remains in the soil. It was devoid of lateral and finer roots. Dark minute

sclerotial bodies can be seen on the roots exposed inside the wood (Nene *et al.*, 1991).

Nene *et al.* (1978) observed continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of dry root rot of chickpea.

Baldev *et al.* (1988) reported symptoms of dry root rot of chickpea which consisted of straw colored leaves and stems. Tap root of infected plants was observed to be dry and devoid of lateral and finer roots. Roots were turned dark and showed signs of rotting. The dead root was observed brittle towards the tip and showed shredding of bark.

Haware (1990) reported that dry root rot of chickpea appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The symptoms induced were drooping of petioles and leaflets which were confined to top of the plants. Shredding of bark in form of flakes was observed.

Rangaswamy (1996) reported field symptoms which include yellowing of plants with drooping of leaves. Due to decay of roots, plants can be easily uprooted. Root portion appears brownish from outside. The stem and root below the region shows rotting with frequently pinkish white mycelial growth. Dried plants scattered throughout the field are indicative of root rot incidence.

Singh (1999) reported that dry root rot of chickpea occurred from flowering to podding stage. Infected plants were suddenly dried in the field. The tap roots turned dark brittle and devoid of lateral roots. Khalid and Ilyas (2000) considered the presence of root lesions and sclerotium plugging in xylem vessels of roots and collar region as symptoms of dry root rot of chickpea for screening of germplasm against this disease.

Singh and Agarwal (2002) observed the withering and drying of chickpea plants in the field due to the infestation of *R. bataticola*.

2.5 Causal organism

Hildebrand *et al.* (1945) differentiated two strains of *M. phaseolina* isolated from soybean, having mean sclerotial diameter of $90.4 \times 75.8 \mu\text{m}$ on host plant and

99.9 × 89.4 μm in culture and from cotton (the Texas strain) whose, sclerotia in culture averaged 85.4 × 73.0 μm and were produced in greater number.

Jharia (1970) also found that *R. bataticola* is associated with seeds of urid both internally and externally. Thus, it seems that *R. bataticola* acts as both soil and seed borne pathogen from Madhya Pradesh.

The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture. On the infected plants sclerotia may be seen in the pith region and outer surface of roots just below the bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape: globose, oval, oblong, elliptical, curve or even forked, varying in size, uniform texture, and more or less loosely packed (Subramanian, 1971).

Dhingra and Sinclair (1973) noticed variation in growth rate among the isolates from the same plant and between isolates from different soybean plants on different media. The stem isolates produced fluffy growth; root isolates showed partially fluffy growth and all other isolates produced appressed growth. Seed and petiole isolates produced concentric alternating zones of sclerotia and this character was more pronounced for the seed isolate. They also collected nine isolates of *M. phaseoli* from different parts of United States and reported variation in growth rate and colony characters.

The cell of hyphae are barrel shaped, branching more or less at right angles and light to dark brown in colour, composed of free thick-walled cells. The species *Rhizoctonia* is distinguished by the light brown-to-brown hyphae with somewhat loose/spongy sclerotia in the form of bulbils. According to Mehrotra and Aneja (1990) both the species *R. bataticola* and *R. solani* have the variable capacity to produce enzymes.

Vishwadhar and Sarbhoy (1993) isolated the pod and seed isolate of soybean causing ashy stem blight and reported the smallest known sclerotia in the world (mean 32 × 24 μm).

During the sclerotial formation, 50–200 individual hyphal cells aggregate to give multicellular bodies called microsclerotia. The microsclerotia were black and variable in size from 50–150 μm depending on the available nutrients of the substrate on which the propagules were produced (Short and Wyllie, 1978).

2.6 : Host range

Fifty nine species have been recorded under genus *Rhizoctonia* (Saksena and Vaartajal, 1961; Hussain and McKeen, 1963). Among them *R. bataticola* is one of the important species, which has a very wide host range and causes root rot, charcoal rot and ashy stem blight in most of the cereals, pulses, oilseeds, fibres, vegetables, ornamental plants, plantation and horticultural crops (Small, 1925 and 1928; Uppal, 1934; Likhite and Kulkarni, 1934).

Singh *et al.* (1990b) identified eleven new hosts, in addition to ten species which were identified as new pycnidial hosts. It was concluded that, *M. phaseolina* was prevalent in all commonly cultivated crops in the semi-arid tropics.

Thakur *et al.* (1992) and Mathur (1993) have identified *Crocus sativum* and *Cyamopsis senesalensin* as new hosts for *M. phaseolina*. Su *et al.* (2001) reported that fungus *M. phaseolina* caused charcoal rot disease on more than 500 plant species worldwide.

2.7 : Inoculation techniques

Amule *et al.* (2014) conducted an experiment to know the best method for screening at chickpea genotypes against collar rot under the net house condition by employing four inoculation techniques. The pathogen was multiplied on four substrates viz., chickpea straw, wheat straw, chickpea grain and wheat grain to see the comparative performance. Four genotypes namely JG 315, JG 74, K 850, JG 62, which are resistant, moderately resistant, late wilter, susceptible and highly susceptible to wilt, respectively were used. In order to find out the comparative performance of four screening techniques viz., wheat straw, chickpea straw, wheat grain and chickpea grain were employed against *S. rolfsii*. Experiment conducted in the net house revealed that maximum seedling mortality after initiation of disease ranged from 8 to 20 days being minimum i.e. 5 days in chickpea grain inoculation technique, where one chickpea grain fully covered with mycelial growth was placed at the collar region of 7 days old seedlings.

Cirulli *et al.* (2008) tested three inoculation methods for olive germplasm for resistance to *Verticillium dahliae* Kleb. compared: (i) dipping soil-free washed roots into inoculum (WRI); (ii) dipping root balls with soil into inoculum (RBI); (iii) direct

introduction of the inoculum into a stem incision (SII). Results indicate that the three screening methods had different sensitivity in identifying variable levels of olive resistance to *Verticillium* wilt. Inoculation of washed roots (WRI) was the most drastic and may be used to identify higher levels of resistance (such as that of cv. Frantoio), but was not sensitive enough to detect lower levels of resistance. Inoculation of root balls with soil (RBI) was less destructive and allowed detection of lower resistance expression (such as that of cv. Coratina), not detected by WRI. Stem incision inoculation (SII) failed to discriminate between resistant (cvs Frantoio and Coratina) and susceptible (cv. Leccino) germplasm.

Gupta (2001) reported reaction of chickpea genotypes against collar rot using potting medium mixed with wheat straw, sand medium @ 50g /2 kg sterilized sand + soil (1:1) mixture. Among the inoculation techniques, the use of inoculum in the form of wheat grains fully impregnated with mycelium of the pathogen was found to be simple, feasible and highly effective (Hussain *et al.*, 2005).

2.8 : Morphological variability of *R. bataticola*

2.8.1 : Hyphal variability

Sobti and Sharma (1992) reported that the width of hyphae varied from 4.16-8.48µm in *R. bataticola* isolates from groundnut.

Thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India were studied for morphological variability. Typical right-angled branching of mycelium was found in isolate MP-1 and it was acute to right angled in isolates MP-2 and MP 3 (Devi and Singh, 1998).

Pathogen isolation was done from T9, PU 30, TPU 4, UK 17, UPU97-10 of black gram plants by Suryawanshi *et al.* (2008) from stem and leaves. The mycelium was superficial, hyaline to brown, septate and tree like in form.

Gupta *et al.* (2012) observed variation in size of hyphal width of 40 *R. bataticola* isolates from chickpea. 18 highly virulent isolates were classified under very thin (5.2-6.5 µm), 17 highly virulent and 2 virulent isolates under thin (6.6 -7.9 µm) and only three highly virulent isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 µm) thick hyphal width.

The hyphal cell size varied from 9.38 x 3.80 µm (Rb10) in Karnataka to 14.88 x 7.50 µm (Rb 63) in Telangana. When the data on length of hyphal cell was subjected to statistical analysis, no significant difference was observed between isolates Rb 10, Rb 27, Rb 3, Rb 2, Rb 25 and Rb 24. Ratio between length and width of hyphal cell varied from 1.35 (Rb 27) to 2.96 (Rb 65) [Srinivas, 2016].

2.8.2 Sclerotial variability

Sclerotial diameter of *R. bataticola* in gram varied from 98.1-119.2 µm (Ghosh and Sen, 1973). Hildebrand *et al.* (1945) observed the sclerotia were black, smooth varying from spherical to oblong to irregular shaped. The sclerotial shape of any isolate varied with substrate on which fungus was grown.

Sobti and Bansal (1988) studied on variation in cultural characteristics of root, leaf and seed isolates of *R. bataticola* on different media. The seed and root isolates had the smallest and largest sclerotia, respectively.

Morphological variability among thirty six isolates of *M. phaseolina* from mung bean collected from different region of North, South, North East and Central India were studied by Devi and Singh (1998) who observed bigger sclerotia in isolate MP-2 (400 x 280 µm).

Twenty-four isolates of *R. solani* (teleomorph: *Thanatophorus cucumeris*) collected from soil, root and collar rot or foliage blight-infected plants from several locations of north India were used for the analysis of variability by using morphological and molecular markers. Among the morphological characters, variation was observed in hyphal cell size. Seventeen isolates produced few to abundant, white to dark brown or black, small to large sclerotia generally in the middle of the colony. (Sharma *et al.*, 2005).

Seven isolates of *M. phaseolina* on maize collected from Ludhiana, Hyderabad, Delhi, Arabhavi, Udaipur, Bangalore and Coimbatore by Shekhar *et al.* (2006) observed Hyderabad isolate produced highest number of sclerotia (180.3 sclerotia 9 mm-1 disc) of bigger size (95.7 µm) compared to other isolates. On the basis of sclerotial morphology, isolates were divided into oblong shape having a regular edges and round with regular edges. On the basis of disease expression, Hyderabad isolate was more virulent as compared to other isolates.

The diversity patterns of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans were analyzed based on morphological characteristics such as shape (round, ovoid and irregular) and production of microsclerotia. Microsclerotia were higher in most Mexican isolates than in non-Mexican isolates. However, no significant differences between Mexican and non-Mexican isolates were found in length, width and length/width ratios of microsclerotia (Fernandez et al., 2006).

M. phaseolina isolation was done from T9, PU 30, TPU 4, UK 17, UPU97-10 cultivars of black gram by Suryawanshi et al. (2008) from stem and leaves. The typically dark black to brown colony observed on PDA medium. The mycelium was superficial, hyaline to brown, septate and tree like in form. The sclerotia produced were round, oblong and irregular in shape.

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their morphological characters. The sclerotia formed in different isolates were dark brown to black and size varied from 40 - 600 μm . The isolates from Bangalore (Rb1) and Faridkot (Rb 5) produced the largest size sclerotia ranged from 200 – 600 μm and 100 – 400 μm respectively. Dumka (Rb19) isolate produced small sized sclerotia which ranged from 43.9- 81 μm . They observed most virulent isolate (Rb1 from Bangalore, Karnataka) was fast growing and produced largest sclerotia. The isolates were highly variable in virulence (Aghakhani and Dubey, 2009).

Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions variable in their characters were categorized into three groups. The first group comprised of all isolates from Bidar except RB7 and exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur and Gulbarga except RG3 and RG7 and also an isolate from Bidar RB7 and showed brownish black with flat sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown coloured sclerotia formed at the center of the culture (Manjunatha and Naik, 2011).

Variations in the 35 isolates of *M. phaseolina* collected from the major sesame producing regions in central China comprising of Hubei, Henan, Anhui, Jiangxi province were studied by Linhai *et al.* (2011). Sclerotia quantity and size had high variations. Pingyu and Xiaogan produced no sclerotia after 24 h of culture and sclerotium size of 7 days old culture varied between 0.10 mm (Wuxue) and 0.16 mm (Jiayu) with average of 0.13 mm.

From 14 Iranian provinces, fifty two isolates of *M. phaseolina* were recovered from 24 host plant species. These isolates were studied for average size and time taken for microsclerotia formation. Isolates from Khorasan and Isfahan kashan had the largest microsclerotia while isolates Yazd and Kerman had the smallest microsclerotia (Mahdizadeh *et al.*, 2011).

Forty isolates of *R. bataticola* from chickpea were studied by Gupta *et al.* (2012) observed sclerotial sizes varied from 72.7- 117.2 x 57.1-106.5 μm . Based on sclerotial size, the isolates were divided into three groups viz. minute size (72.7 - 87.5 x 57.1-73.0 μm), medium size (87.6 - 103.2 x 73.5 - 90.0 μm) and bold (103.3 - 117.2 x 90.1- 106.5 μm) size of sclerotia groups. Bold size of sclerotia was observed in six highly virulent isolates belong to Madhya Pradesh and Maharashtra. Rest of the thirty two highly virulent and two virulent isolates placed in minute and medium size of sclerotial groups. The isolate from Betul, Madhya Pradesh produced maximum number of sclerotia (22.3 sclerotia per microscopic field) and isolate from Dhule, Maharashtra exhibited minimum number of sclerotia (8.1 sclerotia per microscopic field). The shape of sclerotia ranged from oblong (12 isolates) to round (28 isolates).

Eleven isolates of *R. bataticola* from different pulse crops viz., redgram, greengram, cowpea, soybean, blackgram showed variability in sclerotial intensity, shape and intensity of pigment synthesis. All isolates were more aggressive on the original host from which it was isolated than on other hosts (Sundravadana *et al.*, 2012).

Isolates of *M. phaseolina* were collected from semi arid regions of Maharashtra, Andhra Pradesh and Karnataka states of India. Variability in shape, length, width and length/width in morphological characters of microsclerotia were studied and observed AP 21 isolate varied in length/width ratio (483.3/200.0 μm) while no significant differences were observed in others (Prasad *et al.*, 2014).

Iqbal and Mukhtar (2014) collected 65 specimens of charcoal rot caused by *M. phaseolina* in mungbean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan. They classified nine isolates under large sized, 26 under small sized and remaining 30 isolates as medium sized sclerotia. Maximum sclerotial size was observed in isolates MP20 and MP-3 showing 29.00 and 27.33 μm diameter respectively, while the isolates MP-39 and MP-28 were found to be the smallest in size. The average sclerotial size of isolates ranged from 17.00 to 29.00 μm .

Significant variations were also observed among these isolates regarding the size of their sclerotia. Maximum sclerotial size was observed in case of isolate Rb-33 (120.11 μm) while the isolate Rb-40 produced the smallest size sclerotia (42.03 μm). The individual average sclerotial sizes of isolates ranged from 42.03-120.11 μm which are given in Table 2. The size of sclerotia of 16 isolates was above 80 μm and was classified as large sized while 16 isolates with sclerotial size less than 60 μm were rated as small sized. The remaining 8 isolates ranged between 60 and 80 μm sclerotial size and were categorized as medium sized (Gode *et al.*, 2018).

2.9 Cultural variability among *Rhizoctonia bataticola*

2.9.1 Variability in radial growth

Dhingra and Sinclair (1973) collected nine isolates of *M. phaseolina* from various parts of United States reported variation in growth rate and colony characters. Further, the growth rate was correlated with pathogenicity.

Lokesha and Benagi (2004) reported among four isolates of *M. phaseolina* in pigeon pea maximum growth was recorded in ICRISAT isolate which was significantly superior over other isolates (82.7 mm) followed by Bidar isolate (78.35 mm).

The radial growth was found maximum after 24 hours of incubation on PDA (40.3 mm) followed by potato sucrose agar (39.0 mm) and it completely covered the PDA plate by 48 hours of incubation (Chowdary and Govindaiah, 2007).

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in their cultural characters. After 48 hours of incubation, highest radial growth in Rb1 (Bangalore) followed Rb4

(Faridkot) was observed. Rb3 (New Delhi) and Rb5 (Faridkot) were next best with statistically similar colony diameters (83.5 mm). Least and similar colony diameter of 51.3 mm was recorded in Rb7 and Rb10 (Kanpur) isolates (Aghakhani and Dubey, 2009).

Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions were highly variable in their cultural characters. Among these, twelve isolates were fast in growth, another twelve were moderate and the remaining six isolates were slow in growth (Manjunatha and Naik, 2011).

Iqbal and Mukhtar (2014) collected 65 isolates of charcoal rot caused by *M. phaseolina* in mung bean from Punjab and Khyber Pakhtunkhwa provinces of Pakistan were studied for cultural variability. Significant differences were noticed among 65 isolates with sixteen isolate rated as fast growing, 11 as slow growing and the rest as medium growing. Maximum colony diameter of 87.17 and 86.67 mm was observed in isolate MP- 7 (Dera Ghazi Khan) and MP-26 (Layyah) proved to be the fast growing, while isolates MP8, MP-29 and MP-30 showed minimum radial growth and rated as slow growing.

Significant differences among 40 isolates of *R. bataticola* collected from different states were observed on the basis of radial growth. The individual average radial growths of 40 isolates of *R. bataticola* ranged from 75.00 to 90.00 mm. Maximum colony diameters of 90.00 and 85.66 mm were observed in case of isolate Rb-1, Rb-6, Rb-11, Rb-12, Rb-15, Rb22 Rb-23, Rb-29, Rb-37 and Rb-38, and proving to be the fast growing, while isolates Rb-5 and Rb-26 showed the minimum radial growths and were rated as medium growing. (Gode *et al.*, 2018).

2.9.2 Variability in colony texture

Dhingra and Sinclair (1978) noticed fluffy growth in stem isolates, partially fluffy growth in root isolates and appressed growth in all other plant part isolates of *M. phaseolina* on soybean.

Thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India and observed MP-1 isolate had cottony mycelial growth while MP-2 isolate showed 5 μ m appressed mycelial growth (Devi and Singh, 1998).

The diversity patterns of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans were analyzed based on presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth (Fernandez *et al.*, 2006).

Twenty three isolates of *R. bataticola* (*M. phaseolina*) causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in colony texture. The majority of nineteen isolates produced suppressed mycelium whereas, 4 isolates in which three from Kanpur and one from Rohtak produced aerial mycelium (Aghakhani and Dubey, 2009).

Edraki and Banishashemi (2010) studied sixty isolates of *M. phaseolina* collected from different parts of Iran on various crops like soyabean, cucumber and sesame. Isolates were grouped in to four phenotypes viz., fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth.

Thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions were highly variable in their colony texture. These isolates were categorized into three groups. The first group comprised of all isolates from Bidar (except RB7) exhibited light brown with fluffy growth and formed dark brown sclerotia centrally with clustering growth pattern. The second group comprised of all isolates from Raichur, Gulbarga (except RG3 and RG7) and an isolate from Bidar (RB7) exhibited brownish black with flat colony and sclerotia which were uniformly distributed. Third group comprised of only RG3 and RG7 isolates from Gulbarga which showed flat colony and brown colour sclerotia formed at the center of the culture (Manjunatha and Naik, 2011).

Variability among the eleven isolates of *R. bataticola* from different pulse crops (redgram, greengram, cowpea, soybean, blackgram) were categorized into three groups viz., linear, fluffy and linear at the end with fluffy growth at the centre. All isolates were more aggressive on the original host from which it was isolated, which was shown by the variability in pathogenic characters (Sundravadana *et al.*, 2012).

2.9.3 Colony pigmentation

Thirty six isolates of *M. phaseolina* of mung bean collected from different region of North, South, North East and Central India were studied for colony colour. MP-1 isolate had pale grey colonies while brown in MP-2 (Devi and Singh, 1998).

The diversity patterns of 30 isolates of *M. phaseolina* of beans 15 from Mexico and 15 from other countries were analyzed based on morphological characteristics such as colour of colony (grey and black) and observed 22 isolates were grey, 4 isolates were white and 4 isolates were black in colour (Fernandez et al., 2006).

Aboshosha et al. (2007) observed colonies of *Macrophomina* isolates of sunflower charcoal rot were mostly grey, dark grey, black and brown for fourteen stem isolates in frequencies of 4, 3, 5 and 1 on PDA and 4, 5, 2 and 2 on Czapek-Dox Agar while white colony was observed in only one isolate on both media.

Twenty three isolates of *R. bataticola* (*M. phaseolina*) causing dry root rot of chickpea collected from 10 different major chickpea growing states (Karnataka, Haryana, New Delhi, Punjab, Uttar Pradesh, Maharashtra, Jharkhand, Rajasthan, Madhya Pradesh and Chattisgarh) of India were highly variable in pigmentation of isolates. Pigmentation varied from white to dull white, creamy, grey and black. Black was prominent among the isolates (Aghakhani and Dubey, 2009).

Charul and Biswas (2009) reported that the mycelium of *R. bataticola* was varied from fluffy dark brown to partially fluffy mycelium colony, with smooth margin. The potency of infection of different isolates of *R. bataticola* was varied on four different cultivars of pigeon pea viz. Prabhat, Bahar, T-7 and T-21. The maximum disease severity was recorded in the plant had attained the age of 47 days followed by 57 and 67 days old plant. The early age group of plant showed low disease severity as compared to middle and above middle age group.

Subramanian et al. (2011) studied on pathogenic and genetic variability among the eleven isolates of *R. bataticola* (Taub.) Butler from different pulse crops. Based on morphological characters, 11 isolates were categorized into three groups viz., linear, fluffy, and linear at the end with fluffy growth at the center. Isolates also showed variability in sclerotial characters (intensity and shape) and intensity of pigment synthesis.

2.10 : Pathogenic variability

Pande *et al.* (2004) screened 29 chickpea germplasm lines and 10 cultivars against dry root rot caused by *R. bataticola* in chickpea using paper towel method developed by Nene *et al.* (1981). Germplasm line ICC 14395, cultivar ICCV 2 and advanced breeding line (ICCX830203-BH-BH-11H) were found resistant to dry root rot, 22 moderately resistant, 19 susceptible and 3 highly susceptible were observed in the remaining lines. The disease severity in the two susceptible line BG 212 and ICC 12267 used as control was rated highest. The identified genotypes were used as source of resistance to dry root rot.

Forty isolates of *R. bataticola* causing dry root rot of chickpea collected from four major chickpea growing states viz., Madhya Pradesh, Chhattisgarh, Gujarat and Maharashtra of central India were screened for virulence of the isolates on susceptible chickpea variety BG 212 by blotter paper technique. The highly virulent isolates from Madhya Pradesh, Chhattisgarh, and Maharashtra were fast growing isolates and exhibited largest sclerotia. The sclerotial size and number of sclerotia of *R. bataticola* isolates had positive correlation with virulence (Gupta *et al.*, 2012).

Lekhraj *et al.* (2012) conducted pathogenic variability of *M. phaseolina* isolates collected on infected chickpea from 9 various parts of Rajasthan, Hissar and New Delhi. Pathogenicity of these isolates tested on 40 day old chickpea cv. C-235 showed there was no direct relationship between cultural characters and virulence of the isolates. The isolate BKN had scanty, suppressed growth was highly pathogenic. Isolates SGN, NHR, DUR, HAU, DEL and HNG having different types of colony growth were moderately pathogenic and the isolates CHR and UPR with different mycelium characters were less pathogenic. Similarly, there was no apparent correlation between the virulence and growth rate of the fungal colonies.

Fifty isolates of *R. bataticola* collected on chickpea from different agro climatic zones in India were evaluated for the degree variability in pathogenicity test. All the isolates proved pathogenic on chickpea cultivar BG 212. The dry root rot disease severity ranged from 4 to 9 (1-9 severity rating scale). The effect of isolates on severity was significant. The pathogenic and non pathogenic isolates were not concentrated in any one particular state/region (Sharma *et al.*, 2012 a).

Among the 94 isolates of *R. bataticola* collected from different agro ecological zones, 8 isolates from Andhra Pradesh and Madhya Pradesh showed less aggressive on chickpea cultivar BG 212 (Sharma *et al.*, 2012 b)

2.11 : Evaluation of bioagent

Several fungal (*Trichoderma* sp.) and bacterial (*Pseudomonas* sp. and *Bacillus* sp.) antagonists, have been successfully used as biocontrol agents in the control of seed and soil borne pathogens like *Sclerotium rolfsii*, *Rhizoctonia bataticola*, *Fusarium oxysporum* and *Sclerotinia sclerotiorum* in the various crops. (Sharma *et al.*, 1999; Mukhopadhyay *et al.*, 1992; Raguchander *et al.*, 1997; Sankar and Jeyarajan, 1996; Mathew and Gupta, 1998 and Kehri and Chandra, 1991). Application of *Trichoderma viride*, *Pseudomonas* and *Bacillus* sp. have been found to substantially control seedling, root rots and stalk rots of maize caused by *F. graminearum*, when used as seed inoculant (Chen *et al.*, 1999; Chang and Kommedahl, 1986; Kommedahl and Chang, 1975). *P. cepacea* was also found to inhibit a range of soil borne fungal pathogens including *F. graminearum*, *F. moniliforme* and *M. phaseolina* (Hebbar *et al.*, 1992). Incidence of charcoal rot was substantially reduced after seed treatment of mungbean and sunflower by *T. harzianum*, *Gliocladium virens* and *Streptomyces* sp. (Hussain *et al.*, 1990).

Plant growth promoting *Pseudomonas* and *Bacillus* species generally employ an array of mechanisms like antibiosis, site competition, HCN production, siderophore production, fluorescent pigments and/or antifungal volatiles (Weller, 1988; Voisard *et al.*, 1989; Cartwright *et al.*, 1995; Gardener *et al.*, 2000; Pal *et al.*, 2000 ; Validov *et al.*, 2005 ; Singh *et al.*, 2006) to antagonize pathogens.

So far, *Trichoderma* spp. are among the most studied fungal biocontrol agents (BCAs) and commercially marketed as biopesticides, biofertilizers and soil amendments (Harman, 2000; Harman *et al.* 2004; Lorito *et al.* 2004). Depending upon the strain, the use of *Trichoderma* in agriculture claims and can provide numerous advantages

- (i) Colonization of the rhizosphere by the biocontrol agents (“rhizosphere competence”) allowing rapid establishment within the stable microbial communities in the rhizosphere ,
- (ii) Control of pathogenic and competitive/deleterious microflora by using a variety of mechanisms thereby improvement of the plant health and

(iii) Stimulation of root growth (Harman *et al.* 2004).

The success of *Trichoderma* as biocontrol agent (BCA) is due to their high reproductive capacity, ability to survive under quite adverse conditions, efficiency in the utilization of nutrients, capacity to modify the rhizosphere, strong aggressiveness against phytopathogenic fungi and efficiency in promoting plant growth and defence mechanisms. These properties have made *Trichoderma* a ubiquitous genus present in any habitat and at high population density (Misra and Prasad, 2003).

Konde *et al.* (2008) conducted *in vitro* experiment for testing efficacy of bioagent against *R. bataticola*. *T. viride* inhibited the growth of *R. bataticola* to the extent of 96.39 per cent.

Salunke *et al.* (2008) tested the antagonistic activity of *T. viride*, *T. harzianum* and *P. fluorescens* against *R. bataticola* by dual culture method. *T. viride* was found to be most efficient bioagent which suppressed mycelial growth ranging from 71.11 to 75.92% followed by *T. harzianum* 67.29 to 74.93% and *P. fluorescens* 60.37 to 70.00%.

Rajput (2010) in *in vitro* experiment found that *T. harzianum* inhibited the maximum radial growth of *R. bataticola* (89.67%). Kumar *et al.* (2011) also reported *T. harzianum* exhibited the highest mycelial growth inhibition (58.9%) against *R. bataticola*.

Veena *et al.* (2014 a) isolated ten *Trichoderma* spp from chickpea rhizosphere and root endophytic region by using serial dilution technique and purified by single hyphal tip method. Among 10 *Trichoderma* isolates, *Trichoderma* isolate-7 (CT7) showed maximum inhibition of growth of *R. bataticola* (83.33%) followed by *Trichoderma* isolate-4 (81.11%) and *Trichoderma* isolate-10 (74.44%) which were on par with each other (Plate 19). The least inhibition was shown by *Trichoderma* isolate-9 (45.56%).

Bandyopadhyay (2003) reported that strain of *Trichoderma* inhibited the growth of *R. bataticola* by 51.1 per cent under *in vitro* conditions.

Paul *et al.* (2008) reported that among 11 *Trichoderma* isolates tested, *Trichoderma harzianum* showed higher inhibition of the growth of *Rhizoctonia solani* by 77 per cent under *in vitro* conditions.

Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot.

Pan (2009) reported the antagonistic potential of *Trichoderma* isolates through production of volatile and non-volatile substances against *M. phaseolina*.

Choudhary et al. (2010) tested four bioagents viz, *Trichoderma viride*, *Trichoderma harzianum*, *Aspergillus versicolor* and *Bacillus firmus* for the control of dry root rot in mungbean. In dual cultures, *Trichoderma viride*, *Trichoderma harzianum* and *Aspergillus versicolor* were effective in inhibiting the growth of *M. phaseolina* to an extent of 61 to 65%.

Dubey et al. (2009) evaluated six isolates of *Trichoderma* belonging to *T. viride* (IARI P-1), *T. virens* (IARI P-3) and *T. harzianum* (IARI P-4) against two isolates of *R. bataticola* (Bangalore and Delhi) *in vitro*. *T. harzianum* followed by *T. viride* caused maximum growth inhibition of *R. bataticola* in dual culture and by producing volatile and non-volatile compounds. Nine seed dressing and eight soil application formulations (granules and pellets) of *T. harzianum* were evaluated against the highly virulent isolate of *R. bataticola* (Bangalore) in pot culture experiments. Seeds treated with Pusa 20SD, followed by Pusa 26SD and soil applied with Pusa Biopellet 10G supported for maximum seed germination, shoot and root length and number of pods per plant and reduced root rot incidence significantly. These treatments also supported maximum colonization of *Trichoderma* in rhizosphere.

Gowdra et al. (2012) evaluated bioagents *in vitro* against *M. phaseolina* which causes dry root rot in chickpea. bioagents tested, maximum inhibition was recorded in *Trichoderma harzianum* (Th-55) (81.48%) followed by *Bacillus subtilis* (75.85) and *T. viride* (TV-27) (74.07) as compared to control.

Gautam et al. (2015) used antagonists viz., *Trichoderma viride* and *T. harzianum* for antagonize the growth of pathogenic fungi of Mungbean viz. *Fusarium oxysporum* and *R. bataticola*. The inhibition in colony of *F.oxysporum* was found 30.66% by *T.viride* and in the *R. bataticola* was 40%. *T. harzianum* showed inhibition in *F.oxysporum* (42.54%) and in *R. bataticola* (48.14%) after 3 days of inoculation.

2.12: Evaluation of fungicides

Goyal and Mehrotra (1981) found Benlate, Bavistin, Thiophanate methyl and Dyrene to be promising among the nine tested fungicides against *R. bataticola* in chickpea *in vitro*. Benomyl and Bavistin were reported to be most effective in inhibiting fungal growth *in vitro*.

Sarwar and Raju (1985) tested 8 fungicides *in vitro* and found Topsin M-70 was highly effective at 1000, ppm against *R. bataticola*.

Rao (1989) reported that seed treatment of cowpea with 0.2% Mancozeb against seed-borne diseases caused by *Colletotrichum lindemuthianum* and *M. phaseolina* resulted in 14% yield increase and successfully controlled the diseases.

Taya *et al.* (1990) in green house tests, found the control of *R. bataticola* by Carbendazim alone or in combination with thiram as seed treatment, presowing, soil- drench and seed treatment + drenching after sowing treated seed of chickpea.

Singh *et al.* (1992) reported that integrated pesticide spray schedule provided better control of dry root rot (*R. bataticola*) in chickpea than any of the simple treatments. Higher yields were observed in seed treatment with carbendazim +Thiram followed by two sprays of Monocrotophos or Endosulfan.

Peshrey *et al.* (1992) reported that Thiram, Captan, PCNB, Mancozeb Iprodione, Carboxine, Carbendazim, Thiobendazole (all 0.2%) and Tridemorph (0.07%) effectively controlled growth on sclerotial germination of *R. bataticola in vitro*. Carbendazin and Thiobendazole were most effective and inhibited growth at the concentration of 0.006%. In successive cultures, the fungus should increase tolerance to Mancozeb and PCNB and to a lesser extent to Thiram, Captan, Carboxin and Iprodione. There was no reduction in sensitivity towards Carbendazim or Thiobendazole.

Prashanthi *et al.* (2000) conducted *in vitro* experiment for testing efficacy of systemic fungicides (Carbendazim, Metalaxyl MZ and Propiconazole) at 250, 500 and 1000 micro g/ml and non-systemic fungicides (Chlorothalonil SC, Chlorothalonil WP, Copper oxychloride, Captafol and Iprodione) at 1000, 2000 and 3000 micro g/ml against *R. bataticola* [*M. phaseolina*]. Carbendazim (Bavistin) and Propiconazole were highly effective with the mean mycelial growth inhibition of 100 and 93.7%, respectively. Carbendazim showed 100% inhibition at all

concentrations. Among the non-systemic fungicides, Chlorothalonil at 3000 ppm was the most effective (82.45%) for inhibition of mycelial growth under *in vitro*.

Prajapati *et al.* (2002) conducted laboratory experiment to determine the efficacy of 11 fungicides against the causative agent (*R. bataticola* [*M. phaseolina*]) of dry root rot in chickpeas. Carbendazim, Carbendazim + Thiram, Carboxin, and Thiophanate-methyl completely inhibited the growth of *R. bataticola*.

Salunke *et al.* (2008) carried out an experiment to study the efficacy of seven fungicides viz., Companion (0.2%), Zineb (0.2%), Tricyclazole (0.2%), Curzate M8TM (0.2%), Copper oxychloride (0.3%), Carbendazim (0.1%) and Thiram (0.3%) against *R. bataticola* by poisoned food technique. Companion and Zineb each at 0.2% concentration showed complete growth inhibition of all isolates. All isolates showed tolerance to Copper oxychloride, and recorded least efficacy.

Ammajamma and Hegde (2009) evaluated efficacies of systemic and non-systemic fungicides against *R. bataticola* at different concentrations. Among systemic fungicides, Carboxin (0.05 and 0.1) and Hexaconazole, Metalaxyl and Traidemifon at 0.1 per cent completely (100%) inhibited the growth of *R. bataticola* and among the non-systemic fungicides tested Thiram at 0.1, 0.2 and 0.3 per cent concentrations was found effective against *R. bataticola*. Kumar *et al.*, (2011) also reported that Bavistin and Vitavax inhibited 100 per cent of mycelial growth at 50 ppm.

In vitro studies conducted by Konde *et al.* (2008) revealed that Carbendazim + Thiram (0.1+ 0.2%), Propiconazole (0.1%) and Thiophanate-M (0.1%) completely inhibited (100%) the growth of *R. bataticola*.

Gowdra *et al.* (2012) evaluated various fungicides *in vitro* against *M. phaseolina* which causes dry root rot in chickpea. Systemic fungicides, cent per cent inhibition were recorded in Hexaconazole, Carbendazim, Difenoconazole and Propiconazole at 50 ppm. whereas Thiophanate methyl found drastic effect at 500 ppm. Whereas, in contact fungicides and combi-products, cent per cent inhibition was observed in Mancozeb, Carboxin 37.5+ Thiram 37.5 and Carbendazim 12 per cent+ Mancozeb 63 per cent WP at 250 ppm.

Mohan *et al.* (2006) reported in fungicidal trials on management of dry root rot of chickpea caused by *R. bataticola* Carbendazim (0.2%) and Etaconazole

(0.1%) were used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 and 18.2 per cent and highest grain yield of 19.2 q/ha and 18.9 q/ha respectively, during rabi 2001-2002 and 2002-2003 crop seasons. The above treatment recorded 57.2 and 52.4% higher yield over control with per rupee return of 12.78 and 11.80 respectively.

Veena *et al.* (2014 b) tested the efficacy of two non systemic fungicides (Copper oxychloride and Captan), two systemic fungicides (Hexaconazole and Tebuconazole) and one antifungal antibiotic Validamycin each at different concentrations against *R. bataticola*, incitant of dry root rot of chickpea under *in vitro* conditions. The fungicides Copper oxychloride, Captan, Hexaconazole and Tebuconazole were found to be highly effective (100%) in inhibiting the mycelial growth of the highly virulent pathogen at all the concentrations tested.

Maruti *et al.* (2017) evaluated *In vitro* efficacy of non systemic, systemic and combi fungicides against *R. bataticola*. Among contact fungicides tested, Ziram recorded 100 per cent inhibition at all the concentrations (i.e., 0.1, 0.2 and 0.3 %) with the mean inhibition, which was significantly superior over all the treatments whereas, Mancozeb and Thiram showed 100 per cent inhibition at 0.3 per cent concentration. Among systemic fungicides tested, Tebuconazole showed complete inhibition at all the concentration (i.e., 0.05, 0.10 and 0.15), whereas, Propiconazole showed 100 per cent inhibition of *R. bataticola* at 0.10 and 0.15 per cent concentration. Among combi products tested, Carbendazim 12% + Mancozeb 63% WP, Trifloxystrobin 25% + Tebuconazole 50% EC and Carboxin 37.5% + Thiram 37.5% WP showed cent per cent (100%) inhibition at all the concentrations (i.e., 0.10%, 0.20% and 0.30 %).

Ravichandran and Hegde (2017) conducted a lab experiment to evaluate six combiprodukt, five contact and four systemic fungicides against *R. bataticola* causing dry root rot in chickpea. Among combiprodukt evaluated, Carbendazim 25% + Mancozeb 50% (Sprint), Carboxin 37.5% + Thiram 37.5% (Vitavax power 75% WP) and Carbendazim 12% + Mancozeb 63% (Saaf) were found to be most effective with complete inhibition of mycelial growth of *R. bataticola* at all the concentrations tested. Among the contact fungicides tested, Chlorothalonil and Mancozeb at 0.2% were effective in inhibiting cent per cent inhibition. Among the four systemic fungicides evaluated against *R. bataticola*, Carbendazim,

Difenconazole and Tebuconazole were found the best with cent per cent inhibition of mycelial growth at all concentrations tested.

2.13 : Host resistance

Pre-germinated seed of 21 chickpea varieties were sown in *R. bataticola* infested soil. *R. bataticola* caused seed rot within 24h in NEC874 and EG234, Bold 2375, BG209, JG62, JG315, ICC 3357 and JG1133 developed necrotic lesion 3 to 5 cells deep on the hypocotyls region within 7 days, RSG-44, AGC677, NEC41, GL269, JG74, ICC8983 and ICC 5003, developed only superficial necrosis along the hypocotyle region. BGM416, BG416, ICC1376 and ICC113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases (Bhatt and Bhatt, 1993).

Krishnamohan *et al.* (1981) screened 20 chickpea lines under field conditions and artificially in the green house. Although BG205 and BG206 were found moderately susceptible in the green house, however, highly resistant under field conditions against *R. bataticola*.

Singh and Mehrotra (1982) reported resistance to *R. bataticola* was shown by the cultivars BG-203, G-543 and Hare Chhole when grown in infested soil.

Reddy *et al.* (1990) tested resistance to wilt and different root rots of chickpea and found the mortality of J.G.-62 (100%) in comparison of Avrodhi and ICC-48.

Baker and Ahmed (1991) tested the resistance of 90 genotypes of chickpea in field infested with wilt and dry root rot pathogens and found that ICC 12263 was most resistant.

Mishra *et al.* (2005) tested 470 germplasm lines and found KG-86 KWR-4, KWR-108 and KWR-277 as a resistant genotypes.

Chaturvedi and Dua (2009) reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot.

Aghakhani and Dubey (2009) collected twenty three isolates of *R. bataticola* causing dry root rot of chickpea from 10 different major chickpea growing states of India were highly variable in their morphological and cultural characters as well as pathogenicity /virulence. The virulence analysis of the isolates on a set of chickpea cultivars namely ICC12441, ICC1224, ICC12450, Pusa 362, BGD112, Pusa1103,

Pusa212, Pusa1088 and under blotter paper as well as sick soil grouped them into 6 pathotypes . The pathotype groups were related to agro ecological region of the country. The most virulent isolate (RBI from Bangalore, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea.

Khan *et al.* (2012) screened sixty germplasm lines of chickpea for their resistance against dry root rot disease in pot . All the germplasm lines were placed in various categories of resistance and susceptibility. Out of sixty germplasm lines screened only nine lines namely, KGD-1189, KGD-1201, KGD-1209, KGD-1215, KGD-1217, KGD-1220, KGD-1221, KGD-1248 and KGD-1289, were found resistant. Ten lines namely KWR-7, KWR-26, KWR-28, KWR-50, KGR-18, KGR-48, KGR-159, KKG-103, KKG-111 and KGD-1201 were found moderately resistant. Seven lines namely KWR-54, KWR-55, KWR-60, KWR-63, KWR-77, KWR-78, KLWR-79, were found moderately susceptible. 17 lines namely KGD-1238, KGD-1239, C-304, KWR-12, KWR-14, KWR-15, KWR-16, KWR-17, KWR-18, KWR-22, KWR-61, KWR-65, KWR-70, KWR-71 KWR-1111, KWR-1211 were found susceptible and rest 17 lines were found highly susceptible against the disease.

2.14 : Evaluation of culture media

Kulkarni *et al.* (1992) studied the growth of *R. bataticola* on different media and reported that it grew best on cotton root extract agar, PDA, Richards agar and Sabouraud's agar. Sclerotial production was excellent on the last 3 media and on Czapek's agar.

Zhou *et al.* (2002) evaluated 12 culture media for the mycelial growth and sclerotial number of *R. solani*. Results showed that cooked rice straw solution agar, water agar, PDA and maltose agar were the best media for *R. solani* while beef extract peptone agar was not preferred. Richard's agar was the best for sclerotial formation, and the highest sclerotial numbers was obtained.

Singh and Singh (2007) cultured *R. solani* on B-R-Agar, Brown's Agar, Flentze's soil extract Agar, Potato Dextrose agar (PDA), Water agar, and Potato marmite agar. Maximum fungal colony diameter (89.7 mm) was recorded on Potato dextrose agar, which was on par with PDA + rice leaf extract agar. The media supporting growth best were Flentza's soil extract agar, Brown's agar and Potato

marmite agar. Water agar supported the minimum growth of *R. solani*. Maximum sclerotial production was observed on PDA, followed by Richard Agar, soybean decoction sucrose agar and B-R agar. PDA was the best basal medium in terms of maximum sclerotial production and viability.

Csondes (2007) studied six different culture medium viz., potato dextrose, malt extract, Czapek Dox, Sabouraud glucose, maize flour and water agar media against *M. phaseolina*. For all the isolates the most favourable temperature regime was 25 to 35 °C and the most advantageous media was the malt-extract-, Sabouraud-glucose- and potato-dextrose-agar media.

Salunke et al. (2009) carried out an experiment to know the effect of nutritional and physiological factors on growth and sclerotial formation of *R. bataticola* isolates. Out of six media, tested, PDA supported the growth and sclerotial formation of *R. bataticola* followed by Czapek's dox, Richard's, Oat meal and peptone agar media. Sclerotial formation on malt extract medium was moderate with poor mycelia growth whereas, peptone agar medium induced rare or no sclerotial formation. Among six carbon sources, highest growth was obtained in dextrose followed by sucrose, mannitol, starch, maltose and cellulose. Sclerotial production in cellulose was maximum followed by dextrose, sucrose, maltose, starch while, it was minimum in mannitol. Sodium nitrate proved better nitrogen source for growth and sclerotial formation than ammonium nitrate.

Sharma et al. (2013) evaluated five culture medium viz., Asthana and Hawkers, Potato dextrose agar, Czapek's dox agar, Corn meal agar and Richards agar to find out the appropriate growth of *R. solani*. Maximum growth of *R. solani* was found in Potato dextrose agar followed by Czapek's dox agar, Corn meal agar, Richards agar and Asthana and Hawkers.

2.15 : In vitro evaluation of botanicals against *R. bataticola*

Out of eighteen species belonging to eleven families of higher plants screened against *Pythium debaryanum*, *Fusarium oxysporum*, *Rhizoctonia solani* and *Sclerotium rolfsii*, only the juice of *Eupatorium cannabinum* leaves completely inhibited the mycelial growth of all the test pathogens at a minimum dilution of 1:1. The leaf juices of *Saraca indica* and *Adhatoda vasica* exhibited complete toxicity at all dilutions against *P. debaryanum* and *F. oxysporum*. The leaf juices of *A. vasica* and *Crotalaria medicaginea* at 1:1 dilution were toxic to *R. solani* and *S.*

rolfsii, respectively. Shade drying of the leaves had no adverse effect, while oven drying produced an adverse effect on the fungitoxicity of the leaves of *E. Cannabinum* (Kumar and Tripthi, 1990).

Savaliya et al. (2015) evaluated the efficacy of various botanicals against *Macrophomina phaseolina* (Tassi.) Goid causing root rot of sesame. The extract of garlic cloves (*Allium sativum* L.) was proved excellent with maximum inhibiting (77.65 %) mycelial growth and scanty sclerotial formation followed by onion bulb extract (*Allium cepa* L.) (63.98 %).

Sharma et al. (2014) carried out a study to see antifungal properties aqueous and alcoholic extract of *D. stramonium* leaves against four important plant pathogenic fungi (*Rhizoctonia solani*, *R.bataticola*, *Fusarium solani* & *F.moniliforme*). The alcoholic extract of datura leaves showed more antifungal activity then aqueous extract. The ethanoilic extract of *D. stramonium* leaf exhibit maximum inhibition against *Rhizoctonia solani* (88%) followed by *Fusarium solani* (76%).

Datar (1999) studied the effect of garlic (*Allium sativum*) extract on charcoal rot (*M. phaseolina*) in sorghum. Phyto-extracts of eleven plant species against *M. phaseolina* of green gram and revealed that the onion bulb extract produced maximum inhibition followed by extract of acacia, ginger, neem, garlic and karanj. (Tandel et al., 2010).

Oluma et al. (2002) screened eight plants for their fungitoxicity against *M. phaseolina* using Czapeck-Dox-agar plant extract media. The plants were *Prosopis africana*, *Caesalpina pulcherrima*, *Calotropis procera*, *Nauclea latifolia*, *Eucalyptus camladulensis*, *Chromolaena odorata*, *Azadirachta indica* and *Thevetia peruviana*. Out of these, the extracts of pulverized bark of *P. africana* and leaves of *N. latifolia* inhibited both radial mycelial growth and sclerotial formation by 100%. There was no inhibition of radial mycelial growth with extracts of *C. procera*, *A. indica* and *T. peruviana*.

Mandhare and Suryawanshi (2008) reported that *Allium sativum* (15%) inhibited the growth of *F. oxysporum* f. sp. *ciceri* and *S. rolfsii* completely and of *R. bataticola* by 88.8%. The extract of *A. indica* inhibited the growth of *F. oxysporum* f. sp. *ciceri*, *S. rolfsii* and *R. bataticola* by 55.5, 33.3 and 58.8% respectively.

Ammajamma *et al.* (2009) evaluated botanicals and reported that eupatorium was highly effective at 10 per cent (81%) followed by garlic and neem leaf extract (50 & 16.67% respectively).

Mandhare and Suryawanshi (2009) screened the antifungal properties of the extracts (at 10% each) of *A. indica* (leaves), *Ocimum sanctum* [*O. tenuiflorum*] (leaves), *Eucalyptus* sp. (leaves), *Nerium indicum* (leaves), *A. sativum* (cloves) and *Zingiber officinale* (rhizomes) against *R. bataticola* by poisoned food technique. The growth of *R. bataticola* was inhibited by 77.77 and 64.44% by the extracts of *A. sativum* and *A. indica*, respectively. The other extracts were not effective.

Sehajpal *et al.* (2009) evaluated the antifungal effect of 44 plant extracts and 8 plant oils against the pathogen, *R. solani* by disc diffusion method. Out of 44 plants tested, 36 plant extracts showed varied degree of antimicrobial effect at different concentrations against the pathogen whereas 8 plants extracts, viz., *Abrus precatorious*, *Acacia auriculiformis*, *Bougainvillea glabra*, *Convolvulus arvensis*, *Hibiscus rosa-sinensis*, *Morus alba*, *Thevatia peruviana*, and *Withania somnifera* did not exert any effect. Among all the plant extracts, *Allium sativum* exhibited strong fungitoxicity even at the lowest concentration, i.e. 100 ppm, with relative magnitude of inhibition 2.0 mm against the pathogen *R. solani*. With regard to the effect of 8 plant oil samples tested, it was noticed *Syzygium aromaticum* are showed strong inhibition i.e. 7.5 mm at 1000 ppm whereas, *Helianthus annus* and *Oryza sativa* were ineffective.

Sangeetha and Shamarao (2013) *in vitro* study found that *T. viride*, *T. harzianum*, Azadirachtin and Neem oil were more effective in inhibiting the mycelial growth of *R. bataticola*.

Shivpuri *et al.* (1997) tested ethanol extracts of 10 plant species viz., *Allium sativum*, *Allium cepa*, *Azadirachta indica*, *Calotropis procera*, *Tagetes erecta*, *Datura stramonium*, *Ocimum sanctum* (*O. tenuiflorum*), *Polylathia longifolia*, *Vinca rosea* (*Catharanthus roseus*) and (*Withania somnifera*) which showed fungitoxic properties against five pathogenic fungi (*Alternaria brassicicola*) *Colletotrichum capsici*, *Fusarium oxysporum*, *Rhizoctonia solani* and *Sclerotinia sclerotiorum* under laboratory condition at two concentrations (500 and 1000 micro g/ml.). Leaf extracts of *A. indica*, *D. stramonium*, *O. sanctum*, *P. longifolia* and *C. roseas* were the most toxic fungi against all the test fungi at 1000 micro g/ml.

Mdee *et al.* (2009) studied the efficacy of seven common invasive plant species against the pathogens viz., *C. gloeosporioides*, *F. oxysporum*, *T. harzianum*, *Pythium ultimum* and *R. solani* and reported that the extracts of *C. macrocephalum*, *N. glauca* and *C. laevigatum* leaves were the most active with average minimum inhibitory concentrations (MICs) values of 0.33, 0.43 and 0.44 mg/ml. *C. macrocephalum* leaves extract was the most active against *C. Gloeosporioides*.

Jha and Sharma (2008) screened an aqueous autoclaved leaf extract of eighty three plant species *in-vitro* for their fungicidal activity against isolates of *R. bataticola* by paper disc diffusion assay technique. The study revealed that only a few plant extract viz., *Rannunculus scleratus*, *Xanthium stramonium*, *Ipomoea carnea*, *Ocimum basilicum* and *Eclipta alba* showed varied antifungal activities.

Kurucheva *et al.* (1997) screened thirteen plant species under *in-vitro* condition for fungitoxicity against *R. solani*, the causal agent of sheath blight of paddy. Fresh leaf samples of *Allium cepa*, *Azadirachta indica*, *Caesalpinia pulcherrima*, *Eucalyptus globulus*, *Calotropis gigantea*, *Ipomoea carnea*, *Lawsonia inermis*, *Ocimum sanctum*, *Parthenium hysterophorus*, *Piper betel*, *Pongamia glabra*, *Prosopis juliflora* and *Thevetia peruviana* were collected and cold and hot water extracts were prepared. The maximum inhibition of mycelial growth was observed with the cold water extract of *P. juliflora* followed by *T. peruviana*. With hot water extraction, *T. peruviana* was the most fungitoxic followed by *P. juliflora*. Complete inhibition of sclerotial production was recorded with the cold water extracts of *C. pulcherrima*, *E. globulus* and *L. inermis* and in the hot water extracts of *C. gigantea*, *O. sanctum* and *P. juliflora* at all the concentrations tested (2.5, 5 and 10%). The fungitoxicity of *C. pulcherrima* and *E. globulus* was not lost even after air drying or autoclaving. All the plant extracts tested maintained their fungitoxicity following thermal treatment at 70 °C for 10 minutes.

Kagale *et al.* (2011) tested the effect of plant extracts of *Zizyphus jujuba* and the *Ipomoea carnea* on the management of sheath blight of rice (*Rhizoctonia solani*). These extracts enhanced the sheath blight resistance in rice seedlings. This effect can be attributed to the direct inhibitory effects of these leaf extracts as well as their ability to elicit systemic resistance against *R. solani*.

The efficacy of neem (*Azadirachta indica*) leaf, pongamia (*Pongamia glabra*) leaf, garlic bulb, tulsi (*Ocimum sanctum*) leaf and lantana (*Lantana camara*) leaf extracts, each at 2.50, 5.00 and 10.00%, was tested *in vitro* against *F. solani* and *R. solani*. The maximum inhibition of mycelial growth of both pathogens was observed with 10% garlic bulb and neem leaf extracts, whereas, tulsi leaf and pongamia leaf extracts were found effective only against *F. solani* (Mallesh *et al.*, 2008).

Chapter-III

MATERIAL AND METHODS

The present investigation was undertaken to study different aspects of *Rhizoctonia bataticola* (Taub.) Butler causing dry root rot of chickpea. The materials used and methods followed are described below.

A. Materials:

Corning and Borosil make glasswares were used throughout the course of study. Glasswares were cleaned by washing with detergent and finally rinsed with tap water. After drying Glasswares were sterilized in hot air oven at 180°C for two hours. The metallic equipments like forceps, needle and cork borer were sterilized by dipping in alcohol and heating to red hot over flame of a spirit lamp.

Surface sterilization of plant parts and diseased materials were done by dipping them in 0.1% mercuric chloride solution for 30 seconds and washed in sterilized water for three times.

The culture media were sterilized in autoclave at 15 lbs pressure per square inch (1.05 kg/cm²) for 20 minutes. The soil and sand were sterilized at 30 lbs pressure per square inch (3.1 kg/cm²) for two hours.

1.1 The culture Medium

Ten culture media were used in the course of investigation with following ingredients in Table 1.

1.2 Seed:

The seeds of chickpea varieties were obtained from All India Coordinated research project on chickpea, Department of Plant Pathology, College of Agriculture, Rajmata Vijayaraje Scindia Krishi Vishwavidyalaya, Gwalior (M.P.). Seeds of chickpea genotypes were used for their reaction (Resistance level) against dry root rot. List of chickpea cultivars are given in Table 2.

Table 1: List of culture medium

S. No.	Name of Media	Ingredients	Quantity
1.	Potato dextrose agar	Peeled and sliced potato	200.0 g
		Dextrose	20.0g
		Agar-agar	20.0g
		Distilled water	1000 ml
2.	Potato dextrose broth	Peeled and sliced potato	200.0 g
		Dextrose	20.0g
		Agar-agar	20.0g
3.	Chickpea grain agar	Chickpea grain	200g
		Agar-agar	20 g
		Distilled water	1000 ml
4.	Corn meal agar	Cornmeal	20.0 g
		Glucose	20.0 g
		Agar-agar	20.0 g
		Distilled water	1000 m
5	Nutrient agar	Peptone	5 g
		Beef extract	3 g
		Agar-agar	15 g
		NaCl	5 g
		Distilled water	1000 ml
6	Czapek's agar	Sucrose	30.0g
		Sodium nitrate	2.0g
		Di-potassium hydrogen phosphate	4.0g
		Magnesium sulphate	0.01g
		Agar-agar	15.0 g
		Distilled water	1000 ml
7	Richards agar	Potassium nitrate	10 g
		Monopotassium dihydrogen phosphate	5 g
		Magnesium sulphate	2.5 g
		Ferric chloride	0.02 g
		Sucrose	50 g
		Agar-agar	20 g

		Distilled water	1000 ml
8	Rice meal agar	Rice meal	20 g
		Agar-agar	20 g
		Distilled water	1000 ml
9	Sorghum grain agar	Sorghum grain	200 g
		Agar-agar	20 g
		Distilled water	1000 ml
10	<i>Trichoderma</i> selective medium	MgSO ₄ .7H ₂ O	0.2 g
		K ₂ HPO ₄	0.9g
		KCl	0.15g
		NH ₄ NO ₃	1.0 g
		Glucose	3.0g
		Chloramphenicol	0.25g
		Rose Bengal	0.15 g
		Captan	0.15 g
		Apron 35-SD	0.025 g
		Agar-agar	20 g
		Distilled water	1000 ml

1.3: Botanicals

Six botanicals viz., *Azadirachta indica* leaf, *Azadirachta indica* seed, *Datura stramonium* leaf, *Eucalyptus globulus* leaf, *Calotropis procera* leaf and *Parthenium hysterophorus* leaf were collected from College of Agriculture, Gwalior Campus, and three botanicals viz., *Allium sativum* (bulb), *Zingiber officinale* (rhizome) and *Allium cepa* (bulb) were purchased from market. The botanicals were evaluated for their fungitoxic properties which can be used against the test pathogen. The list of botanicals is given in Table 3.

Table 2: List of chickpea genotypes

S.No.	Genotypes	S.No.	Genotypes
1	GNG 469	32	GJG 1202
2	GNG 2171	33	AKG 1106
3	GNG 1581	34	NBeG 453
4	GNG 1958	35	H 11-58
5	Phule G 0405	36	JG 41
6	DKG 876	37	KGD 2011-1
7	NBeG 452	38	DIBG 202
8	AKG 1108	39	IPC 2011-138
9	GNG 2207	40	GNG 2249
10	GL 10023	41	BGD 1088
11	RSG 888	42	Phule G 08108
12	BDNG 804	43	H 12-63
13	DKG 964	44	HC 5
14	RSG 931	45	NBeG 780
15	GAG 1107	46	BGD 1082
16	Phule G 0408	47	GNG 2215
17	RSG 963	48	Pusa 547
18	BG 3044	49	GJG 0831
19	PBG 5	50	GNG 2219
20	IPC 2010-14	51	H 10-22
21	BG 3043	52	PG 0109
22	GL 10006	53	IPC 2010-62
23	BAUG 15	54	BG 3046
24	BDNG 2013-1	55	CSJ 769
25	DBGV 102	56	JG 14
26	Phule G 12110	57	BG 372
27	GNG 2264	58	GNG 2259
28	DCP 92-3	59	GJG 1208
29	BG 3051	60	IPC 2007-28
30	GJG 1209	61	GNG 2261
31	Phule G 12107	62	BG 3054
63	IPC 2010-134	95	HK 4

64	H 11-41	96	BG 1053
65	NBeG 454	97	PKV Kabuli-2
66	Phule G 0609-15	98	IPCK 2010-92
67	NDG 13-21	99	CSJK 77
68	BGD 1091	100	PKV Kabuli-4
69	IPC 2010-72	101	CSJK 4
70	H 10-41	102	JGK 1
71	GAG 1107	103	GNG 2285
72	GNG 2226	104	IPCK 2009-165
73	IPC 2009-191	105	NBeG 439
74	JG 16	106	GNG 2281
75	GNG 2263	107	IPCK 2002-29
76	DBGV 101	108	JGK 28
77	Phule G 0611	109	Phule G 12310
78	GJG 1205	110	HK 11-104
79	IPC 2010-112	111	JGK 29
80	BG 3055	112	HK 4
81	GJG 1207	113	NBeG 460
82	NBeG 47	114	DBGV 151
83	Phule G 0302	115	BG 3057
84	BG 3056	116	HK 10-124
85	H 11-22	117	IPCK 2009-79
86	DIBG 201	118	JSC 26
87	GNG 2258	119	NBeG 177
88	GNG 2237	120	NBeG 179
89	BG 1003	121	IPCK 2010- 124
90	HK 09-211	122	Phule G 12407
91	PG 071	123	NBeG 399
92	GNG 2228	124	BG 3059
93	HK 10-103	125	Phule G 12404
94	GNG 1969	126	BGD 1097

1.4: Fungicides

Eleven systemic fungicides namely Pyroclostrobin, Thiophanate methyl, Carbendazim, Tricyclazole, Azoxystrobin, Difenconazole, Hexaconazole, Propiconazole, Tebiconazole, Kitazin, Carboxin and three non-systemic namely Mancozeb, Copper oxychloride and Chlorothalonil were used during study. The list of fungicides is given in Table 4.

1.5: Test pathogen

The pathogen, *Rhizoctonia bataticola* was isolated from infected roots of chickpea cv. L-550.

1.3 Experimental site:

All the field experiments were conducted at the experimental field of Department of Plant Pathology, College of Agriculture, Gwalior during *Rabi* season 2014-15 and 2015-16 and laboratory work was done in the Department of Plant Pathology, College of Agriculture, Gwalior. It is situated in the northern part of Madhya Pradesh at an elevation of 211.52 m from the sea level in between 26° 14' North and 78° 15' East latitude and longitude, respectively.

1.4: Climate and weather conditions

Gwalior is having a typical sub tropical climate with extreme of temperature both in summer (maximum temperature more than 46 °C) and winter (minimum temperature less than 2°). Severe frost is of rare occurrence but cold waves are experienced from the middle of December to the end of February. The average annual rainfall is between 760 to 1060 mm. which occurs in the months of July, August and September with scanty rains during winter months of November to January. This experiment was conducted during the winter season.

The mean weekly values of important weather parameters as recorded at the meteorological observatory, Central farm, R.V.S.K.V.V., Gwalior during the period of trial is given in Table 5 and 6 (Fig. 1 and 2).

Table 3: List of botanicals:

S. No.	Botanical name	Plant part used	Per cent/ 100 ml	Common name	Family	Constituents
1.	<i>Azadirachta indica</i>	Leaf	20 %	Margosal/ Neem	Meliaceae	Mineral like Ca, Mg, I, & P, protein, vitamin A, crude fiber, carbohydrates, fat and alkaloids like quercetin, nimbosterol, azadirachtin, nimboflavone, and nomicinol.
2.	<i>Azadirachta indica</i>	Seed	20 %	Margosal/ Neem	Meliaceae	Ca, Mg, P, an amorphous morgosopicrin, diacetylnimbin, thionemone, solanin, vilasimin, azadirechtol, nimbine, nimbanidol, nomolicinol, solannotide.
3.	<i>Datura stramonium</i>	Leaf	20 %	Dhatura	Solanaceae	Atropin, hyosine, hyoscyamine.
4.	<i>Eucalyptus globules</i>	Leaf	20 %	Lemon scented eucalyptus	Myrtaceae	Eucalyptol, tannin, cerylic alcohol, fatty acid, resin.
5.	<i>Calotropis procera</i>	Leaf	20 %	Crown flower, giant milkweed	Apocynaceae	calotropeol, fattyacids, β -amyrin, sharin, gigantint, β -tetracyclic triterpene.

6.	<i>Zingiber officinale</i>	Rhizome	20 %	Zinger, Adu	Scitamineaceae	Zingiberine, gingerol, camphene, phellandrene, cineol & borneol, gingerin.
7.	<i>Allium sativum</i>	Clove	20 %	Garlic	Liliaceae	Allicin, di-allyl disulphide, di-allyl tri-sulfide allyl propyl, di-allyl polysulfide.
8.	<i>Parthenium hysterophorus</i>	Leaf	20 %	Gajar Ghans, Congress Grass	Asteraceae	parthenolide, artemorin santamarin, reynosin, magnolialide, parthenin, cornopolin, artecanin, balchanin poxyartemorin.
9.	<i>Allium cepa</i>	Bulb	20 %	Onion	Liliaceae	Alliin, allicin, E/Zajoene, sterols, flavones and polyphenolcarboxylic acids

Table 4: List of fungicides

S.No	Coined name	Labelled name	Chemical name	Manufacturer's name
1	Insure	Pyroclostrobin	Methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxyethyl} phenyl} (N-methoxy) carbamate	BASF (India) Ltd. Tiecicon House, Dr. E. Moses road, Mumbai - 400 001, India.
2	Acronis	<i>Thiophanate-methyl</i>	Dimethyl 4,4'-o-phenylene bis(3-thioallophanate)	BASF (India) Ltd. Tiecicon House, Dr. E. Moses road, Mumbai - 400 001, India.
3	Bavistin	Carbendazim	Methyl-1-2-benzimidazole carbamate (MBC)	BASF (India) Ltd. Tiecicon House, Dr. E. Moses road, Mumbai - 400 001, India.
4	Blastin	Tricyclazole	8-methyl-[1,2,4]triazolo[3,4-b][1,3]benzothiazole	Biostadt India Limited Poonam Chambers, 'A' Wing, 6th Floor, Dr. A. B. Road, Worli, Mumbai - 400 018, India.
5	Amistar	Azoxystrobin	methyl (E)-2-[2-[6-(2-cyanophenoxy)pyrimidin-4-yl]oxyphenyl]-3-methoxyprop-2-enoate	Syngenta UK Limited CPC4 Capital Park Fulbourn Cambridgeshire, United Kingdom
6	Score	Difenconazole	1-[[2-[2-chloro-4-(4-chlorophenoxy)phenyl]-4-methyl-1,3-dioxolan-2-yl]methyl]-1,2,4-triazole	Syngenta UK Limited CPC4 Capital Park Fulbourn Cambridgeshire, United Kingdom
7	Contaf	Hexaconazole	2-(2,4-dichlorophenyl)-1-(1,2,4-triazol-1-yl)hexan-2-ol	Rallis India Limited, 156/157, 15th Floor, Nariman Bhavan, 227, Nariman Point, Mumbai 400 021
8	Tilt	Propiconazole	1-[[2-(2,4-dichlorophenyl)-4-propyl-1,3-dioxolan-2-yl]methyl]-1,2,4-triazole	Syngenta UK Limited CPC4 Capital Park Fulbourn Cambridgeshire, United Kingdom
9	Folicur	Tebiconazole	1-(4-chlorophenyl)-4,4-dimethyl-3-(1,2,4-triazol-1-ylmethyl)pentan-3-ol	Bayer Crop Science House Central Avenue, Hiranandani Estate Thane (West) – 400 607.

10	Kitazin	Kitazin	Diethoxyphosphorylsulfanylmethylbenzene	PI Industries Limited Unit No 206, Second Floor, IRIS Tech Park, Sector 48 Sohna Road, Gurugram, Haryana Sushant Lok Phase 1 Gurgaon - 122018 Haryana, India
11	Vitavax	Carboxin	5,6-dihydro-2-methyl-1,4-oxathiin-3-carboxamide	Arysta LifeScience Australia Pty Ltd Level 7, 435 King William Street Adelaide SA 5000
12	Indofil M-45	Mancozeb	zinc;manganese(2+);N-[2-(sulfidocarbothioylamino)ethyl]carbamoithioate	Indofil Industries Limited Off Swami Vivekananda Road, Azad Nagar, Sandoz Baug P. O. Thane - 400607 Maharashtra, India
13	Dhanucop	Copper oxycchloride	dicopper;chloride;trihydroxide	Dhanuka Agritech Limited, 14th Floor, Building 5A, Cyber City, DLF Phase III Gurgaon, Haryana - 122002, India.
14	Kavach	Chlorothalonil	2,4,5,6-tetrachlorobenzene-1,3-dicarbonitrile	Syngenta UK Limited CPC4 Capital Park Fulbourn Cambridgeshire, United Kingdom

Table 5 : Meteorological data 2014-15

Date of observation	Temperature °C		Relative humidity (per cent)		Rainfall (mm)	Evaporation (mm)
	Max.	Mini.	Max.	Mini.		
Nov. 12-18	29.1	10.4	91.8	44.5	0.0	2.2
Nov. 19-25	29.8	8.6	93.5	51.4	0.0	2.8
Nov. - Dec. 26-2	30.1	10.2	89.8	46.5	0.0	3.0
Dec.3-9	28.1	8.2	88.8	49.1	0.0	3.6
Dec.10-16	22.3	9.1	96.1	69.1	35.0	1.5
Dec.17-23	19.2	5.1	97.5	73.7	0.0	1.2
Dec.24-31	18.8	4.0	99.2	74.1	0.0	1.0
Jan. 1-7	18.7	10.8	94.8	78.1	3.0	1.1
Jan. 8-14	18.8	5.6	97.5	74.7	0.0	1.3
Jan. 15-21	17.0	7.5	97.0	84.0	1.4	0.8
Jan. 22-28	16.0	8.7	97.0	92.2	24.0	0.8
Jan.-Feb. 29-4	21.5	6.6	95.5	70.8	0.0	2.2
Feb. 5-11	22.5	8.7	96.0	74.8	4.2	1.8
Feb. 12-18	26.9	10.5	94.7	64.7	0.0	3.0
Feb. 19-25	30.1	13.9	93.4	67.8	2.4	2.6
Feb.-Mar. 26-4	26.0	12.3	91.0	73.0	18.4	2.1
Mar. 5-11	26.8	10.5	92.7	65.4	2.8	3.2
Mar. 12-18	28.2	14.7	93.2	67.2	70.4	3.0

Source : Meteorological observatory, Central farm, Gwalior (M.P.)

Table 6 : Meteorological data 2015-16

Date of observation	Temperature °C		Relative humidity (per cent)		Rainfall (mm)	Evaporation (mm)
	Max.	Mini.	Max.	Mini.		
Nov. 12-18	31.9	11.6	85.2	28.4	0.0	2.5
Nov. 19-25	29.2	10.3	91.4	28.3	0.0	2.0
Nov. - Dec. 26-2	29.3	12.0	84.7	44.4	0.0	1.9
Dec.3-9	28.2	9.0	93.0	56.3	0.0	1.7
Dec.10-16	29.7	9.2	93.1	24.4	0.0	1.4
Dec.17-23	21.9	3.9	93.7	33.2	0.0	1.1
Dec.24-31	23.6	5.6	88.8	28.7	0.0	2.0
Jan. 1-7	27.0	7.7	94.2	33.1	0.0	1.6

Jan. 8-14	26.6	8.9	89.8	38.5	0.0	1.5
Jan. 15-21	18.9	7.0	95.0	69.4	8.0	0.8
Jan. 22-28	23.0	5.6	92.4	39.5	0.0	1.4
Jan.-Feb. 29-4	25.7	8.4	91.4	40.7	0.0	2.4
Feb. 5-11	25.0	7.6	91.4	36.7	0.0	2.6
Feb. 12-18	27.1	11.1	84.1	39.4	0.0	2.8
Feb. 19-25	29.2	11.7	87.0	31.1	0.0	2.8
Feb.-Mar. 26-4	32.0	13.0	87.1	32.3	0.0	4.1
Mar. 5-11	31.5	16.1	84.4	39.1	22.0	3.1
Mar. 12-18	32.1	15.6	76.5	30.1	0.0	5.1

Source : Meteorological observatory, Central farm, Gwalior (M.P.)

B. Methods:

3.1: Symptomatology:

The crop of chickpea was regularly observed at research farm and appearance of symptoms were regularly recorded and described.

3.2: Isolation, purification and identification of the pathogen:

3.2.1: Isolation:

Samples collected during the survey were used for isolation. The diseased chickpea root showing dry root rot symptoms were washed properly with tap water and infected tissue showing disease characteristics and bearing fungal sclerotia were selected. The tissue was cut into 5 mm long and 2 to 3 mm thick pieces. These pieces were surface sterilized with 0.1% HgCl₂ solution for about one minute followed by three changes in sterilized water to remove traces of HgCl₂. Sterilized roots were cut into two-three pieces with the help of sterilized blade in sterilized petri plate. The pieces were dried on sterilized blotting papers and then transferred to Potato dextrose agar (PDA) seeded Petri plates and incubated at 25 ± 1 °C.

3.2.2 : Purification:

After 48 hrs of incubation, hyphal tips of the growing mycelium were marked on the underside of the Petri dish with a glass marker by viewing through a light microscope. The hyphal tips from margins of resulting colonies were cut with the help of sterilized 2mm cork borer and transferred to Petri

dish containing PDA (Dhingra and Sinclair, 1985). The pure cultures of all the isolates grown on PDA slants were stored at $4 \pm 1^{\circ}\text{C}$ for further studies.

3.2.3 : Identification

Isolated fungus was identified based on its morphological characters according to Barnett and Hunter (1986). *R. bataticola* was isolated and the pure culture was maintained by sub-culturing once in a fortnight on PDA slants.

3.3 Survey

An intensive roving survey was conducted during *Rabi* season 2014-15 and 2015-16 in the farmer's field of the districts of Gwalior, Malwa and Nimar region of Madhya Pradesh and presented in Table 7 to find out the incidence of root rot in chickpea. In each district, a minimum of five villages and in each village five fields were examined. A total of 40 fields covering 10 districts were surveyed during flowering stage and information on soil type, cultivars grown, disease incidence and irrigation condition followed were recorded (Plate 1).

Four 1m^2 quadrants were randomly selected in each field and infected plants were counted in each quadrant. Based on infected and total number of plants, per cent disease incidence was calculated. Chickpea plants showing the typical dry root rot symptoms were collected from surveyed areas (Plate 1), packed in labelled paper bags and brought to the laboratory for isolation of the pathogen.

$$\text{Per cent disease incidence} = \frac{\text{Diseased plants}}{\text{Total number of plants}} \times 100$$

3.4: Evaluation of culture media for growth of *Rhizoctonia bataticola*:

Eight culture media viz., Potato dextrose agar, Corn meal agar, Rice meal agar, Nutreint agar medium, Czapek's agar medium, Richard's agar medium, Sorghum grain medium, Chickpea grain medium **were** used to compare the growth rate of *R. bataticola*. The culture medium were prepared by the standardized method and autoclaved at 121.6°C , 15 psi pressure for 20 minutes. Mycelial discs of the seven days old culture of *R. bataticola* (5 mm diameter) were placed on the middle of each pre poured medium and incubated at $25 \pm 1^{\circ}\text{C}$ (Dela Paz et al., 2006). The experiment laid out in complete randomized design and replicated four times.

The diameter of the growth of the fungus was measured inoculation after 3, 5 and 7 days of the incubation and growth was calculated in mm with the help of a scale.

Table-7.: List of surveyed districts of Madhya Pradesh

S.No	District	Name of the village visited	S.No	District	Name of the village visited
1	Gwalior	Ghatigaon	21	Khargone	Khargone
2	Gwalior	Mohana	22	Khargone	Segaon
3	Shivpuri	Lukwasa	23	Khandwa	Deshgaon
4	Shivpuri	Goda	24	Khandwa	Khandwa
5	Shivpuri	Shivpuri	25	Burhanpur	Asirgarh
6	Shivpuri	Kolarash	26	Burhanpur	Bhurhanpur
7	Shivpuri	Badarwas	27	Barwani	Jhulwania
8	Guna	Binaganj	28	Barwani	Sendwa
9	Guna	Guna	29	Barwani	Rajpur
10	Guna	Raghogarh	30	Barwani	Barwani
11	Guna	Rudhiyai	31	Dhar	Kukshi
12	Rajgarh	Biaora	32	Dhar	Bagh
13	Rajgarh	Sarangpur	33	Dhar	Sardarpur
14	Sajapur	Sajapur	34	Dhar	Dhar
15	Sajapur	Makshi	35	Dhar	Ghatabillor
16	Sehore	Panchor	36	Dhar	Betma
17	Dewas	Dewas	37	Jhabua	Bhabra
18	Indore	Mau	38	Jhabua	Jhabua
19	Khargone	Barwaha	39	Alirajpur	Mangode
20	Khargone	Bhikangaon	40	Alirajpur	Alirajpur

3.5 : Evaluation of inoculation techniques to find out the most effective technique for artificial inoculation :

Four different inoculation techniques viz., soil inoculation method (Russell, 1978), root dip technique (Phipps and Stipes, 1973), towel paper method (Nene *et al.*, 1981) and seed soak method (Sakar *et al.*, 1982)

was evaluated to find out the most effective technique for artificial inoculation. The experiment laid out in complete randomized design with three replications. All the inoculation technique tested at 1 %, 2 %, 3 % and 4 % inoculum loads of *R. bataticola*.

3.5.1: Soil inoculation method :

Surface sterilized plastic pots (15 cm diameter) were filled (1.5 kg/pot) with 1.5 kg mixture of sterilized river sand and soil (1:2). The soil is sterilized for three consecutive days at 1.1 kg/cm² for 1 hrs. thr sterilized mixture of soil is inoculated with 30 g inoculums of *R. bataticola* multiplied on sorghum grains. After washing, sorghum grains were soaked in tap water for 12 hrs, and filled into polythene bags (300 g/bag). The grains were autoclaved for 2 subsequent days at 1.1 kg/cm² for 30 minutes and inoculated with 3-day old culture of *R. bataticola*. Inoculated polythene bags were incubated at 25 ± 1⁰ C for 15 days. Ten seeds of highly susceptible cultivar L-550 were sown in each pot in four replications and observed for symptom development.

3.5.2: Root dip technique :

Chickpea seedlings of susceptible genotypes L-550 was grown in plastic pots (12 cm diameter) filled with the mixture of sterilized river sand and soil (1:2) in a greenhouse. Plastic pots were filled up to 2/3 of its volume with sterilized river sand and soil mixture. Ten seeds of L-550 was surface sterilized using 2% sodium hypochlorite for 2 minutes, rinsed in sterile water 2-3 times in order to wash off sodium hypochlorite. These seeds were sown in plastic pots and allow to grow for 8 days. Pure culture of *R. bataticola* was prepared from infected chickpea plants and purified by single hyphal tip isolation on potato dextrose agar (PDA). The trays were surface sterilized with alcohol and placed in a sterilized plastic bags tied with rubber band. The trays were filled with 300 ml potato dextrose broth. The trays were inoculated with 7 mm disk of actively growing culture of *R. bataticola*. The inoculated trays were incubated at 25°C in a incubator for 7 days. The entire contents of trays were diluted with sterilized distilled water (300 ml). Eight days old seedlings grown in sterilized sand soil mixture were uprooted, and cleaned under running tap water. Roots dipped for 1-2 min in inoculum taken in a beaker with a concentration as given above. Inoculated seedlings were transplanted in pre-

irrigated and sterilized mixture of sterilized river sand and soil (1:2) filled in plastic pots. The inoculated pots were incubated in a green house maintained at $25 \pm 1^{\circ}\text{C}$. Inoculated seedlings are observed for the incubation period, latent period and disease mortality up to 60 days after inoculation.

3.5.3: Towel paper method :

Seedlings of cultivar L-550 of chickpea was raised in plastic pots (12 cm diameter) filled with autoclaved mixture of sand and soil (2:1, 1.5 kg/ pot). Apparently healthy looking surface-sterilized (5 minutes in 2.5 percent sodium hypochlorite) 20 seeds were sown in each pot. *R. bataticola* was multiplied on 300 ml potato dextrose broth in trays at $25 \pm 1^{\circ}\text{C}$ for 7 days. Mycelial mats were taken out from the trays at the end of incubation period. After sieving, these mats were ground separately in 300 ml sterilized distilled water with pestle and mortar and placed in a beaker of 250 ml. 10-day-old seedlings of the L-550 were uprooted, washed in running tap water and rinsed in sterilized distilled water. The entire root system of seedlings was dipped in the inoculum for 1-2 seconds. The excess inoculum was removed by touching the edge of the beaker. These seedlings were placed side by side on a sterilized blotter paper so that only cotyledon and roots were covered and green tops of the seedlings remain outside the blotter paper. The blotter paper was folded and moistened adequately, but not excessively with sterilized distilled water. The inoculated seedlings were placed in a tray and incubated at $25 \pm 1^{\circ}\text{C}$ for 8 days provided 12 hrs artificial light and moistened the blotters adequately everyday. Control was maintained by dipping the seedlings of chickpea in sterilized water.

3.5.4: Seed soak method :

Pure culture of *R. bataticola* was multiplied on 300 ml potato dextrose broth in trays at $25 \pm 1^{\circ}\text{C}$ for 7 days. Mycelial mats were taken out from the trays at the end of incubation period. These mats were ground separately in 100 ml sterilized distilled water with pestle and mortar and placed in a beaker of 250 ml. Twenty seeds of cultivar L-550 soaked in pre-prepared inoculum for 2 hours and after that the seed was sown in plastic pots filled with a mixture of sterilized river sand and soil mixture (1:2) in a greenhouse. Inoculated seedlings were observed for the incubation period, latent period and disease mortality up to 60 days after inoculation.

3.5: Variability among *Rhizoctonia bataticola* isolates

3.5.1 : Morphological and cultural characterization of *R. bataticola*

The isolates of *R. bataticola* collected from survey of Madhya Pradesh (Table 8) were characterized for morphological and cultural characteristics. The mycelial discs of 5 mm diameter were cut from the edge of a three days old culture and transferred aseptically to 90 mm Petri dish containing 20 ml PDA. These plates were incubated at 27°C with 12hrs photoperiod. Each treatment was replicated thrice.

The colonies of isolates were characterized for growth rate at 72hrs after incubation. Seven days old cultures were used to record texture, colour and presence or absence of aerial mycelium. At the seventh day after incubation length and width of hyphal cells, number of sclerotia/microscopic field (10x), sclerotia size and shape were recorded. All the cultures were observed daily for recording the time taken for sclerotial initiation.

Table-8: List of isolates of *R. bataticola* collected during the survey

S. No.	District	Place	Isolates code
1	Gwalior	Gwalior	Rb-1
2	Shivpuri	Lukwasa	Rb-2
3	Guna	Guna	Rb-3
4	Rajgarh	Biaora	Rb-4
5	Sajapur	Makshi	Rb-5
6	Sehore	Panchor	Rb-6
7	Sehore	Dewas	Rb-7
8	Indore	Mau	Rb-8
9	Khargone	Khargone	Rb-9
10	Khargone	Segaon	Rb-10
11	Khandwa	Deshgaon	Rb-11
12	Khandwa	Khandwa	Rb-12
13	Barwani	Jhulwania	Rb-13
14	Barwani	Kukshi	Rb-14
15	Dhar	Dhar	Rb-15
16	Jhabua	Jhabua	Rb-16

3.5.2: Characterization of *R. bataticola* isolates for pathogenicity

Pathogenicity test for all the sixteen isolates (Table 8) was performed in laboratory on a susceptible genotype L-550 by paper towel technique (Nene *et al.*, 1981). Inoculum was prepared from the seven days old culture of *Rhizoctonia bataticola* grown on potato dextrose broth medium (PDB). The culture of each isolate was grinded in a blender by adding 50ml of sterile distilled water to each fungal mat. Seven days old seedlings of L-550 grown in sterilized sand soil mixture (1:2) were uprooted, washed under running water and were inoculated by dipping in the inoculum of *R.bataticola* for 1-2 min. Seedlings inoculated with sterile deionized water served as control. Ten inoculated seedlings were placed in paper towel with the shoot left outside, folded, moistened and placed in trays. Trays were transferred to incubators maintained at $25 \pm 1^{\circ}\text{C}$ with 12hrs photoperiod and regularly moistened with sterile deionized water for seven days. The inoculated seedlings were observed daily for symptom expression and the data on incubation period was recorded as and when the symptoms were observed. Four days after inoculation, the data on disease severity was recorded using 1 to 9 disease severity rating scale (Table 9) developed by Nene *et al.* (1991). Those isolates which showed 1-5 rating on disease scale were considered the least virulent while those with 6 rating as virulent and 7-9 rating as highly virulent.

Table 9: Disease rating scale for dry root rot in chickpea (0-9 scale)

Rating	Observation
1	No infection on roots
>1 - ≤3	Very few small lesions on roots
>3 - ≤5	Lesions on roots clear but small, new roots free from infection
>5 - ≤7	Lesions on roots many, new roots generally free from lesions
>7 - 9	Roots infected and completely discoloured

3.6 : Field evaluation of chickpea germplasm against dry root rot under sick micro-plots :

In the investigation, 126 chickpea germplasm/varieties listed in table-2 were evaluated to find out the resistance reaction against dry

root rot under dry root rot sick micro-plots. The experiment was laid out in randomized block design which was replicated twice during 2014-15 and 2015-16. The soil sterilized with formaldehyde (2 per cent) was filled in cemented pits. The *R. bataticola* inoculated sorghum seeds were placed in each line. The seeds of the test line were sown in infested soil. It is essential to pre-treat the pit soil in order to destroy the propagules of disease, harmful moulds and insect pests present in soil. Chemical treatment is the oldest and best method widely used in india for sterilizing the soil. Formaldehyde 2 per cent (2 litres of formalin (40% a.i. in 40 litres water) is used for fumigation. The soil is spread over a concrete floor to a depth of 5 cm and treated with 3 litres of formalin/cubic meter of soil. The second, third and fourth 15 cm layers of soil are turned, similarly treated and made into a pile and covered with a plastic sheet and allowed to stand for three days. For proper result the soil was made free from lumps and clods and drenched thoroughly. The treatment of the soil was given two weeks before its use. After treatment the soil was thoroughly aerated to remove all traces of formalin. The observation was recorded after 45 days. Root rot incidence was recorded as per the following formula :

$$\text{Root rot incidence (\%)} = \frac{\text{Infected plants}}{\text{Total plant}} \times 100$$

Table 10: Disease rating scale for dry root rot of chickpea

Reaction	Per cent mortality
Resistant (R)	0-10
Moderately resistant (MR)	10.1-20
Moderately susceptible	20.1-30
Susceptible (S)	30.1-50
Highly susceptible (H.S)	Above 50

3.7: Management of dry root rot:

3.7.1: *In vitro* evaluation of botanicals for the management of *R. bataticola*

The experiment was carried out at Department of Plant Pathology, College of Agriculture, R.V.S.K.V.V., Gwalior. The effect of phyto extracts of various plant species as listed in Table 4 were tested *in vitro* by poisoned food technique to know their inhibitory effect on the growth of *R. bataticola* (*Macrophomina phaseolina*). The experiment was laid out in complete randomized design and replicated three times. Healthy fresh plant parts *i.e.*, leaves, bulbs, seeds or rhizomes were taken, washed thoroughly with fresh water and finally rinsed with sterilized distilled water and air dried on news paper. Fifty gram of fresh plant parts were cut into small pieces and minced with the help of a grinder by adding 50 ml sterilized distilled water. The phyto extracts were filtered through double-layered muslin cloth in 150ml conical flasks and plugged with non-absorbent cotton. These filtered extracts were autoclaved at 1.2 kg cm⁻² pressure for 20 minutes. Autoclaved extract were individually added into molten sterilized Potato dextrose agar (PDA) @ 20 per cent and mixed thoroughly at the time of pouring in the previously sterilized Petri plates. The Petri plates were inoculated aseptically after solidification by placing 5 mm diameter mycelial disc at the centre, 10 days old pure culture of *R. bataticola*. Four repetitions of each treatment were maintained. The plate without phyto extract served as control. The Petri plates were incubated at 25 ± 1 °C temperature. The observations were taken 3, 5 and 7 days after inoculation. The per cent growth inhibition (PGI) of the pathogen was worked out by using formula given by Vincent (1947) :

$$\text{PGI} = \frac{C-T}{C} \times 100$$

where,

PGI = Per cent growth inhibition
 C = Growth in control
 T = Growth in botanical

3.7.2: *In-vitro* evaluation Fungicides against *R. bataticola*

In laboratory experiment, the efficacy of 16 fungicides (table-4) was assayed against *R. bataticola* by poisoned food technique (Nene and Thapliyal, 1979). All the systemic fungicides namely, Pyroclostrobin,

Thiophanate methyl, Carbendazim, Tricyclazole, Azoxystrobin, Difenconazole, Hexaconazole, Propiconazole, Tebiconazole, Kitazin and Carboxin were evaluated at 0.1 % concentration, while non-systemic fungicides namely, Mancozeb, Copper oxychloride, Chlorothalonil, Mancozeb + Copper oxychloride and Mancozeb + Chlorothalonil were evaluated at 0.3 % concentration. The experiment was laid out in complete randomized design and replicated four times. Effective fungicides were evaluated again at lower concentration (Systemic fungicides at 0.05 % concentration and non-systemic fungicides at 0.2 % concentration). Required quantity of individual fungicide was added separately into molten and cooled sterilized Potato dextrose agar so as to get the desired concentration of fungicides. Later, 20 ml of the poisoned medium was poured into sterile Petriplates. Mycelial discs of 7 mm size from actively growing culture of the fungus were placed at the centre of each plate. Control was maintained without adding any fungicide to the medium. These plates were incubated at $25 \pm 1^{\circ}\text{C}$ and radial colony growth was measured 7 days after inoculation. The efficacy of a fungicide was expressed as per cent inhibition of mycelial growth over control that was calculated by using the formula suggested by Vincent (1947) :

$$\text{PGI} = \frac{\text{C}-\text{T}}{\text{C}} \times 100$$

where,

PGI = Per cent growth inhibition
 C = Growth in control
 T = Growth in fungicides

3.7.3: Evaluation of *Trichoderma* isolates against *R. bataticola*

Random samples were taken from about 20 cm depth of rhizosphere zone of viz., cotton, chilli, chickpea, greengram, pigeonpea, blackgram, sorghum, soybean, maize, tomato for the isolation of *Trichoderma* spp. by means of a soil auger. The samples were placed into clean polythene bags. Three samples were taken from each place. The sampling of the rhizosphere soil was done separately. For rhizosphere analysis, only the soil adhering to the root system was used. The soil samples were taken about 10 cm away

from the root. The composite samples were, mixed thoroughly and screened through 2 mm sieve (Baruah and Dutta, 1978). Soil samples were collected from different crops viz., cotton, chilli, chickpea, greengram, pigeonpea, blackgram, sorghum, soybean, maize, tomato for the isolation of *Trichoderma* spp.. Samples were brought to laboratory and stored at 4°C until used. Five-fold serial dilutions of each soil sample were prepared in sterilized distilled water by using serial dilution method and 0.5 ml diluted sample was poured on the surface of *Trichoderma* specific medium (TSM) [Elad et al., 1982]. Plates were incubated at 25 ± 1°C for 96 hrs. Isolated *Trichoderma* spp. were evaluated by dual culture method (Broadbent et al., 1971).

3.7.4 : Pot evaluation of selected botanicals, bio control agents and chemicals for the management of dry root rot disease

The management of dry rot of chickpea disease under field condition (pot) through chemicals, botanicals and bio-agents selected after *in-vitro* studies were used. The effective treatments were taken for their efficacy against *R. bataticola*.

The experiment was conducted in the pots during rabi season 2014-15 and 2015-16. The seeds of the susceptible cultivar (L-550) were sown in pots and replicated thrice. Seed of L-550 were treated individually with the fungicides and bio-agents while the seeds were soaked in extract of all the botanicals (Table 11). The experiment was laid out in complete randomized design with three replications. Soil was infested by inoculum of *R. bataticola* before sowing. Sorghum seeds were water-soaked overnight, air-dried under room temperature, and placed in conical flasks. The mouth of each flask was plugged with cotton wool, wrapped in aluminium foil, and autoclaved at 15 psi (121° C) for 20 minutes. After cooling, the seeds in flasks were inoculated with 7 mm mycelial disc from 10 days old culture of *R. bataticola* and incubated at 25 ± 1°C for 15 days. The flasks were shaken at alternate days for uniform colonization of the grains. The inoculum produced was used in pot assay. Pots without any treatment served as controls. The pots were placed in a greenhouse. The pots were watered as and when required. Observations on disease incidence were recorded after 60 days after sowing. Details of treatments were given in Table 11.

Table 11: Doses of fungicides, bio-agents and botanicals evaluated against *R. bataticola*

Treatments	Dose
Carbendazim	1 g/kg seed
Difenconazole	1 g/kg seed
Propiconazole	1 g/kg seed
Tebiconazole	1 g/kg seed
Carboxin	1 g/kg seed
Mancozeb	2 g/kg seed
<i>Trichoderma</i> isolate T3	8 g/kg seed
<i>Trichoderma</i> isolate T5	8 g/kg seed
<i>Trichoderma</i> isolate T6	8 g/kg seed
<i>Trichoderma</i> isolate T7	8 g/kg seed
<i>Zingiber officinale</i> rhizome	20%
<i>Datura stramonium</i> leaf	20%
<i>Allium sativum</i> clove	20%
<i>Eucalyptus globus</i> leaf	20%
Control	-

Chapter-IV

RESULTS

The present investigations on various aspects of dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi.) Goid.] included survey on disease incidence, collection of pathogen isolates, diversity among isolates with respect to cultural, morphological and pathological characteristics, evaluation of culture media against growth of *R. bataticola*, evaluation of inoculation techniques, screening of chickpea cultivar, evaluation of botanicals, fungicides and native isolates of *Trichoderma* against *R. bataticola* *in-vitro* and selected botanicals, bio control agents and chemicals were evaluated by pot experiment. The results obtained on these aspects are presented hereunder.

4.1: Survey

A survey was conducted in 2014-2015 *Rabi* cropping season to obtain information on the distribution and incidence of dry root rot of chickpea in respect of soil type, cultivars and irrigation condition.

4.1.1: Disease and regions

During the survey, 40 places (Table-12) were visited and disease incidence was recorded. Total 40 places visited under 15 districts viz., Gwalior (2), Shivpuri (5), Guna (4) Rajgarh (2), Sajapur (2), Sehore (1), Dewas (1), Indore (1), Khargone (4), Khandwa (2) Burhanpur (2), Barwani (4), Alirajpur (2), Jhabua (2) and Dhar (6).

Dry root rot was found at all the sites and the incidence ranged from 2.54 to 20.89 % (Table 12). The mean maximum dry root rot incidence was noticed in Lukwasa (20.89 %) followed by Shivpuri (17.67%), Kolarash (15.88 %), Ruthiyai (15.75 %), Mau (15.34 %), Raghogarh (14.34 %), Panchor (12.07 %) and the least (2.54 %) in Khargone followed by Rajpur (3.20 %), Barwani (3.45 %), Deshgaon (3.52 %), Dewas (3.96 %) and Kukshi (4.14 %).

Table 12 : Survey of chickpea diseases in Madhya Pradesh

S.No	District	Name of the village visited	Rainfed/ Irrigated	Soil type	Dry root rot incidence (%)	
					Range	Average
1	Gwalior	Ghatigaon	Rainfed	Alluvial Soil	8-15%	10.57
2	Gwalior	Mohana	Rainfed	Alluvial Soil	6-9%	7.65
3	Shivpuri	Lukwasa	Rainfed	Alluvial Soil	15-40%	20.89
4	Shivpuri	Goda	Rainfed	Alluvial Soil	2-11%	7.53
5	Shivpuri	Shivpuri	Rainfed	Alluvial Soil	9-21%	17.67
6	Shivpuri	Kolarash	Rainfed	Alluvial Soil	12-19%	15.29
7	Shivpuri	Badarwas	Rainfed	Alluvial Soil	4-17%	10.95
8	Guna	Binaganj	Rainfed	Alluvial Soil	5-11%	9.27
9	Guna	Guna	Irrigated	Alluvial Soil	2-7%	4.58
10	Guna	Raghogarh	Rainfed	Alluvial Soil	8-17%	14.34
11	Guna	Rudhiyai	Rainfed	Alluvial Soil	13-17%	15.88
12	Rajgarh	Biaora	Irrigated	Medium black soil	5-13%	7.67
13	Rajgarh	Sarangpur	Rainfed	Medium black soil	6-11%	9.34
14	Sajapur	Sajapur	Rainfed	Medium black soil	4-13%	8.94
15	Sajapur	Makshi	Rainfed	Medium black soil	4-10%	6.57
16	Sehore	Panchor	Rainfed	Medium black soil	10-13%	12.07
17	Dewas	Dewas	Irrigated	Medium black soil	3-5%	3.96
18	Indore	Mau	Rainfed	Deep black soil	12-18%	15.34
19	Khargone	Barwaha	Irrigated	Deep black soil	4-7%	5.10
20	Khargone	Bhikangaon	Rainfed	Deep black soil	4-9%	6.68
21	Khargone	Khargone	Irrigated	Deep black soil	1-4%	2.54
22	Khargone	Segaon	Rainfed	Deep black soil	3-7%	5.69
23	Khandwa	Deshgaon	Irrigated	Deep black soil	2-5%	3.52
24	Khandwa	Khandwa	Rainfed	Deep black soil	11-13%	11.09
25	Burhanpur	Asirgarh	Irrigated	Medium black soil	5-8%	6.45
26	Burhanpur	Bhurhanpur	Rainfed	Medium black soil	7-11%	9.15
27	Barwani	Jhulwania	Rainfed	Medium black soil	8-11%	10.05
28	Barwani	Sendwa	Rainfed	Medium black soil	4-7%	5.17

29	Barwani	Rajpur	Irrigated	Medium black soil	2-5%	3.20
30	Barwani	Barwani	Irrigated	Deep black soil	1-7%	3.45
31	Dhar	Kukshi	Irrigated	Deep black soil	3-5%	4.14
32	Dhar	Bagh	Rainfed	Deep black soil	4-11%	7.84
33	Dhar	Sardarpur	Rainfed	Medium black soil	5-8%	7.05
34	Dhar	Dhar	Rainfed	Medium black soil	5-9%	7.25
35	Dhar	Ghatabillor	Rainfed	Medium black soil	2-13%	9.38
36	Dhar	Betma	Irrigated	Medium black soil	3-8%	5.26
37	Jhabua	Bhabra	Irrigated	Deep black soil	4-7%	5.29
38	Jhabua	Jhabua	Rainfed	Medium black soil	10-12%	10.86
39	Alirajpur	Mangode	Irrigated	Medium black soil	3-7%	4.69
40	Alirajpur	Alirajpur	Rainfed	Deep black soil	6-14%	10.45

Table 13 : Survey of dry root rot of chickpea in districts of Madhya Pradesh

S. No.	District	Per cent disease incidence
1	Gwalior	9.11
2	Shivpuri	14.47
3	Guna	11.02
4	Rajgarh	8.51
5	Sajapur	7.76
6	Sehore	12.07
7	Dewas	3.96
8	Indore	15.34
9	Khargone	5.00
10	Khandwa	7.31
11	Burhanpur	7.80
12	Barwani	5.47
13	Dhar	6.82
14	Jhabua	8.08
15	Alirajpur	7.57

Data presented in Table 13 showed that maximum disease incidence (15.34 %) was recorded in Indore district (Fig. 3, Plate 10) followed by Shivpuri (14.47 %), Sehore (12.07 %), Guna (11.02 %), Gwalior (9.11 %), Rajgarh (8.51 %) and Jhabua (8.08 %), while minimum disease incidence was recorded in Dewas (3.96 %) followed by Khargone (5.00 %), Barwani (5.47 %), Dhar (6.82 %), Khandwa (7.31 %), Alirajpur (7.57 %), Sajapur (7.76 %), and Burhanpur (7.80 %).

4.1.2: Disease and irrigation condition

During the survey, 27 places found in rainfed condition, while 13 places were found in irrigated condition (Table 14, Fig. 4). Data presented in Table 14 revealed that the percent disease incidence in the irrigated locations varied from 2.54 (Khargone) to 7.67 (Biaora) whereas in the locations where no irrigation facility was available there, it varied from 5.17 (Sendwa) to 20.89 (Lukwasa). Comparatively the disease incidence was less in irrigated fields (4.59 %) over unirrigated ones (10.48 %).

Table 14: Occurrence of dry root rot in different soil types and irrigation condition

Irrigation conditions	Dry root rot incidence (%)	
	Range	Average
Rainfed	5.17-20.89	10.48
Irrigated	2.54-7.67	4.59
Soil types	Dry root rot incidence (%)	
	Range	Average
Alluvial	4.58-20.89	12.24
Medium black	3.20-12.07	7.47
Deep black	2.54-15.34	6.75

4.1.3: Disease and soil type

Three types of soil were recorded in which chickpea crop cultivated. Maximum 17 places recorded under medium black soil followed by deep black soil (12 places) and alluvial (11 places) (Table 14, Fig. 4). The percent disease incidence varied from 4.58 (Guna) to 20.89 (Lukwasa) in locations

where the chickpea crop grown in alluvial soil while it was from 3.20 (Rajpur) to 12.07 (Panchor) in locations where the chickpea crop grown in medium black soil whereas in the locations where the chickpea crop grown in deep black soil, it varied from 2.54 (Khargone) to 15.34 (Mau). Comparatively the disease incidence was less in Deep black (6.75 %) over medium black soil (7.47 %) and Alluvial soil (12.24 %).

4.2: Symptomatology

Symptoms of dry root rot are most commonly observed in chickpea during post-flowering stage. The symptoms of the disease were yellowing of the leaves within a day or such leaves drop (Plate-10) and plant completely dry within a week after the appearance of the first symptom. The seedlings can also get infected. Drooping of petioles and leaflets is confined to those at the top of the plant. Sometimes when the rest of the plant is dried, the topmost leaves are chlorotic. The leaves and stems of affected plants are usually straw coloured, (Plate 11) but in some cases the lower leaves and stems are brown. If the plant are pulled out from the soil and examined the basal stem and main root system of diseased plant, show extensive rooting with most of the lateral roots destroyed (Plate 12). The tissues were weakened and break off easily. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark.

4.3 : Identification of pathogen

On the basis of morphological, cultural characters and pathogenicity test the fungus responsible for dry root rot of chickpea was identified as *Rhizoctonia bataticola*. The growth colony characters and sclerotia of pathogenic isolates of chickpea are presented in Plate 13.

On PDA at 25°C the culture of *R. bataticola* was black with appressed growth. Hypha measured 14.32 x 6.45 µm and sclerotia were irregular measured 160.35 x 151.87 µm in diameter, black in colour.

4.4 : Evaluation of culture medium

In the study, eight culture media viz., Potato dextrose agar, Corn meal agar, Rice meal agar, Nutreint agar medium, Czapek's agar

medium, Richard's agar medium, Sorghum grain medium and Chickpea grain medium were evaluated to find out the appropriate growth of *R. bataticola*. Mycelial growth was recorded at 3, 5 and 7 days after inoculation. The results is presented in Table 15 (Plate 14 and Fig 5).

At 3 days after inoculation, significantly maximum mycelial growth was observed in Potato dextrose agar (29.33 mm) which was found at par with Sorghum grain medium (28.33 mm) and Chickpea grain medium (27.33 mm) while minimum mycelial growth of *R. bataticola* was recorded in Rice meal agar (20.67 mm) followed by Richard's agar medium (24.67 mm), Corn meal agar (26.00 mm), Nutreint agar medium (26.33 mm) and Czapek's agar medium (27.00 mm). Statistically Richard's agar medium, Corn meal agar and Nutreint agar medium were found at par with each other at 3 days after inoculation.

Table 15: Growth of *R. bataticola* on culture medium

Treatments	Mycelial growth (mm) (Days)		
	Three	Five	Seven
Potato dextrose agar	29.33	56.33	85.00
Corn meal agar	26.00	47.67	68.00
Rice meal agar	20.67	34.67	60.33
Nutreint agar medium	26.33	49.67	70.00
Czapek's agar medium	27.00	51.00	73.00
Richard's agar medium	24.67	48.67	67.00
Sorghum grain medium	28.33	53.00	77.00
Chickpea grain medium	27.33	51.67	72.67
S.Em. $\pm$	0.74	0.76	0.83
C.D. at 5%	2.21	2.29	2.50

Significantly maximum mycelial growth of *R. bataticola* was recorded in Potato dextrose agar (56.33 mm) at 5 days after inoculation followed by Sorghum grain medium (53.00 mm) which was found at par with Chickpea grain medium (51.67 mm), however, minimum mycelium growth was observed in Rice meal agar (34.67 mm) followed by Corn meal agar (47.67 mm), Richard's agar medium (48.67 mm), Nutreint agar medium

(49.67 mm) and Czapek's agar medium (51.00 mm). Statistically, Richard's agar medium, Nutreint agar medium and Czapek's agar medium found similar at 5 days after inoculation.

At 7 days after inoculation, significantly maximum mycelial growth was observed in Potato dextrose agar (85.00 mm) followed by Sorghum grain medium (77.00 mm), Czapek's agar medium (73.00 mm) which was found at par with Chickpea grain medium (72.67 mm). Minimum mycelium growth was observed in Rice meal agar (60.33 mm) followed by Richard's agar medium (67.00 mm), Corn meal agar (68.00 mm) and Nutreint agar medium (70.00 mm). Statistically, Czapek's agar medium was found at par with Chickpea grain medium, however, Richard's agar medium and Corn meal agar were found similar at 7 days after inoculation.

4.5: Performance of different inoculation techniques:

In order to find out the comparative performance of four screening techniques viz., seed soaked method, root dip technique, towel paper method and soil inoculation method were employed against dry root rot of chickpea caused by *R. bataticola*.

At 1 % inoculum load, maximum seedling mortality (37.50 %) was recorded in soil inoculation method (Table-16, fig.-4.4) followed by seed soak method (32.50 %) and towel paper method (25.33 %), while minimum seedling mortality was recorded in root dip technique (20.00 %). Statistically, soil inoculation method and seed soak method were found at par with each other.

At 2 % inoculum load, maximum seedling mortality was recorded in soil inoculation method (55.00 %) which was found at par with seed soak method (45.00 %), however, minimum seedling mortality was observed in root dip technique (27.50 %) followed by towel paper method (35.56 %).

Maximum seedling mortality was recorded in soil inoculation method (75.00 %) at 3 % inoculum load which was found at par with seed soak method (67.50 %), while minimum seedling mortality was observed in towel paper method (49.33 %) and root dip technique (37.50 %).

Table-16 : Comparative performance of inoculation techniques at different inoculum loads of *R. bataticola*

Techniques employed	Seedling mortality (%)			
	Inoculum loads			
	1%	2%	3%	4%
Soil inoculation method	37.50 (37.44)*	55.00 (47.88)*	75.00 (60.64)*	87.50 (69.53)*
Root dip technique	20.00 (26.19)	27.50 (31.39)	37.50 (37.44)	50.00 (45.00)
Towel paper method	25.33 (30.12)	35.56 (36.52)	57.50 (49.33)	72.75 (58.59)
Seed soak method	32.50 (34.34)	45.00 (41.83)	67.50 (56.03)	77.50 (62.14)
Control	0.00	0.00	0.00	0.00
SE.m ±	2.42	3.19	3.29	3.01
C.D. at 5 %	7.30	9.62	9.92	9.07

At 4 % inoculum load, maximum seedling mortality was recorded in soil inoculation method (87.50 %) (Fig.-6) which was found at par with seed soak method (77.75 %) followed by towel paper method (72.75 %) , while minimum seedling mortality (50.00 %) was observed in root dip technique.

4.6 : Studies on variability in *Rhizoctonia bataticola*

4.6.1 Cultural variability

Variability in the cultural characters of sixteen isolates of *R. bataticola* was studied by growing on potato dextrose agar medium. The colony growth of the *R. bataticola* isolates measured at 3 DAI, 5 DAI and 7 DAI. Observations on various cultural characters were recorded as described in material and methods and the results obtained are presented in Table 17 to Table 19.

4.6.1.1 Colony diameter

The data on colony diameter presented in the Table 17 revealed that there was a significant difference in the colony which varied from 25.33 to 30.00 mm at three days after inoculation (Table 17, fig. 7). Isolate Rb-5 showed highest colony growth (30.00 mm) which was found at par with Rb-12 (29.67 mm), Rb-15 (29.00 mm) and Rb-11 (29.00 mm) while the least colony diameter was observed with the isolate Rb-7 (25.33 mm) followed by Rb-10 (25.67 mm), Rb-3 (26.33 mm) and Rb-4 (26.33 mm).

At five days after inoculation, maximum mycelium growth was recorded in Isolate Rb-5 (56.00 mm) which was found at par with Rb-12 (55.00 mm), Rb-1 (54.33 mm) and Rb-11 (54.00 mm), however, minimum mycelium growth was recorded in Isolate Rb-10 (46.67 mm) followed by Rb-7 (47.00 mm), Rb-4 (49.00 mm) and Rb-8 (49.00 mm).

Highest mycelium growth was recorded in Isolate Rb-5 (85.67 mm) which was found at par with Rb-12 (84.67 mm), Rb-1 (83.33 mm) and Rb-11 (82.67 mm) at seven days after inoculation (Plate 15), while, the least colony diameter was observed with the isolate Rb-10 (72.00 mm) followed by Rb-7 (72.33 mm), Rb-4 (75.00 mm), Rb-3 (75.33 mm) and Rb-8 (75.33 mm).

Table 17 : Mycelial growth of isolates of *R. bataticola* at different days after inoculation.

Place	Isolates	Mycelial growth (mm) at		
		3 DAI	5 DAI	7 DAI
Gwalior	Rb-1	28.00	54.33	83.33
Lukwasa	Rb-2	27.33	51.00	78.33
Guna	Rb-3	26.33	49.33	75.33
Biaora	Rb-4	26.33	49.00	75.00
Makshi	Rb-5	30.00	56.00	85.67
Panchor	Rb-6	27.00	51.67	79.00
Dewas	Rb-7	25.33	47.00	72.33
Mau	Rb-8	26.67	49.00	75.33
Khargone	Rb-9	26.33	49.67	76.33
Segaon	Rb-10	25.67	46.67	72.00
Deshgaon	Rb-11	29.00	54.00	82.67
Khandwa	Rb-12	29.67	55.00	84.67
Jhulwania	Rb-13	26.67	49.67	76.00
Kukshi	Rb-14	27.33	52.00	79.67
Dhar	Rb-15	29.00	53.33	82.00
Jhabua	Rb-16	27.00	50.67	77.67
	Mean	27.35	51.15	78.46
	SEm±	0.55	0.92	1.14
	C.D. at 5 %	1.57	2.65	3.29

4.6.1.2 : Colony colour

Based on visual observation on colony colour, the cultures were divided into four groups. Black colour colony was observed in 4 isolates (Rb-1, Rb-7, Rb-8, Rb-11) while black with grey aerial mycelium was recorded in 7 isolates (Rb-2, Rb-3, Rb-5, Rb-9, Rb-14, Rb-15, Rb-16). Dark brown (Rb-6, Rb-12, Rb-13) and grey colour (Rb-4, Rb-10) was observed in 3 and 2 isolates (Table 18, Table 19 and Plate 15), respectively.

Table 18: Variability in cultural characteristics of *Rhizoctonia bataticola* isolates

Place	Isolates code	Colony texture	Pigmentation	Aerial mycelium
Gwalior	Rb-1	Appressed	Black	Absent
Lukwasa	Rb-2	Fluffy	Black with grey aerial mycelium	Present
Guna	Rb-3	Fluffy	Black with grey aerial mycelium	Present
Biaora	Rb-4	Velvety	Grey	Absent
Makshi	Rb-5	Fluffy	Black with grey aerial mycelium	Present
Panchor	Rb-6	Fluffy	Dark brown	Present
Dewas	Rb-7	Velvety	Black	Absent
Mau	Rb-8	Velvety	Black	Present
Khargone	Rb-9	Fluffy	Black with grey aerial mycelium	Present
Segaon	Rb-10	Velvety	Grey	Absent
Deshgaon	Rb-11	Appressed	Black	Absent
Khandwa	Rb-12	Appressed	Dark brown	Absent
Jhulwania	Rb-13	Appressed	Dark brown	Absent
Kukshi	Rb-14	Fluffy	Black with grey aerial mycelium	Present
Dhar	Rb-15	Fluffy	Black with grey aerial mycelium	Present
Jhabua	Rb-16	Fluffy	Black with grey aerial mycelium	Present

4.6.1.3 : Colony texture

Isolates categorized into three groups on the basis of colony texture viz., fluffy, appressed and velvety (Table 18, Table 19 and Plate 15). Maximum of 8 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-9, Rb-14 Rb-15, Rb-16) produced fluffy colony while appressed (Rb-1, Rb-11, Rb-12, Rb-13) and velvety growth (Rb-4, Rb-7, Rb-8, Rb-10) produced by 4 isolates each respectively.

Table 19: Grouping of isolates on the basis of cultural characteristics.

Character	Type	Isolates code	No. of isolates
Radial growth (mm)	Less then ≤ 75	Rb-4, Rb-7, Rb-10	3
	>75 to 80	Rb-2, Rb-3, Rb-6, Rb-8, Rb-9, Rb-13, Rb-14, Rb-16	8
	>80	Rb-1, Rb-5, Rb-11, Rb-12, Rb-15	5
Colony texture	Appressed	Rb-1, Rb-11, Rb-12, Rb-13	4
	Velvety	Rb-4, Rb-7, Rb-8, Rb-10	4
	Fluffy	Rb-2, Rb-3, Rb-5, Rb-6, Rb-9, Rb-14 Rb-15, Rb-16	8
Pigmentation	Dark brown	Rb-6, Rb-12, Rb-13	3
	Black	Rb-1, Rb-7, Rb-8, Rb-11	4
	Grey	Rb-4, Rb-10	2
	Black with grey	Rb-2, Rb-3, Rb-5, Rb-9, Rb-14, Rb-15, Rb-16	7
Aerial mycelium	Present	Rb-2, Rb-3, Rb-5, Rb-6, Rb-8, Rb-9, Rb-14, Rb-15, Rb-16	9
	Absent	Rb-1, Rb-4, Rb-7, Rb-10, Rb-11, Rb-12, Rb-13	7

4.6.1.4 : Aerial mycelium

Maximum number of 9 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-8, Rb-9, Rb-14, Rb-15, Rb-16) produced aerial mycelium and 7 isolates (Rb-1, Rb-4, Rb-7, Rb-10, Rb-11, Rb-12, Rb-13) did not develop any aerial mycelium (Table 18, Table 19 and Plate 15).

4.6.2 Morphological variability in *Rhizoctonia bataticola***4.6.2.1 : Sclerotial size**

Data presented in Table 20 showed that the size of sclerotia varied from 101.15 x 98.85 μm (Rb-15) to 181.35 x 170.72 μm (Rb-6) (Plate-16). Maximum sclerotial size was observed in case of isolate Rb-6 (181.35 x

170.72 μm) followed by Rb-7 (178.23 x 172.28 μm), Rb-12 (171.89 x 157.63 μm), Rb-5 (165.52 x 163.27 μm), Rb-1 (160.35 x 151.87 μm), Rb-11 (157.63 x 149.54 μm), while the isolate Rb-15 produced the smallest size sclerotia (101.15 x 98.85 μm) followed by Rb-9 (112.37. x 107.95 μm), Rb-8 (115.75 x 112.24 μm), Rb-2 (123.55 x 116.79 μm), Rb-14 (133.36 x 118.51 μm), Rb-3 (136.82 x 128.23 μm), Rb-13 (140.15 x 129.77 μm), Rb-4 (140.33 x 132.15 μm).

4.6.2.2 : Length : width ratio of sclerotia

The length : width ratio varied from 1.01 (Rb-5) to 1.13 (Rb-14) [Table 20]. Rb-5 (1.01) found least length : width ratio followed by Rb-15 (1.02), Rb-7 (1.03), Rb-8 (1.03), Rb-9 (1.04), Rb-1 (1.05), Rb-16 (1.05), while maximum length : width ratio Rb-14 (1.13), Rb-10 (1.12), Rb-12 (1.09), Rb-13 (1.08), Rb-3 (1.07).

4.6.2.3 : Sclerotial shape

The isolates were categorized into irregular, round and ovoid groups based on shape of sclerotia. Irregular shaped sclerotia were observed in 7 isolates (Rb-1, Rb-3, Rb-6, Rb-9, Rb-13, Rb-14, Rb-16) [Table 20, Plate 16] while round shaped sclerotia were observed in 5 isolates (Rb-4, Rb-5, Rb-7, Rb-10, Rb-11). Ovoid shape of sclerotia was observed in 4 isolates (Rb-2, Rb-8, Rb-12, Rb-15).

4.6.2.4 : Sclerotial intensity

The number of sclerotia per microscopic field when observed through 10x objective varied from 16.60 (Rb-12) to 44.60 (Rb-15) [Table 20, Plate 16 and Fig., 8]. Maximum number of sclerotia was produced by Rb-15 (44.60 sclerotia) followed by Rb-4 (29.80 sclerotia), Rb-2 (26.00 sclerotia), Rb-10 (26.00 sclerotia), Rb-16 (25.80 sclerotia), Rb-9 (24.20 sclerotia), Rb-3 (22.40 sclerotia), Rb-6 (22.40 sclerotia), Rb-8 (22.20 sclerotia), however, minimum number of sclerotia was produced by Rb-12 (16.60 sclerotia), Rb-13 (16.80 sclerotia), Rb-7 (18.40 sclerotia), Rb-1 (18.60 sclerotia), Rb-5 (18.60 sclerotia), Rb-11 (18.80 sclerotia), Rb-14 (18.80 sclerotia).

Table 20 : Variability in morphological characteristics of *Rhizoctonia bataticola* isolates

Place	Isolates code	Sclerotia				Hyphal cell	
		Length x Width (μm)	Length/ Width	Intensity per 10x microscopic field	Shape	Length x Width (μm)	Length/ Width
Gwalior	Rb-1	160.35 x 151.87	1.05	18.60	Irregular	14.32 x 6.45	2.22
Lukwasa	Rb-2	123.55 x 116.79	1.06	26.00	Ovoid	10.95 x 6.24	1.75
Guna	Rb-3	136.82 x 128.23	1.07	22.40	Irregular	10.35 x 5.10	2.03
Biaora	Rb-4	140.33 x 132.15	1.06	29.80	Round	14.65 x 7.40	1.98
Makshi	Rb-5	165.52 x 163.27	1.01	18.60	Round	13.98 x 6.22	2.25
Panchor	Rb-6	181.35 x 170.72	1.06	22.40	Irregular	12.42 x 5.62	2.21
Dewas	Rb-7	178.23 x 172.28	1.03	18.40	Round	11.79 x 6.25	1.89
Mau	Rb-8	115.75 x 112.24	1.03	22.20	Ovoid	13.41 x 7.20	1.86
Khargone	Rb-9	112.37. x 107.95	1.04	24.20	Irregular	13.22 x 6.05	2.19
Segaon	Rb-10	146.39 x 130.24	1.12	26.00	Round	12.46 x 5.13	2.43
Deshgaon	Rb-11	157.63 x 149.54	1.05	18.80	Round	11.50 x 6.28	1.83
Khandwa	Rb-12	171.89 x 157.63	1.09	16.60	Ovoid	13.58 x 7.15	1.90
Jhulwania	Rb-13	140.15 x 129.77	1.08	16.80	Irregular	14.22 x 6.55	2.17
Kukshi	Rb-14	133.36 x 118.51	1.13	18.80	Irregular	10.65 x 6.86	1.55
Dhar	Rb-15	101.15 x 98.85	1.02	44.60	Ovoid	10.47 x 4.88	2.15
Jhabua	Rb-16	140.67 x 133.65	1.05	25.80	Irregular	14.34 x 7.26	1.97

4.6.2.5 : Hyphal cell

The hyphal cell size varied from 10.35 x 5.10 μm (Rb-3) to 14.65 x 7.40 μm (Rb-4) [Table 20]. Maximum hyphal cell size was recorded in Rb-4 (14.65 x 7.40 μm) followed by Rb-16 (14.34 x 7.26 μm), Rb-1 (14.32 x 6.45 μm), Rb-13 (14.22 x 6.55 μm), Rb-5 (13.98 x 6.22 μm), Rb-12 (13.58 x 7.15 μm), Rb-8 (13.41 x 7.20 μm), Rb-9 (13.22 x 6.05 μm), Rb-10 (12.46 x 5.13 μm), Rb-6 (12.42 x 5.62 μm), while minimum hyphal cell size was recorded in Rb-3 (10.35 x 5.10 μm) followed by Rb-15 (10.47 x 4.88 μm), Rb-14 (10.65 x 6.86 μm), Rb-2 (10.95 x 6.24 μm), Rb-11 (11.50 x 6.28 μm), Rb-7 (11.79 x 6.25 μm).

Ratio between length and width of hyphal cell ranged from 1.55 (Rb-14) to 2.43 (Rb-10). Data presented in Table 20 showed that minimum ratio between length and width of hyphal cell was observed in Rb-14 (1.55), followed by Rb-2 (1.75), Rb-11 (1.83), Rb-8 (1.86), Rb-7 (1.89), Rb-12 (1.90), Rb-16 (1.97), Rb-4 (1.98), Rb-3 (2.03), Rb-15 (2.15), Rb-13 (2.17), while maximum ratio between length and width of hyphal cell was observed in Rb-10 (2.43) followed by Rb-5 (2.25), Rb-1 (2.22), Rb-6 (2.21), Rb-9 (2.19).

4.6.3: Pathogenicity test

Pathogenicity of the sixteen isolates of *R. bataticola* obtained from different districts of Madhya Pradesh was tested by artificial root dip inoculation using paper towels. The data pertaining to incubation period and disease severity are presented in the Table 21. Among sixteen isolates, four isolates viz., Rb-1, Rb-2, Rb-5 and Rb-12 showed highly virulent reaction, while seven isolates showed virulent reaction viz., Rb-4, Rb-6, Rb-8, Rb-10, Rb-11, Rb-15 and Rb-16, however five isolates viz., Rb-3, Rb-7, Rb-9, Rb-13 and Rb-14 showed least virulent reaction.

The incubation period ranged from 2.80 days (Rb-5) to 5.00 days (Rb-7). Shortest incubation period was recorded in Rb-5 (2.80 days) followed by Rb-2 (3.00 days), Rb-6 (3.00 days), Rb-12 (3.00 days), Rb-1 (3.60 days) and Rb-15 (3.80 days), while longest incubation period was recorded in Rb-7 (5.00 days) followed by Rb-16 (4.80 days), Rb-13 (4.60 days), Rb-14 (4.60 days), Rb-10 (4.40 days) and Rb-3 (4.40 days).

Table 21: Identification of the isolates along with their disease severity rate and incubation period

Place	Isolates code	Disease severity rating (1-9 rating scale)	Reaction	Incubation period (days)
Gwalior	Rb-1	7.00	Highly virulent	3.60
Lukwasa	Rb-2	7.80	Highly virulent	3.00
Guna	Rb-3	5.40	Least virulent	4.40
Biaora	Rb-4	6.20	Virulent	3.80
Makshi	Rb-5	8.20	Highly virulent	2.80
Panchor	Rb-6	5.80	Virulent	3.00
Dewas	Rb-7	5.00	Least virulent	5.00
Mau	Rb-8	5.80	Virulent	4.00
Khargone	Rb-9	4.60	Least virulent	4.20
Segaon	Rb-10	5.60	Virulent	4.40
Deshgaon	Rb-11	6.20	Virulent	4.00
Khandwa	Rb-12	7.80	Highly virulent	3.00
Jhulwania	Rb-13	5.40	Least virulent	4.60
Kukshi	Rb-14	5.20	Least virulent	4.60
Dhar	Rb-15	6.40	Virulent	3.80
Jhabua	Rb-16	6.20	Virulent	4.80

4.7: Evaluation of chickpea entries against dry root rot caused by *R. bataticola*

In the experiment, total 126 chickpea entries were screened against dry root rot during rabi season 2014-15 and 2015-16. Data on disease incidence was tabulated in Table 22, Table 23, Table 24 and Table 25.

4.7.1: Reaction of chickpea entries during 2014-15

In 2014-16, 126 entries of chickpea were evaluated against dry root rot during 2014-15. Out of 126 entries, twenty four entries viz., Phule G 0405, GL 10023, RSG 888, BAUG 15, DBGV 102, GJG 1209, GJG 1202, NBeG 453, JG 41, Phule G 08108, BGD 1082, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, GJG 1207, NBeG 439, HK 4, NBeG 460, HK 10-124, NBeG 399, BG 3059 showed resistance reaction [Table 22, Table 23]. Forty nine entries viz., GNG 469, GNG 2171, RSG

931, Phule G 0408, PBG 5, GL 10006, DCP 92-3, Phule G 12107, AKG 1106, H 11-58, IPC 2011-138, GNG 2249, NBeG 780, Pusa 547, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, JG 14, GJG 1208, IPC 2010-134, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, Phule G 0611, IPC 2010-112, BG 3055, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, JGK 1, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, DBGV 151, JSC 26, NBeG 179, BGD 1097 found moderately resistant reaction. Thirty two entries have been found moderately susceptible viz., DKG 876, NBeG 452, GNG 2207, DKG 964, RSG 963, BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, H 10-22, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, Phule G 0302, BG 3056, H 11-22, DIBG 201, GNG 2228, , PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177. Sixteen entries viz., GNG 1581, GNG 1958, BDNG 804, GAG 1107, BG 3051, HC 5, BG 3046, JG 16, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010-124, Phule G 12407, Phule G 12404 showed susceptible reaction and five entries have been found to show highly susceptible reaction viz., AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4.

4.7.2: Reaction of chickpea entries during 2015-16

In 2015-16, 126 entries of chickpea were evaluated against dry root rot during 2015-16. Out of 126 entries, seventeen entries viz., GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059 were found resistant [Table 22, Table 24]. Forty two entries viz., GNG 2171, RSG 931, DCP 92-3, Phule G 12107, IPC 2011-138, GNG 2249, NBeG 780, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, GJG 1208, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, IPC 2010-112, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, JSC 26, NBeG 179, Phule G 0405, DBGV 102, NBeG 453, Phule G 08108, BGD 1082, GJG 1207, NBeG 439, HK 4 found moderately resistant reaction. Forty two entries have

been found moderately susceptible viz., NBeG 452, DKG 964, RSG 963, BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, BG 3056, H 11-22, DIBG 201, GNG 2228, , PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177, GNG 469, Phule G 0408, PBG 5, GL 10006, AKG 1106, H 11-58, Pusa 547, GNG 2219, JG 14, IPC 2010-134, Phule G 0611, BG 3055, JGK 1, DBGV 151, BGD 1097. Seventeen entries viz., GNG 1581, GNG 1958, BDNG 804, BG 3051, BG 3046, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010-124, Phule G 12407, Phule G 12404, DKG 876, GNG 2207, H 10-22, Phule G 0302 showed susceptible reaction and eight entries have been found to show highly susceptible reaction viz., AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4, GAG 1107, HC 5, JG 16.

4.7.3: Mean performance of chickpea entries (average of 2014-15 and 2015-16)

Average of two year data was taken for all the entries and according to disease incidence the entries were classified in five groups viz., resistant, moderately resistant, moderately susceptible, susceptible and highly susceptible. 126 entries of chickpea were evaluated against dry root rot during 2014-15 and 2015-16. Out of 126 entries, twenty two entries viz., GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059, NBeG 439 were found resistant [Table 22, Table 25].. Forty three entries viz., GNG 2171, RSG 931, DCP 92-3, Phule G 12107, IPC 2011-138, GNG 2249, NBeG 780, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, GJG 1208, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, IPC 2010-112, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, JSC 26, NBeG 179, Phule G 0405, DBGV 102, NBeG 453, Phule G 08108, BGD 1082, GJG 1207, HK 4, Phule G 0408, H 11-58, Pusa 547, GNG 2219, DBGV 151, BGD 1097 found moderately resistant reaction. Thirty seven entries have been found moderately susceptible viz., NBeG 452, DKG 964, RSG 963,

BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, BG 3056, H 11-22, DIBG 201, GNG 2228, PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177, GNG 469, PBG 5, GL 10006, AKG 1106, JG 14, IPC 2010-134, Phule G 0611, BG 3055, JGK 1, GNG 2207. Seventeen entries *viz.*, GNG 1581, GNG 1958, BDNG 804, BG 3051, BG 3046, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010- 124, Phule G 12407, Phule G 12404, DKG 876, H 10-22, Phule G 0302, JG 16 showed susceptible reaction and seven entries have been found to show highly susceptible reaction *viz.*, AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4, GAG 1107, HC 5.

Table 22 : Field evaluation of chickpea genotypes against dry root rot under disease sick micro-field during 2014-15 and 2015-16.

		Per cent disease incidence and reaction of chickpea entries against dry rot root					
S. No.	Name of entry	2014-15		2015-16		Average	
		Incidence per cent	Reaction	Incidence per cent	Reaction	Incidence per cent	Reaction
1	GNG 469	19.38	MR	21.34	MS	20.36	MS
2	GNG 2171	14.65	MR	16.38	MR	15.51	MR
3	GNG 1581	39.61	S	42.59	S	41.10	S
4	GNG 1958	43.75	S	46.94	S	45.34	S
5	Phule G 0405	9.59	R	11.07	MR	10.33	MR
6	DKG 876	29.17	MS	31.63	S	30.40	S
7	NBeG 452	21.11	MS	23.17	MS	22.14	MS
8	AKG 1108	52.27	HS	55.89	HS	54.08	HS
9	GNG 2207	27.78	MS	30.17	S	28.97	MS
10	GL 10023	6.25	R	7.56	R	6.91	R
11	RSG 888	3.85	R	5.04	R	4.44	R
12	BDNG 804	45.45	S	48.73	S	47.09	S
13	DKG 964	27.50	MS	29.88	MS	28.69	MS

14	RSG 931	16.23	MR	18.05	MR	17.14	MR
15	GAG 1107	49.49	S	52.97	HS	51.23	HS
16	Phule G 0408	18.33	MR	20.25	MS	19.29	MR
17	RSG 963	21.11	MS	23.17	MS	22.14	MS
18	BG 3044	27.50	MS	29.88	MS	28.69	MS
19	PBG 5	19.19	MR	21.15	MS	20.17	MS
20	IPC 2010-14	27.50	MS	29.88	MS	28.69	MS
21	BG 3043	21.11	MS	23.17	MS	22.14	MS
22	GL 10006	19.87	MR	21.87	MS	20.87	MS
23	BAUG 15	4.55	R	5.77	R	5.16	R
24	BDNG 2013-1	54.55	HS	58.27	HS	56.41	HS
25	DBGV 102	9.40	R	10.87	MR	10.14	MR
26	Phule G 12110	24.35	MS	26.57	MS	25.46	MS
27	GNG 2264	23.30	MS	25.46	MS	24.38	MS
28	DCP 92-3	12.69	MR	14.33	MR	13.51	MR
29	BG 3051	41.48	S	44.55	S	43.01	S
30	GJG 1209	8.00	R	9.40	R	8.70	R
31	Phule G 12107	17.69	MR	19.58	MR	18.63	MR
32	GJG 1202	4.17	R	5.38	R	4.77	R

33	AKG 1106	18.33	MR	20.25	MS	19.29	MS
34	NBeG 453	9.13	R	10.58	MR	9.86	MR
35	H 11-58	19.09	MR	20.55	MS	19.82	MR
36	JG 41	7.74	R	9.13	R	8.43	R
37	KGD 2011-1	25.82	MS	28.12	MS	26.97	MS
38	DIBG 202	21.97	MS	24.07	MS	23.02	MS
39	IPC 2011-138	12.69	MR	14.33	MR	13.51	MR
40	GNG 2249	15.26	MR	17.02	MR	16.14	MR
41	BGD 1088	26.50	MS	28.82	MS	27.66	MS
42	Phule G 08108	9.82	R	11.31	MR	10.57	MR
43	H 12-63	24.04	MS	26.24	MS	25.14	MS
44	HC 5	48.61	S	52.04	HS	50.33	HS
45	NBeG 780	15.00	MR	16.75	MR	15.88	MR
46	BGD 1082	9.56	R	11.04	MR	10.30	MR
47	GNG 2215	4.00	R	5.20	R	4.60	R
48	Pusa 547	18.33	MR	20.25	MS	19.29	MR
49	GJG 0831	13.64	MR	15.32	MR	14.48	MR
50	GNG 2219	18.33	MR	20.25	MS	19.29	MR
51	H 10-22	28.75	MS	31.19	S	29.97	S

52	PG 0109	15.56	MR	15.33	MR	15.44	MR
53	IPC 2010-62	17.69	MR	17.58	MR	17.63	MR
54	BG 3046	35.83	S	36.63	S	36.23	S
55	CSJ 769	17.79	MR	17.68	MR	17.73	MR
56	JG 14	10.51	MR	9.04	R	9.78	R
57	BG 372	5.00	R	4.75	R	4.88	R
58	GNG 2259	24.75	MS	24.98	MS	24.87	MS
59	GJG 1208	13.33	MR	13.00	MR	13.17	MR
60	IPC 2007-28	28.13	MS	28.53	MS	28.33	MS
61	GNG 2261	7.14	R	7.00	R	7.07	R
62	BG 3054	9.17	R	8.63	R	8.90	R
63	IPC 2010-134	19.87	MR	21.87	MS	20.87	MS
64	H 11-41	13.64	MR	15.32	MR	14.48	MR
65	NBeG 454	16.67	MR	18.00	MR	17.33	MR
66	Phule G 0609-15	16.67	MR	18.50	MR	17.58	MR
67	NDG 13-21	9.09	R	9.86	R	9.48	R
68	BGD 1091	23.23	MS	25.39	MS	24.31	MS
69	IPC 2010-72	12.70	MR	14.33	MR	13.52	MR
70	H 10-41	20.10	MS	22.10	MS	21.10	MS

71	GAG 1107	17.79	MR	19.68	MR	18.73	MR
72	GNG 2226	11.90	MR	13.50	MR	12.70	MR
73	IPC 2009-191	21.33	MS	23.40	MS	22.36	MS
74	JG 16	47.86	S	51.25	HS	49.55	S
75	GNG 2263	10.42	MR	11.94	MR	11.18	MR
76	DBGV 101	37.50	S	40.38	S	38.94	S
77	Phule G 0611	19.23	MR	21.19	MS	20.21	MS
78	GJG 1205	7.67	R	9.05	R	8.36	R
79	IPC 2010-112	16.67	MR	18.50	MR	17.58	MR
80	BG 3055	20.00	MR	22.00	MS	21.00	MS
81	GJG 1207	9.72	R	11.21	MR	10.47	MR
82	NBeG 47	22.50	MS	24.63	MS	23.56	MS
83	Phule G 0302	29.86	MS	32.35	S	31.11	S
84	BG 3056	25.40	MS	27.67	MS	26.53	MS
85	H 11-22	21.67	MS	23.75	MS	22.71	MS
86	DIBG 201	23.21	MS	25.38	MS	24.29	MS
87	GNG 2258	17.79	MR	19.68	MR	18.73	MR
88	GNG 2237	15.56	MR	17.33	MR	16.44	MR
89	BG 1003	36.67	S	39.50	S	38.08	S

90	HK 09-211	35.42	S	38.19	S	36.80	S
91	PG 071	68.18	HS	69.59	HS	68.89	HS
92	GNG 2228	23.33	MS	25.50	MS	24.42	MS
93	HK 10-103	14.29	MR	16.00	MR	15.14	MR
94	GNG 1969	50.61	HS	54.14	HS	52.37	HS
95	HK 4	62.50	HS	66.63	HS	64.56	HS
96	BG 1053	10.10	MR	11.61	MR	10.85	MR
97	PKV Kabuli-2	11.11	MR	12.67	MR	11.89	MR
98	IPCK 2010-92	12.18	MR	13.79	MR	12.98	MR
99	CSJK 77	45.00	S	48.25	S	46.63	S
100	PKV Kabuli-4	22.50	MS	24.63	MS	23.56	MS
101	CSJK 4	23.33	MS	25.50	MS	24.42	MS
102	JGK 1	19.93	MR	21.93	MS	20.93	MS
103	GNG 2285	15.97	MR	17.76	MR	16.87	MR
104	IPCK 2009-165	16.67	MR	18.50	MR	17.58	MR
105	NBeG 439	8.89	R	10.33	MR	9.61	R
106	GNG 2281	18.33	MR	19.75	MR	19.04	MR
107	IPCK 2002-29	11.51	MR	13.09	MR	12.30	MR
108	JGK 28	40.66	S	43.69	S	42.18	S

109	Phule G 12310	14.84	MR	16.58	MR	15.71	MR
110	HK 11-104	16.67	MR	18.50	MR	17.58	MR
111	JGK 29	25.25	MS	27.51	MS	26.38	MS
112	HK 4	10.00	R	11.50	MR	10.75	MR
113	NBeG 460	5.00	R	6.25	R	5.63	R
114	DBGV 151	18.33	MR	20.25	MS	19.29	MR
115	BG 3057	23.81	MS	26.00	MS	24.90	MS
116	HK 10-124	4.17	R	5.38	R	4.77	R
117	IPCK 2009-79	21.67	MS	23.75	MS	22.71	MS
118	JSC 26	14.58	MR	16.31	MR	15.45	MR
119	NBeG 177	24.04	MS	26.24	MS	25.14	MS
120	NBeG 179	16.23	MR	18.05	MR	17.14	MR
121	IPCK 2010- 124	34.72	S	37.46	S	36.09	S
122	Phule G 12407	35.12	S	37.88	S	36.50	S
123	NBeG 399	5.00	R	6.25	R	5.63	R
124	BG 3059	8.33	R	9.75	R	9.04	R
125	Phule G 12404	44.16	S	47.36	S	45.76	S
126	BGD 1097	18.33	MR	20.25	MS	19.29	MR

Table 23: Reaction of chickpea entries against dry root rot during 2014-15

Reaction	No. of entries	Entries
Resistant (R)	24	Phule G 0405, GL 10023, RSG 888, BAUG 15, DBGV 102, GJG 1209, GJG 1202, NBeG 453, JG 41, Phule G 08108, BGD 1082, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, GJG 1207, NBeG 439, HK 4, NBeG 460, HK 10-124, NBeG 399, BG 3059,
Moderately resistant (MR)	49	GNG 469, GNG 2171, RSG 931, Phule G 0408, PBG 5, GL 10006, DCP 92-3, Phule G 12107, AKG 1106, H 11-58, IPC 2011-138, GNG 2249, NBeG 780, Pusa 547, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, JG 14, GJG 1208, IPC 2010-134, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, Phule G 0611, IPC 2010-112, BG 3055, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, JGK 1, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, DBGV 151, JSC 26, NBeG 179, BGD 1097
Moderately susceptible (MS)	32	DKG 876, NBeG 452, GNG 2207, DKG 964, RSG 963, BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, H 10-22, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, Phule G 0302, BG 3056, H 11-22, DIBG 201, GNG 2228, , PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177
Susceptible (S)	16	GNG 1581, GNG 1958, BDNG 804, GAG 1107, BG 3051, HC 5, BG 3046, JG 16, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010- 124, Phule G 12407, Phule G 12404
Highly susceptible (HS)	5	AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4

Table 24: Reaction of chickpea entries against dry root rot during 2015-16

Reaction	No. of entries	Entries
Resistant (R)	17	GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059,
Moderately resistant (MR)	42	GNG 2171, RSG 931, DCP 92-3, Phule G 12107, IPC 2011-138, GNG 2249, NBeG 780, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, GJG 1208, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, IPC 2010-112, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, JSC 26, NBeG 179, Phule G 0405, DBGV 102, NBeG 453, Phule G 08108, BGD 1082, GJG 1207, NBeG 439, HK 4,
Moderately susceptible (MS)	42	NBeG 452, DKG 964, RSG 963, BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, BG 3056, H 11-22, DIBG 201, GNG 2228, , PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177, GNG 469, Phule G 0408, PBG 5, GL 10006, AKG 1106, H 11-58, Pusa 547, GNG 2219, JG 14, IPC 2010-134, Phule G 0611, BG 3055, JGK 1, DBGV 151, BGD 1097,
Susceptible (S)	17	GNG 1581, GNG 1958, BDNG 804, BG 3051, BG 3046, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010- 124, Phule G 12407, Phule G 12404, DKG 876, GNG 2207, H 10-22, Phule G 0302,
Highly susceptible (HS)	8	AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4, GAG 1107, HC 5, JG 16,

Table 25: Reaction of chickpea entries against dry root rot (Average of 2014-15 and 2015-16)

Reaction	No. of entries	Entries
Resistant (R)	22	GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059, NBeG 439,
Moderately resistant (MR)	43	GNG 2171, RSG 931, DCP 92-3, Phule G 12107, IPC 2011-138, GNG 2249, NBeG 780, GJG 0831, GNG 2219, PG 0109, IPC 2010-62, CSJ 769, GJG 1208, H 11-41, NBeG 454, Phule G 0609-15, IPC 2010-72, GAG 1107, GNG 2226, IPC 2010-112, GNG 2258, GNG 2237, HK 10-103, BG 1053, PKV Kabuli-2, IPCK 2010-92, GNG 2285, IPCK 2009-165, GNG 2281, IPCK 2002-29, Phule G 12310, HK 11-104, JSC 26, NBeG 179, Phule G 0405, DBGV 102, NBeG 453, Phule G 08108, BGD 1082, GJG 1207, HK 4, Phule G 0408, H 11-58, Pusa 547, GNG 2219, DBGV 151, BGD 1097,
Moderately susceptible (MS)	37	NBeG 452, DKG 964, RSG 963, BG 3044, IPC 2010-14, BG 3043, Phule G 12110, GNG 2264, KGD 2011-1, DIBG 202, BGD 1088, H 12-63, GNG 2259, IPC 2007-28, BGD 1091, H 10-41, IPC 2009-191, NBeG 47, BG 3056, H 11-22, DIBG 201, GNG 2228, , PKV Kabuli-4, CSJK 4, JGK 29, BG 3057, IPCK 2009-79, NBeG 177, GNG 469, PBG 5, GL 10006, AKG 1106, JG 14, IPC 2010-134, Phule G 0611, BG 3055, JGK 1, GNG 2207,
Susceptible (S)	17	GNG 1581, GNG 1958, BDNG 804, BG 3051, BG 3046, DBGV 101, BG 1003, HK 09-211, CSJK 77, JGK 28, IPCK 2010- 124, Phule G 12407, Phule G 12404, DKG 876, H 10-22, Phule G 0302, JG 16,
Highly susceptible (HS)	7	AKG 1108, BDNG 2013-1, PG 071, GNG 1969, HK 4, GAG 1107, HC 5,

Table 26: *In-vitro* evaluation of botanicals against *Rhizoctonia bataticola*

Botanicals	Mycelium growth (mm) and per cent mycelium inhibition at					
	3 DAI		5 DAI		7 DAI	
	Growth (mm)	Inhibition (%)	Growth (mm)	Inhibition (%)	Growth (mm)	Inhibition (%)
<i>Azadirachta indica</i> leaf	16.25	38.68	39.00	36.33	64.50	24.34
<i>Azadirachta indica</i> seed	15.75	40.57	37.75	38.37	59.50	30.21
<i>Datura stramonium</i> leaf	12.75	51.89	32.25	47.35	53.00	37.83
<i>Eucalyptus globulus</i> leaf	14.75	44.34	36.25	40.82	56.50	33.72
<i>Zingiber officinale</i> rhizome	12.00	54.72	29.75	51.43	48.25	43.40
<i>Allium sativum</i> clove	14.25	46.23	35.50	42.04	54.50	36.07
<i>Parthenium hysterophorus</i> leaf	21.75	17.92	53.50	12.65	74.50	12.61
<i>Allium cepa</i> bulb	22.25	16.04	53.25	13.06	76.50	10.26
<i>Calotropis procera</i> leaf	23.00	13.21	55.00	10.20	78.75	7.62
Control	26.50	-	61.25	-	85.25	0.00
SEm $\pm$	0.65		1.35		2.09	
C.D. at 5 %	1.87		3.90		6.04	

4.8: Effect of botanicals on the growth *R. bataticola*

*In the present investigation, efficacy of nine botanicals was evaluated against the growth of *R. bataticola*. Data on mycelial growth was recorded at three, five and seven days after inoculation.*

At three days after inoculation, all the botanical significantly inhibited the mycelial growth of *R. bataticola* over control. Among the botanicals, maximum mycelium inhibition was recorded in *Z. officinale* rhizome (54.73 %) [Table 26 and Fig., 9] which was found at par with *D. stramonium* leaf (51.89 %) followed by *A. sativum* clove (46.23 %) and *E. globulus* leaf (44.34 %), the minimum mycelium inhibition was recorded in *C. procera* leaf (13.21 %) which was found at par with *A. cepa* bulb (16.46 %) followed by *P. hystrophorus* leaf (17.92 %). Statistically *A. indica* leaf and *A. indica* seed were found at par with each other.

At five days after inoculation, all tested botanicals significantly inhibited and reduced the mycelial colony diameter over the control. Data presented in Table 26 showed that significantly maximum mycelium growth inhibition was recorded in *Z. officinale* rhizome (51.43 %) which was found at par with *D. stramonium* leaf (47.35 %) followed by *A. sativum* clove (42.04 %) and *E. globulus* leaf (40.82 %), whereas, minimum mycelium inhibition was recorded in *C. procera* leaf (10.20 %) followed by *P. hystrophorus* leaf (12.65 %). and *A. cepa* bulb (13.06 %). Statistically *D. stramonium* leaf and *A. sativum* clove were found at par with each other.

Data recorded on mycelium inhibition of *R. bataticola* at seven days after inoculation showed that all the screened botanicals significantly inhibited and reduced mycelial colony diameter over the control. Significantly maximum mycelium growth inhibition was recorded in *Z. officinale* rhizome (43.40 %) [Plate 17] followed by *D. stramonium* leaf (37.83 %), *A. sativum* clove (36.07 %) and *E. globulus* leaf (33.72 %), whereas, minimum mycelium inhibition was recorded in *C. procera* leaf (7.62 %) followed by *A. cepa* bulb (10.26 %) and *P. hystrophorus* leaf (11.14 %).

Table 27: In-vitro evaluation of fungicides against *Rhizoctonia bataticola*

Fungicides	Mycelial growth (mm)	Inhibition (per cent)
Pyroclostrobin @ 0.1 %	31.67	64.81
Thiophanate methyl @ 0.1 %	26.33	70.74
Carbendazim @ 0.1 %	0.00	100.00
Tricyclazole @ 0.1 %	0.00	100.00
Azoxystrobin @ 0.1 %	27.67	69.26
Difenconazole @ 0.1 %	0.00	100.00
Hexaconazole @ 0.1 %	0.00	100.00
Propiconazole @ 0.1 %	0.00	100.00
Tebiconazole @ 0.1 %	0.00	100.00
Kitazin @ 0.1 %	8.00	91.11
Carboxin @ 0.1 %	0.00	100.00
Mancozeb @ 0.3 %	0.00	100.00
Copper oxychloride @ 0.3 %	0.00	100.00
Chlorothalonil @ 0.3 %	42.33	52.96
Mancozeb + Copper oxychloride @ 0.3 %	0.00	100.00
Mancozeb + Chlorothalonil @ 0.3 %	0.00	100.00
Control	90.00	-
SEm ±	0.83	
C.D. at 5 %	2.39	

4.9.1: Effect of fungicides on the growth *R. bataticola*

Efficacy of 11 systemic and five non systemic fungicides were tested at 0.1 % and 0.3 % concentrations respectively in the laboratory against *Rhizoctonia bataticola* (Table 27, Fig., 10). The results indicated that there was significant difference among the fungicides in inhibiting the mycelial growth of *R. bataticola*. Among the fungicides evaluated, 100 per cent mycelium inhibition was recorded in Carbendazim @ 0.1 %, Tricyclazole @ 0.1 %, Difenconazole @ 0.1 %, Hexaconazole @ 0.1 %, Propiconazole @ 0.1 %, Tebiconazole @ 0.1 %, Carboxin @ 0.1 %, Mancozeb @ 0.3 %, Copper oxychloride @ 0.3 %, Mancozeb + Copper oxychloride @ 0.3 % and Mancozeb + Chlorothalonil @ 0.3 % (*Plate-18*), while maximum mycelium growth in

Chlorothalonil @ 0.3 % (42.33 mm) followed by Pyroclostrobin @ 0.1 % (31.76 mm), Azoxystrobin @ 0.3 % (27.67 mm) and Thiophanate methyl @ 0.3 % (26.33 mm).

Table 28: Standarization of concentration of effective fungicides against *Rhizoctonia bataticola*

Fungicides	Mycelial growth (mm)	Inhibition (%)
Carbendazim @ 0.05 %	0.00	100.00
Tricyclazole @ 0.05 %	23.67	73.70
Difenconazole @ 0.05 %	0.00	100.00
Hexaconazole @ 0.05 %	11.33	87.41
Propiconazole @ 0.05 %	0.00	100.00
Tebiconazole @ 0.05 %	0.00	100.00
Carboxin @ 0.05 %	0.00	100.00
Mancozeb @ 0.2%	0.00	100.00
Copper oxycloiride @ 0.2 %	13.33	85.19
Mancozeb + Copper oxycloiride @ 0.2 %	16.67	81.48
Mancozeb + Chlorothalonil @ 0.2 %	22.33	75.19
Control	90.00	

4.9.2: Standarization of concentration of effective fungicides against *Rhizoctonia bataticola*

Seven systemic fungicides namely Carbendazim, Tricyclazole, Difenconazole, Hexaconazole, Propiconazole, Tebiconazole, Carboxin at 0.1 % concentration and four non systemic fungicides namely Mancozeb, Copper oxycloiride, Mancozeb + Copper oxycloiride and Mancozeb + Chlorothalonil at the concentration 0.3 % were found significantly superior over control under *in-vitro* condition. After the experiment all the selected fungicides were evaluated again at lower concentration. Among the fungicides, 100 per cent mycelium inhibition was recorded in Carbendazim @ 0.05 %, Difenconazole @ 0.05 %, Propiconazole @ 0.05 %, Tebiconazole @ 0.05 %, Carboxin @ 0.05 % and Mancozeb @ 0.2% (Table-28, Plate-19 and Fig.-11). least mycelium inhibition was recorded in Tricyclazole @ 0.05 % (73.70 %)

followed by Mancozeb + Chlorothalonil @ 0.2 % (75.19 %) and Mancozeb + Copper oxychloride @ 0.2 % (81.48 %).

4.10.1: Isolation of *Trichoderma* isolates from soils and rhizospheric zones of different crops in districts of M.P.

Fifty four soil samples were collected from rhizosphere of five different crops and seven districts of Madhya Pradesh. Out of these seven strains of *Trichoderma* were isolated through serial dilution technique. The details of the districts and crop rhizospheres from which strains of *Trichoderma* isolated were given in detail in the Table 29. The result indicated that *Trichoderma* spp. could grow and survive in various kinds of soil conditions viz., black soil, alluvial soil and medium black medium.

Table-29 : Isolation of *Trichoderma* spp. from soil of different districts and rhizospheric zones of different crops of Madhya Pradesh

Trichoderma isolates	Place	Crop	Soil type
T-1	Khandwa	Cotton	Black soil
T-2	Bharwani	Chilli	Black soil
T-3	Dhar	Chickpea	Alluvial soil
T-4	Harda	Greengram	Medium black soil
T-5	Indore	Chickpea	Alluvial soil
T-6	Khargone	Pigeonpea	Black soil
T-7	Dewas	Chickpea	Medium black soil

4.10.2: *In vitro* evaluation antagonistic mycoflora (*Trichoderma* spp.) against *R. bataticola* in dual culture technique

All isolates obtained from different crop from different district were evaluated against *R. bataticola* and designated them as T1, T2, T3, T4, T5, T6 and T7. All the *Trichoderma* isolates showed significant reduction in mycelial growth of *R. bataticola* when compared to control. The data pertaining to per cent inhibition of mycelial growth of *R. bataticola* due to *Trichoderma* isolates are presented in Table 30 and Fig.,12. Among the *Trichoderma* isolates, T-6 showed maximum

inhibition of growth of *R. bataticola* (67.22 %) [Plate 20] followed by T-7 (63.61%), T-3 (59.72 %) and T-5 (57.50 %). The least inhibition was shown by T-1 (50.83 %) followed by T-4 (52.78%) and T-2 (54.72 %). Statistically, T-6 and T-7 were found at par with each other, while T-1 and T-4 showed similar effect on mycelium inhibition of *Rhizoctonia bataticola*.

Table 30: *In-vitro* evaluation of *Trichoderma* isolated from soil of different districts and rhizospheric zones of five crops

<i>Trichoderma</i> isolates	Mycelium growth (mm) and per cent mycelium inhibition	
	Growth (mm)	Inhibition per cent
T-1	44.25	50.83
T-2	40.75	54.72
T-3	36.25	59.72
T-4	42.50	52.78
T-5	38.25	57.50
T-6	29.50	67.22
T-7	32.75	63.61
Control	90.00	-
SEm ±	1.38	
C.D. at 5 %	4.04	

4.11: Pot evaluation of selected botanicals, bio control agents and chemicals for the management of dry root rot disease

Six fungicides, four botanicals and four strains of *Trichoderma* were selected for pot experiment after *In vitro* screening and evaluated against dry root rot caused by *R. bataticola*.

During 2014-15, all the treatments significantly reduced the incidence of dry root rot over untreated control. The data (Table 31, Fig.,13) depict that minimum disease incidence was recorded in Carbendazim (26.67 %) which was found at par with Propiconazole (30.00 %) Carboxin (30.00 %), Mancozeb, (33.33 %) and Difenconazole (33.33 %), while significantly maximum disease incidence was recorded in untreated

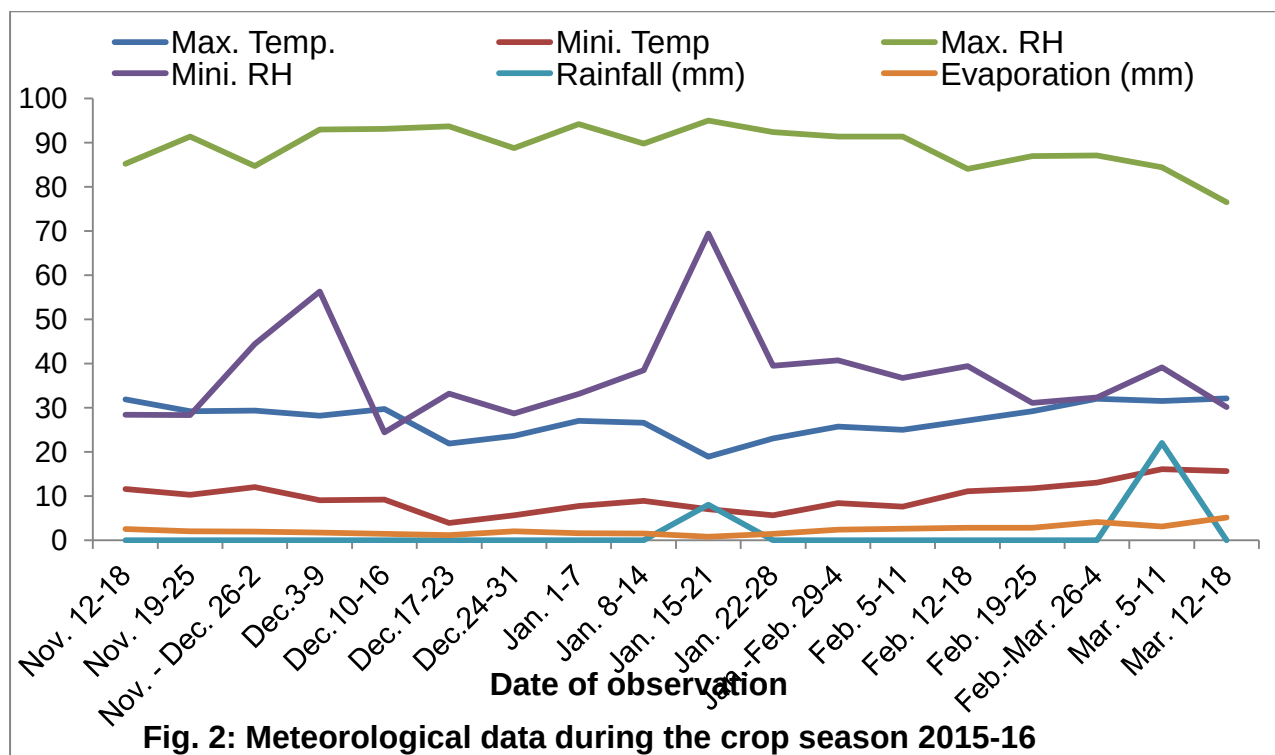
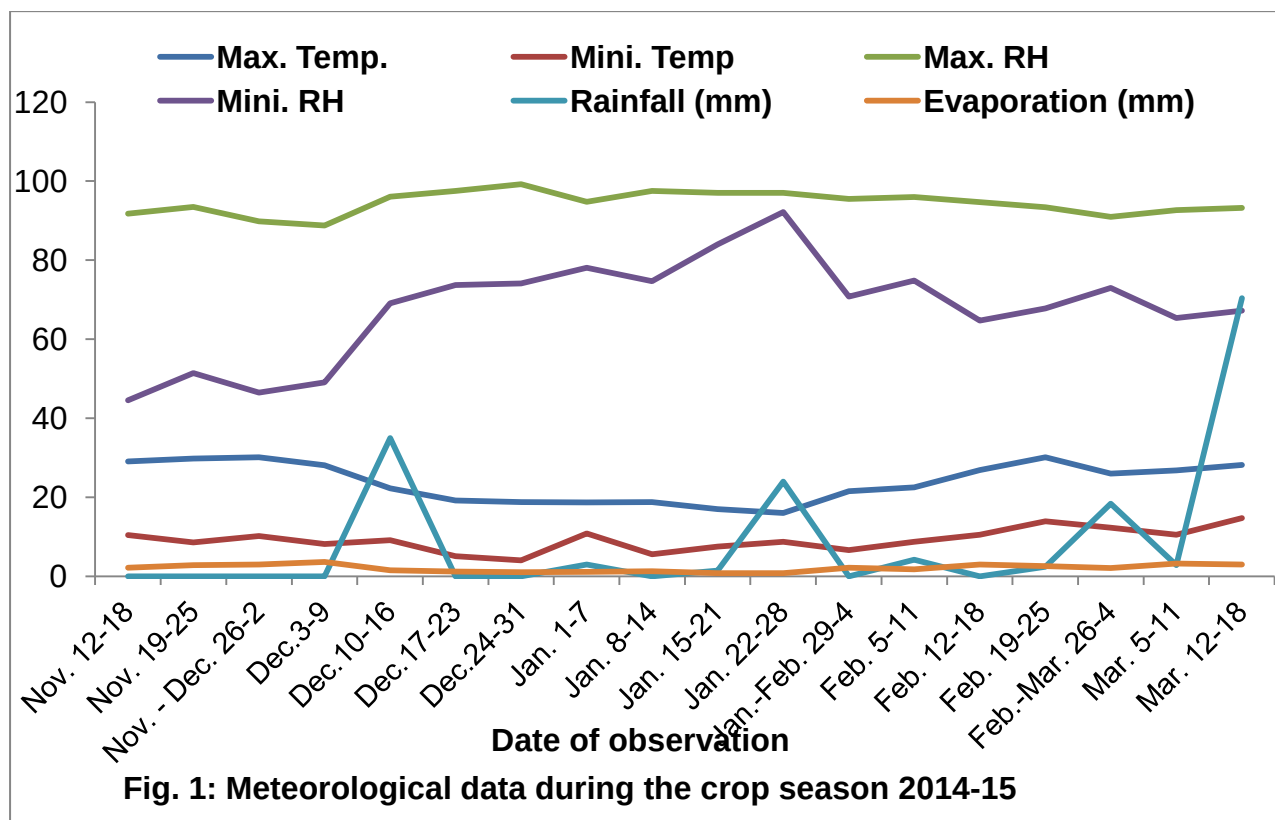
control (70.00 %) followed by *Eucalyptus globulus* leaf (56.67 %) which was found at par with *Trichoderma* isolate T5 (53.33 %).

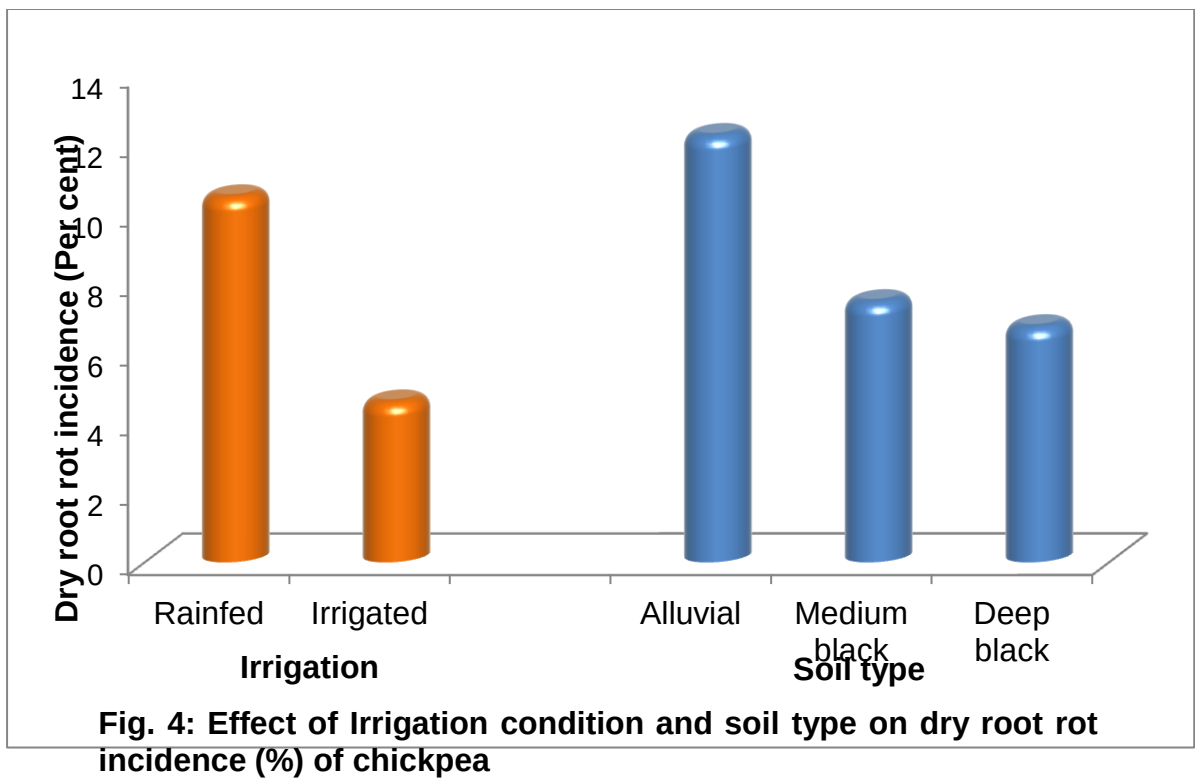
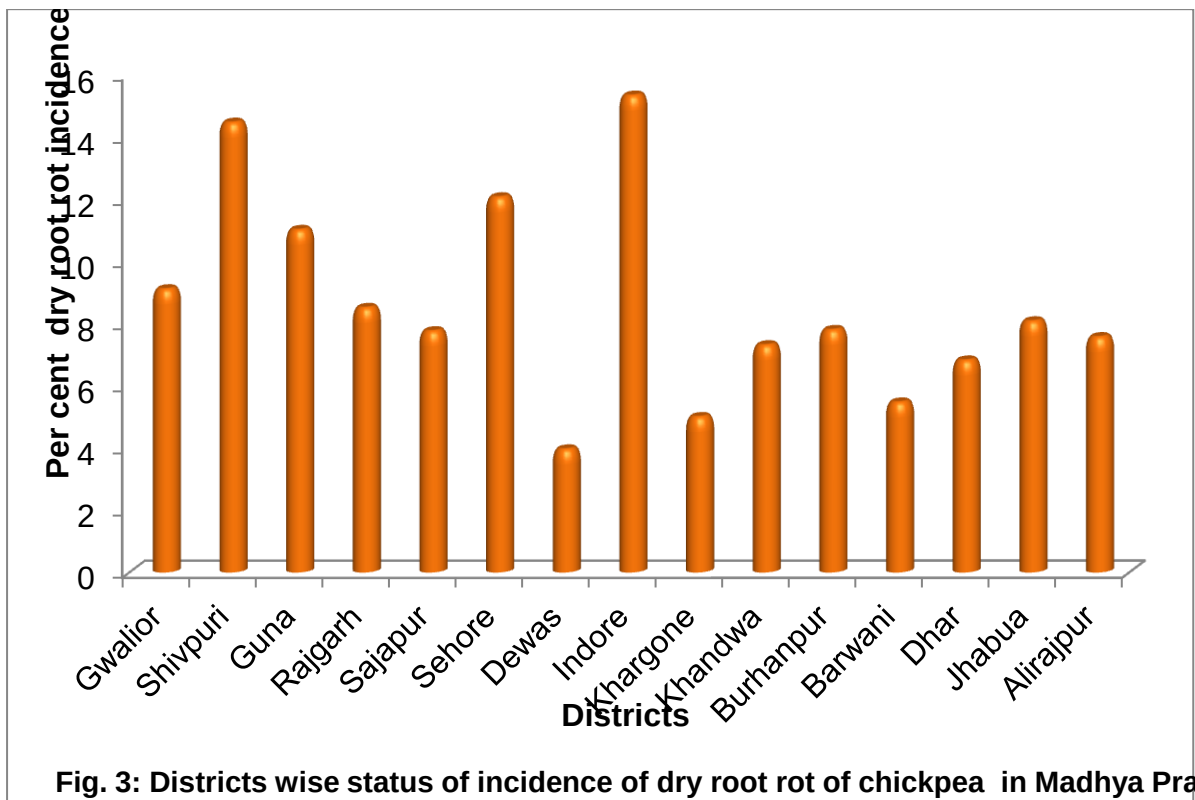
During 2015-16, all the treatments significantly reduced the incidence of dry root rot over untreated control. Minimum disease incidence was recorded in Carbendazim (30.00 %) and Difenconazole (30.00 %), which was found at par with Carboxin (33.33 %), *Trichoderma* isolate T7 (33.33 %), Mancozeb, (36.67 %) and Propiconazole (36.67 %) over untreated control (76.67 %) followed by *Eucalyptus globulus* leaf (60.00 %) which was found at par with *Trichoderma* isolate T5 (50.00 %).

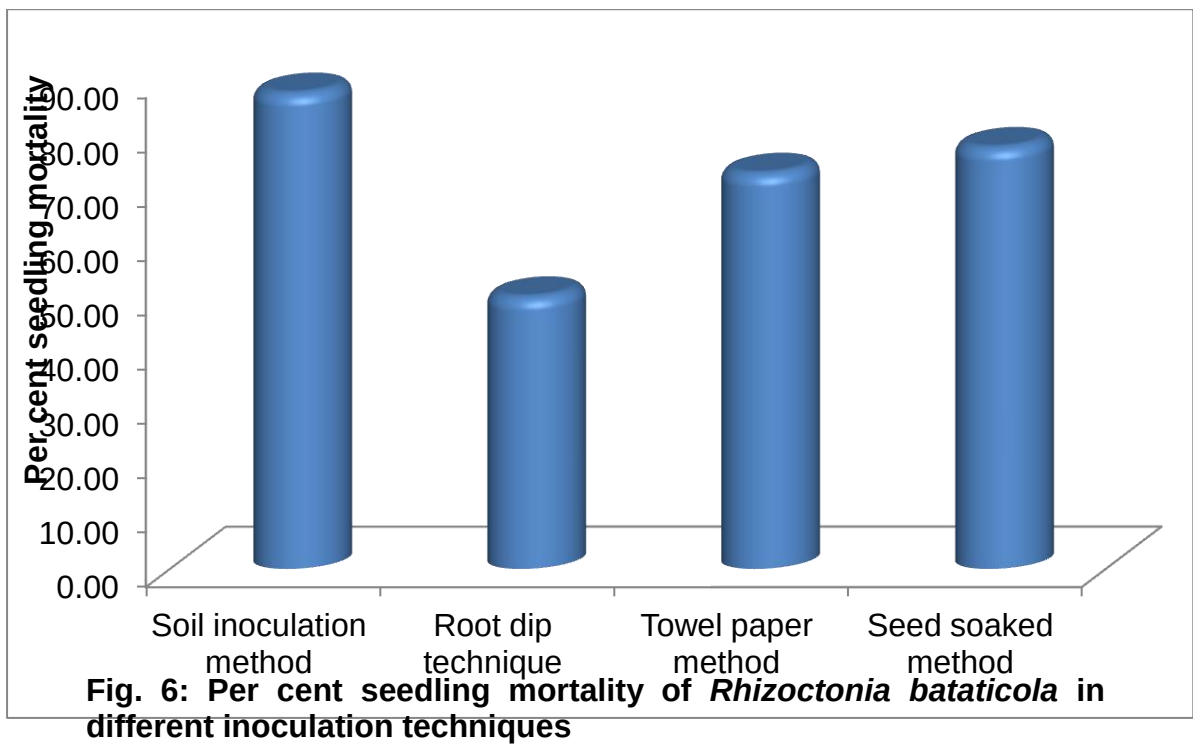
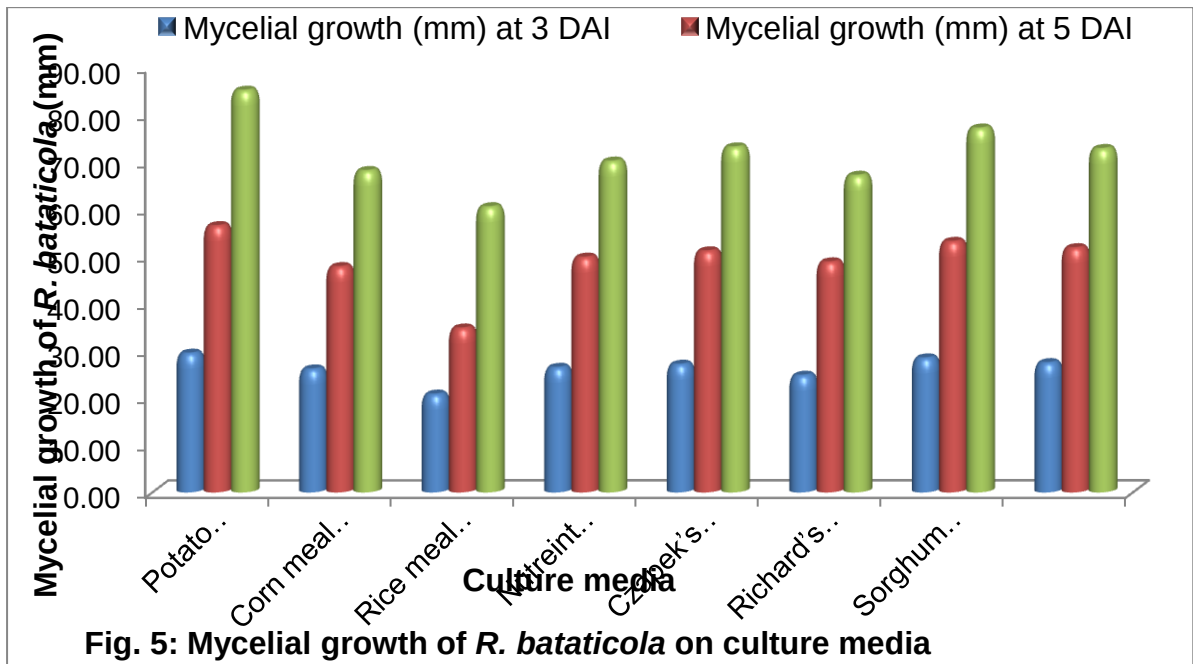
On the basis of average incidence of dry root rot, it was found that all the tested fungicides, botanicals and bio-agents significantly reduced the incidence of dry root rot over untreated control. Data presented in Table 31 showed that minimum disease incidence was recorded in Carbendazim (28.33 %) which was found at par with Difenconazole (31.67 %), Carboxin (31.67 %) and Propiconazole (33.33 %) over untreated control (73.33 %) followed by *Eucalyptus globulus* leaf (58.33 %), *Trichoderma* isolate T5 (51.67 %), *Allium sativum* clove (45.00 %) and *Trichoderma* isolate T6 (43.33 %)

Table 31: Effect of fungicides, botanicals and bio-agents on incidence of dry root rot of chickpea

Treatments	Dose	2014-15		2015-16		Mean	
		Disease incidence (Per cent)	Reduction over control	Disease incidence (Per cent)	Reduction over control	Disease incidence (Per cent)	Reduction over control
Carbendazim	1 g/kg seed	26.67 (31.00)	61.90	30.00 (33.00)	60.87	28.33 (32.00)	61.39
Difenconazole	1 g/kg seed	33.33 (35.22)	52.39	30.00 (33.00)	60.87	31.67 (34.11)	56.63
Propiconazole	1 g/kg seed	30.00 (33.21)	57.14	36.67 (37.22)	52.17	33.33 (35.22)	54.66
Tebiconazole	1 g/kg seed	36.67 (37.22)	47.61	40.00 (39.15)	47.83	38.33 (38.19)	47.72
Carboxin	1 g/kg seed	30.00 (33.21)	57.14	33.33 (35.22)	56.53	31.67 (34.21)	56.84
Mancozeb	2 g/kg seed	33.33 (35.22)	52.39	36.67 (37.22)	52.17	35.00 (36.22)	52.28
<i>Zingiber officinale</i> rhizome	20%	36.67 (37.22)	47.61	40.00 (39.23)	47.83	38.33 (38.23)	47.72
<i>Datura stramonium</i> leaf	20%	40.00 (39.23)	42.86	43.33 (41.15)	43.49	41.67 (40.19)	43.17
<i>Allium sativum</i> clove	20%	43.33 (41.15)	38.10	46.67 (43.08)	39.13	45.00 (42.12)	38.61
<i>Eucalyptus globulus</i> leaf	20%	56.67 (48.93)	19.04	60.00 (50.85)	21.74	58.33 (49.89)	20.39
<i>Trichoderma</i> isolate T3	8 g/kg seed	40.00 (39.15)	42.86	43.33 (41.15)	43.49	41.67 (40.15)	43.17
<i>Trichoderma</i> isolate T5	8 g/kg seed	53.33 (47.01)	23.81	50.00 (45.00)	34.79	51.67 (46.00)	29.30
<i>Trichoderma</i> isolate T6	8 g/kg seed	40.00 (39.23)	42.86	46.67 (43.08)	39.13	43.33 (41.15)	40.99
<i>Trichoderma</i> isolate T7	8 g/kg seed	40.00 (39.15)	42.86	33.33 (35.22)	56.53	36.67 (37.18)	49.69
Untreated control	-	70.00 (56.79)	-	76.67 (61.22)	-	73.33 (59.00)	-
S.Em. ±		1.73		2.02		1.33	
C.D. at 5%		4.99		5.84		3.76	







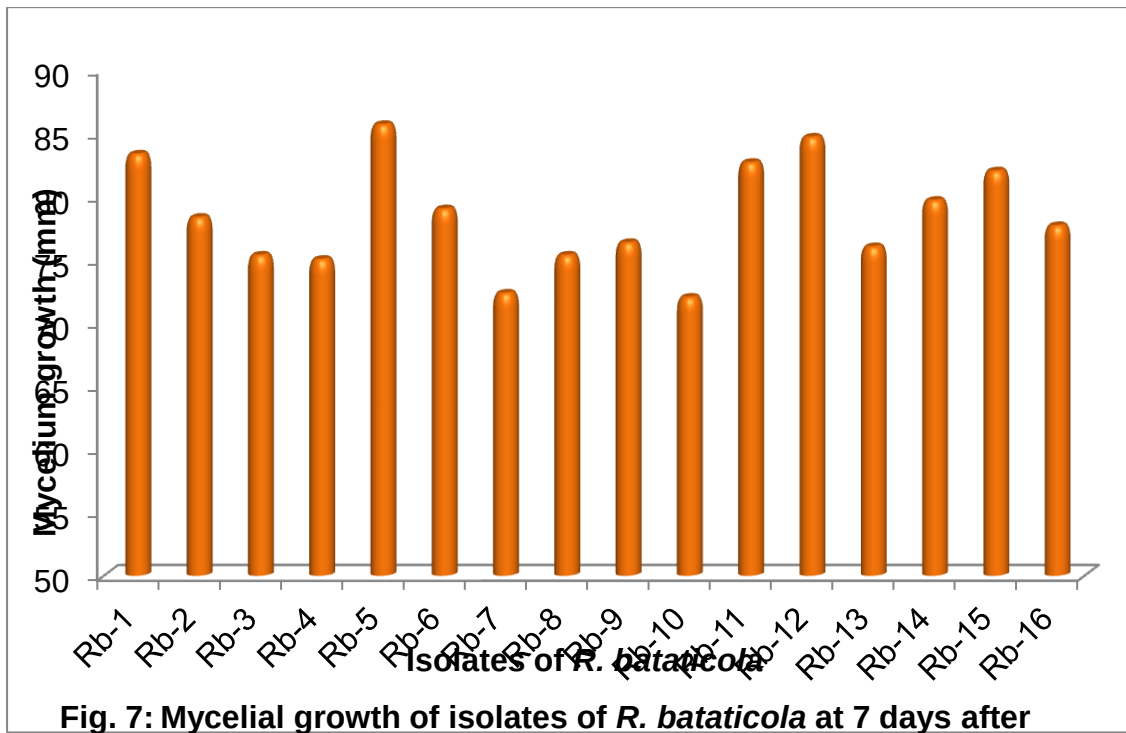


Fig. 7: Mycelial growth of isolates of *R. bataticola* at 7 days after inoculation.

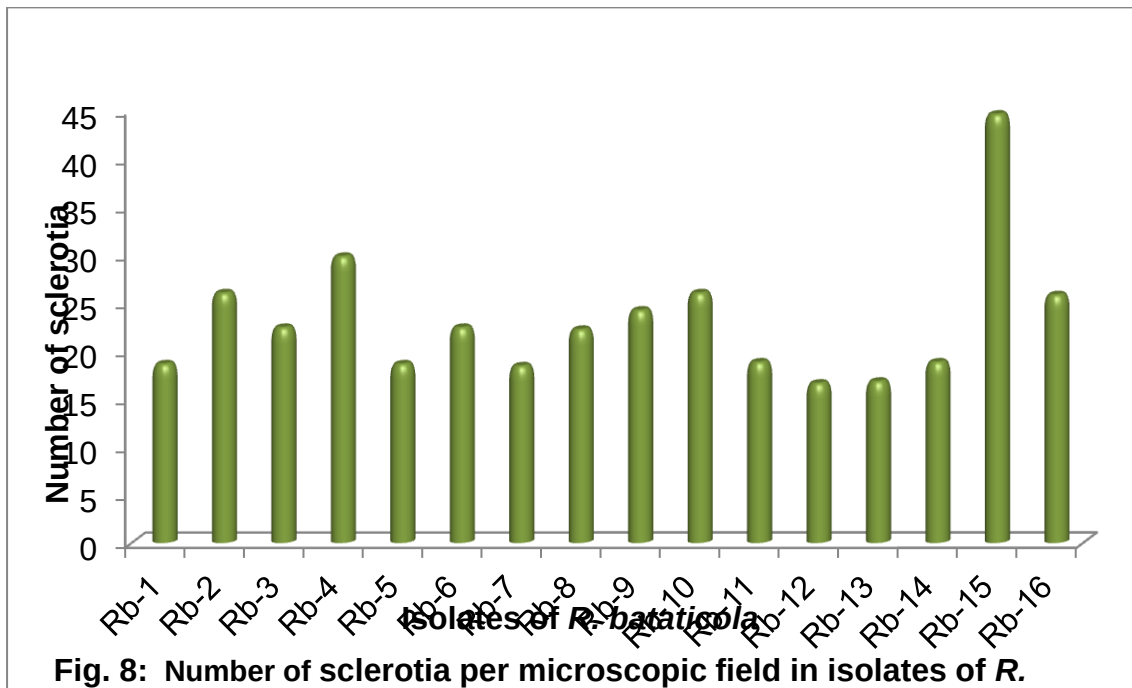


Fig. 8: Number of sclerotia per microscopic field in isolates of *R. bataticola*

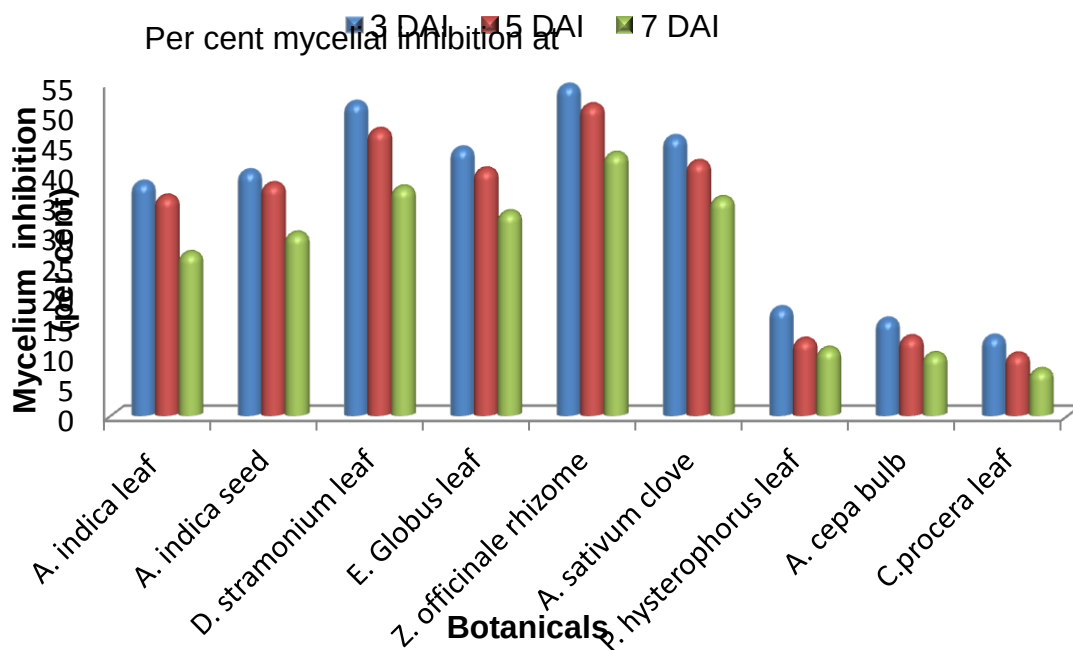


Fig. 9: Per cent mycelial inhibition of *R. bataticola* by botanicals at three, five and seven days after inoculation

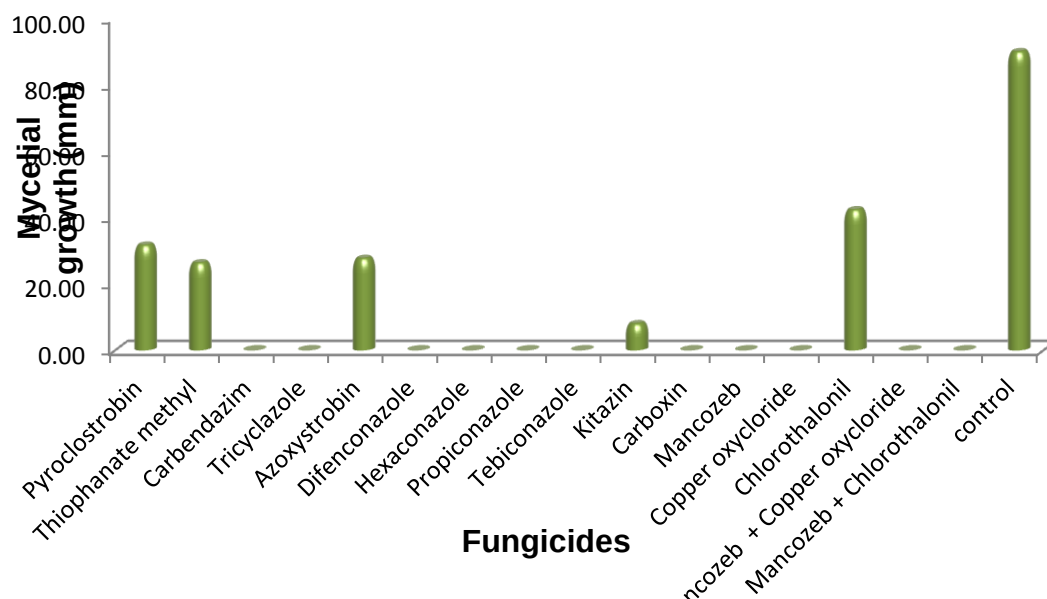


Fig. 10 : Evaluation of chemicals against *R. bataticola*

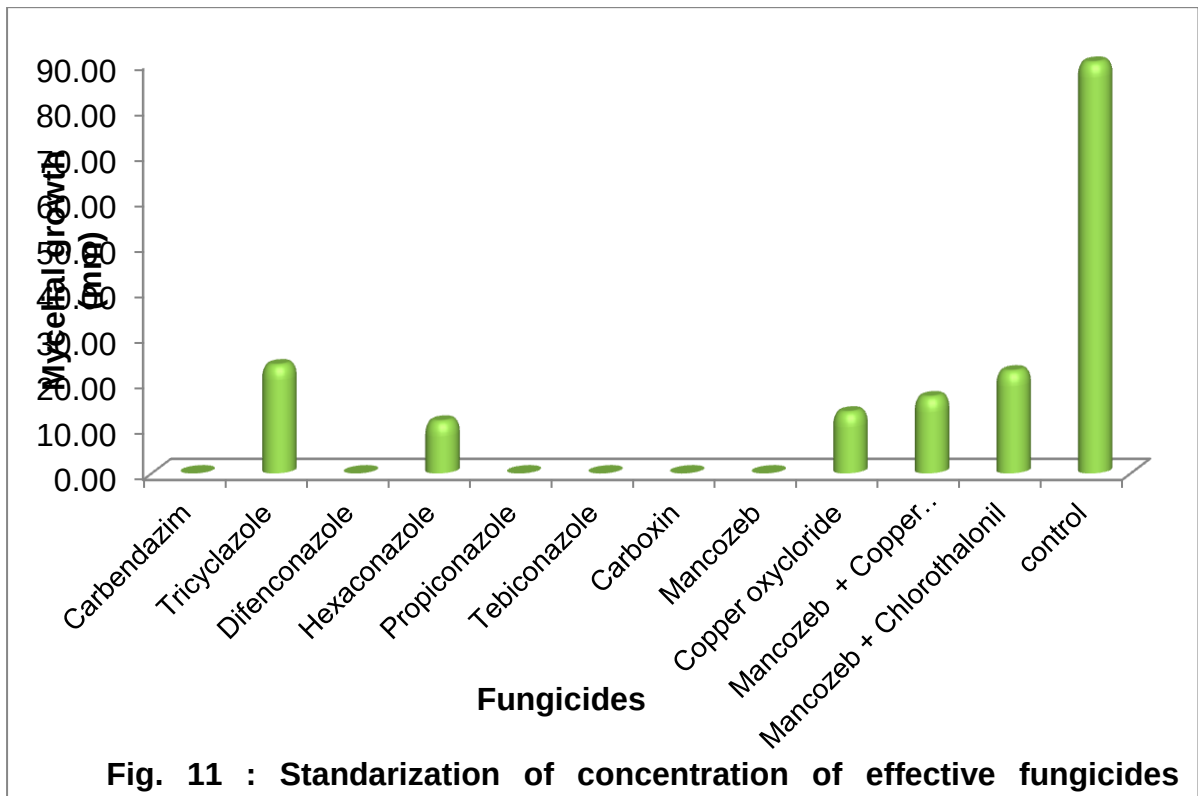


Fig. 11 : Standardization of concentration of effective fungicides against *Rhizoctonia bataticola*

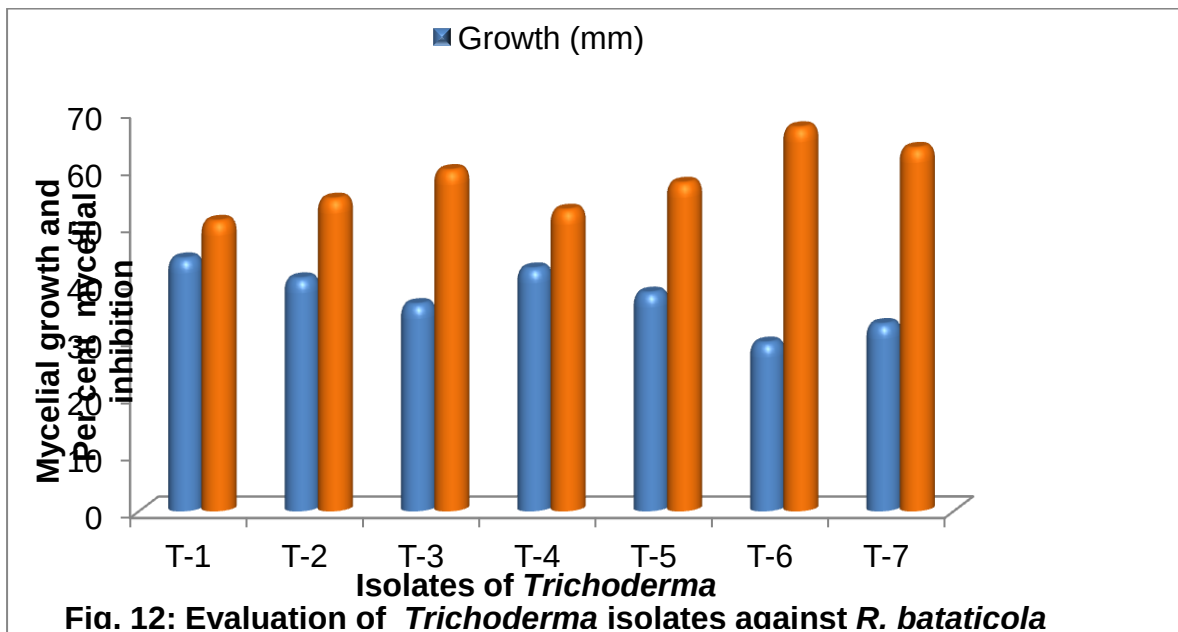


Fig. 12: Evaluation of *Trichoderma* isolates against *R. bataticola*

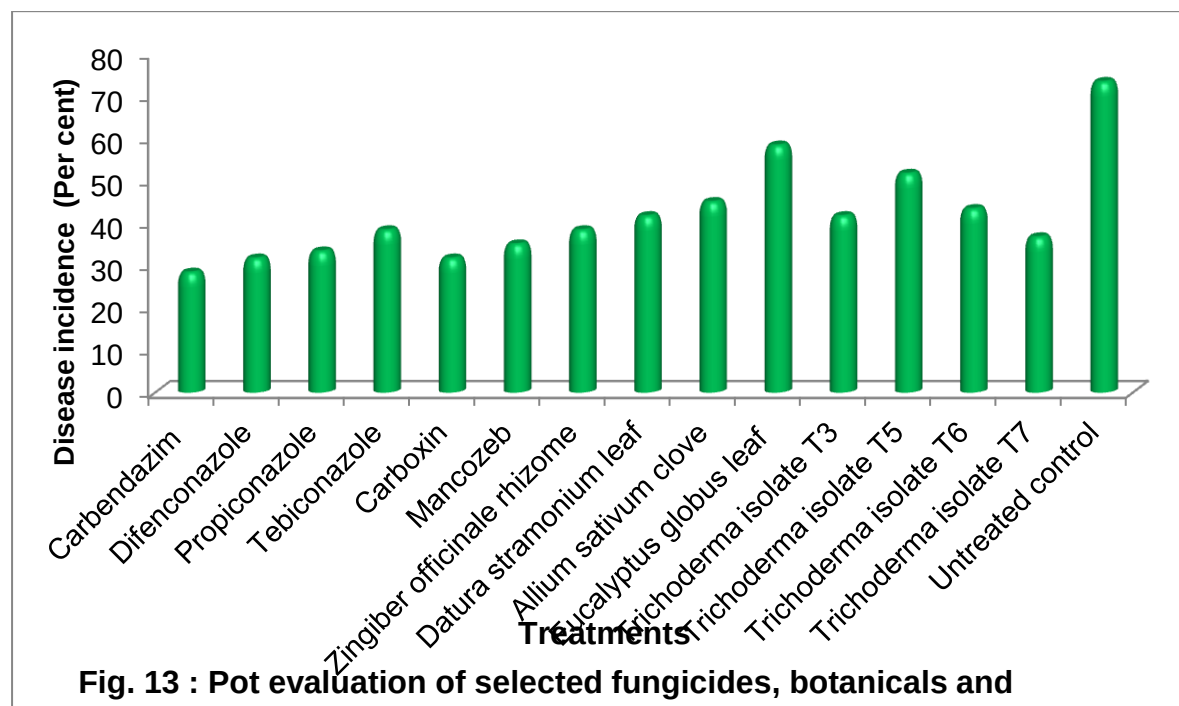


Fig. 13 : Pot evaluation of selected fungicides, botanicals and *Trichoderma* isolates against *R. bataticola*

Chapter-V

DISCUSSION

Chickpea cultivation is often subjected to significant yield losses due to insects and diseases ranging from 5-10% in temperate and 50-100% in tropical regions (Van Emden *et al.*, 1988). Currently chickpea is attacked by 172 pathogens viz., 67 fungi, 3 bacteria, 22 viruses and mycoplasma, and 80 nematodes reported from 55 countries. Among the several constraints affecting the productivity of chickpea, 10-25 per cent loss in yields are due to wilt and dry root rot diseases (Mahendra Pal, 1998). Among them, dry root rot of chickpea caused by necrotrophic fungus, *Rhizoctonia bataticola* (Taub.) Butler has emerged as a serious threat to the chickpea production worldwide (Pande and Sharma, 2010,; Ali and Shivkumar, 2001). The first report of occurrence of root rot in chickpea along with wilt was made by Padwick (1948). *R. bataticola* is responsible for seriousness disease.

The results obtained on these aspects are presented here under and discussed critically based on the previous information available on these aspects.

An intensive roving survey was conducted during 2014-2015 *Rabi* cropping season in the farmer's field at Gwalior, Malwa and Nimar regions of Madhya Pradesh to assess the status of dry root rot incidence and to collect diseased samples infected by *Rhizoctonia bataticola*. Irrigation condition and soil types were also observed at all the places. During the investigation 40 places were visited and disease incidence was recorded. Dry root rot disease of chickpea crop was found in all the surveyed sites. The mean maximum dry root rot incidence was noticed in Lukwasa followed by Shivpuri and Kolarash while, the least dry root rot incidence in Khargone followed by Rajpur and Barwani. Gurha and Trivedi (2008) observed dry root rot incidence in Gulberga and Raichur up to 60%. Similarly, survey conducted by Trivedi and Gurha (2006) in different blocks of Bhundelkhand region of Uttar Pradesh found infection of *R. bataticola* on chickpea ranged from 5-22%. Intensive roving survey conducted by Manjunatha *et al.* (2011) during November to December, 2008 in the farmers' fields at Raichur, Gulbarga and Bidar districts in Karnataka. The maximum dry root rot incidence in chickpea was noticed in Gulbarga district (9.8%) followed by Raichur (7.6%) and the least (6.18 %) in Bidar. Khan *et al.* (2012) conducted survey for the occurrence and severity of chickpea dry

root rot was made during crop season 2010-11 in Jammu and Kashmir. The highest incidence was up to 40% in Shangus and lowest up to 4.11% in Naina of Jammu and Kashmir.

Soil moisture plays an important role in the development of the crop. It also plays an important role in the survival and spread of the disease like dry root rot caused by *R. bataticola* which favours low moisture conditions. In the present investigation irrigation condition of all the places were recorded. Percent disease incidence in the irrigated locations varied from 2.54 (Khargone) to 7.67 (Biaora) whereas in the locations where no irrigation facility was available there, it varied from 5.17 (Sendwa) to 20.89 (Lukwasa). Comparatively the disease incidence was less in irrigated fields over unirrigated ones. [Sharma and Pande \(2013\)](#) reported that dry root rot Incidence and severity was more in plants subjected to moisture stress (40% and 60% SMC) as compared to non-stress (80% and 100% SMC). Similar results were also observed by [Bajpal et al., \(1999\)](#) as hot and dry weather in vertisols favoured more disease development compared to alfisols. This observations are agreement with the [Hussain et al., \(1997\)](#) and [Manjunath et al. \(2011\)](#) where they reported that chickpea plants to moisture stress conditions evidenced during rabi season, which ultimately led to more production of sclerotia of *Rhizoctonia* sp. on chickpea plants roots ([Pande et al., 2004](#)). Later, [Wokocha \(2000\)](#) found that, the disease incidence of dry root rot of soybean was significantly lower when the soil moisture was 60-70 per cent, but it was 5.0 per cent at 10-20 per cent soil moisture. [Mina and Dubey \(2010\)](#) reported that soil moisture was found negatively correlated with wilt incidence of chickpea. [Pande et al. \(2010\)](#) reported that unpredictable moisture stress and higher temperatures in central and southern parts of India probably predisposed the chickpea crop to favorable conditions for dry root rot development.

Three types of soil were recorded in which chickpea crop was cultivated. The percent disease incidence varied from 4.58 (Guna) to 20.89 (Lukwasa) in locations where the chickpea crop grown in alluvial soil while it was from 3.20 (Rajpur) to 12.07 (Panchor) in locations where the chickpea crop grown in medium black soil whereas, in the locations where the chickpea crop grown in deep black soil, it varied from 2.54 (Khargone) to 15.34 (Mau). Comparatively the disease incidence was less in deep black over medium black soil and alluvial soil. A survey conducted by [Manjunath et al \(2011\)](#) during *Rabi* season of 2008 revealed that dry root rot of

chickpea varied from locality to locality, obviously due to different soil conditions (black/red soil conditions). Taya *et al.* (1988) agreed with this finding. Rajkrishan *et al.* (1999) studied the effect of edaphic factors and moisture regime on the incidence of *Macrophomina* root rot in sesame. Amongst the various soil types, sandy soil supported maximum disease (78.33%) as compared with clay soil (51.56%).

Symptoms of dry root rot are most commonly observed in chickpea during post-flowering stage. The symptoms of the disease are yellowing of the leaves within a day and such leaves drop and plant show completely dried symptoms within a week after the appearance of the first symptom. The seedlings can also get infected. Drooping of petioles and leaflets is confined to those at the very top of the plant. Sometimes when the rest of the plant is dry, the topmost leaves are chlorotic. The leaves and stems of affected plants are usually straw colored, but in some cases the lower leaves and stems are brown. If the plants are pulled out from the soil and examined the basal stem and main root system of diseased plant, show extensive rooting with most of the lateral roots destroyed. The tissues were weakened and break off easily. Dark minute sclerotial bodies can be seen on the roots exposed and inner side of the bark. Baldev *et al.* (1988) reported symptoms of dry root rot of chickpea which consisted of straw colored leaves and stems. Tap root of infected plants was observed to be dry and devoid of lateral and finer roots. Roots were turned dark and showed signs of rotting. The dead root was observed brittle towards the tip and showed shredding of bark. Haware (1990) reported that dry root rot of chickpea appears around flowering and podding time in the form of scattered dried plants but seedlings are also infected. The symptoms induced were drooping of petioles and leaflets which were confined to top of the plants. Shredding of bark in form of flakes was observed. The symptoms were resembled with symptoms as described by Dastur (1935); Nene *et al.*, 1991 and Mehrotra and Aneja (1990).

On PDA at 25°C the culture of *R. bataticola* was black with appressed growth. Hypha measured 14.32 x 6.45 µm and sclerotia were irregular measured 160.35 x 151.87 µm in diameter, black in colour. . The microsclerotia were black and variable in size from 50–150 µm depending on the available nutrients of the substrate on which the propagules were produced (Short and Wyllie, 1978). Hildebrand *et al.* (1945) differentiated two strains of *M. phaseolina*

isolated from soybean, having mean sclerotial diameter of $90.4 \times 75.8 \mu\text{m}$ on host plant and $99.9 \times 89.4 \mu\text{m}$ in culture and from cotton (the Texas strain) whose, sclerotia in culture averaged $85.4 \times 73.0 \mu\text{m}$ and were produced in greater number.

Culture media is of fundamental importance for most pathological tests: to obtain pure cultures, to grow and to count microbial cells, and to cultivate and select microorganisms. Without high quality media then the possibility of achieving accurate, reproducible and repeatable test results is reduced (Sandle, 2010). An experiment was conducted in laboratory to find the suitable culture media for the growth of *R. bataticola*. In the study, eight different culture media viz Potato dextrose agar, Corn meal agar, Rice meal agar, Nutreint agar medium, Czapek's agar medium, Richard's agar medium, Sorghum grain medium and Chickpea grain medium were evaluated to find out the appropriate growth of *R. bataticola*. Mycelial growth was recorded at 3, 5 and 7 days after inoculation.

At 7 days after inoculation, the mycelial growth differed significantly in different treatment varied from 60.33 mm to 85.00 mm. Maximum mycelial growth was observed in Potato dextrose agar which was significantly highest than rest of treatments. The minimum mycelial growth was found in Rice meal agar which was significantly less than rest of treatments. Kulkarni *et al.* (1992) studied the different media and reported *R. bataticola* grew best on cotton root extract agar, PDA, Richards agar and Sabouraud's agar. Sclerotial production was excellent on the last 3 media and on Czapek's agar. Salunke *et al.* (2009) evaluated six different media and reported that PDA supported the growth and sclerotial formation of *R. bataticola* followed by Czapek's dox, Richard's, Oat meal and peptone agar media. Singh and Singh (2007) cultured *R. solani* on seven culture media and reported that maximum fungal colony diameter (89.7 mm) was recorded on Potato dextrose agar, which was on par with PDA + rice leaf extract agar. Water agar supported the minimum growth of *R. solani*.

Different screening techniques have been employed by the scientists to find out the resistant sources. Comparative performance of the screening techniques regarding the effective and consuming less time for identification of resistant sources is lacking, Hence, the present investigation was conducted to find out the comparative performance of four screening techniques viz., soil inoculation method,

root dip technique, towel paper method and seed soak method were employed against dry root rot of chickpea caused by *R. bataticola*.

At 1 % inoculum load, maximum seedling mortality was recorded in soil inoculation method followed by Seed soak method and towel paper method, while minimum seedling mortality was recorded in root dip technique. Statistically, soil inoculation method and Seed soak method were found at par with each other.

At 2 % inoculum load, maximum seedling mortality was recorded in soil inoculation method which was found at par with Seed soak method, however minimum seedling mortality was observed in root dip technique followed by towel paper method.

Maximum seedling mortality was recorded in soil inoculation method at 3 % inoculum load which was found at par with Seed soak method, while minimum seedling mortality was observed in towel paper method and root dip technique.

At 4 % inoculum load, maximum seedling mortality was recorded in soil inoculation method which was found at par with Seed soak method followed by towel paper method, while minimum seedling mortality was observed in root dip technique.

Experiment conducted in the net house revealed that maximum seedling mortality was recorded in soil inoculation method followed by seed soaked method, while minimum seedling mortality was observed in root dip technique followed by towel paper method. Among inoculation load, maximum seedling mortality was observed in 4 % concentration followed by 3 % concentration, while minimum seedling mortality was recorded in lowest concentration. Pande *et al.* (2010) observed that most of lines found resistant in sick-plot to dry root rot were susceptible under controlled environment screening. Amule *et al.* (2014) reported that maximum collar root incidence after initiation of disease ranged from 8 to 20 days being minimum i.e. 5 days in chickpea grain inoculation technique, where one chickpea grain fully covered with mycelial growth was placed at the collar region of 7 days old seedlings. Gupta (2001) reported reaction of chickpea genotypes against collar rot using potting medium mixed with wheat straw, sand medium @ 50g /2 kg sterilized sand + soil (1:1) mixture. Among the inoculation techniques, the use of inoculum in the form of wheat grains fully impregnated with mycelium of the pathogen was found to be simple, feasible and highly effective (Hussain *et al.*,

2005). Pathogenicity test **was** conducted by **Veena et al. (2014 b)** through soil infestation method and observed pre-emergence rot of seedlings and the survived plants showed stunted growth followed by wilting and drying of leaves and stems. When the infected plants were pulled out the tap root was blackened and devoid of lateral as well as finer roots. Several workers have found soil inoculation method as most suitable in establishing the disease caused by *Rhizoctonia bataticola* (Katariya et al, 2007) in chickpea.

Cultural variability among sixteen isolates of *R. bataticola* was studied on the basis of the colony diameter, colour, texture and aerial mycelium

Significant differences among sixteen isolates of *R. bataticola* collected from districts of Madhya Pradesh were observed on the basis of radial growth. The individual average radial growths of sixteen isolates of *R. bataticola* ranged from 72.00 mm to 85.67 mm. Maximum colony diameters was recorded in Isolate Rb-5 which was found at par with Rb-12, Rb-1 and Rb-11, while the least colony diameter was observed in isolate Rb-10 followed by Rb-7, Rb-4, Rb-3 and Rb-8. Similar studies on cultural and morphological characters of *R. bataticola* were conducted by **Sharma et al. (2012)** and **Gupta et al. (2012)**. **Gade et al. (2018)** reported that the individual average radial growths of 40 isolates of *R. bataticola* ranged from 75.00 to 90.00 mm. Maximum colony diameters of 90.00 and 85.66 mm were observed in case of isolate Rb-1, Rb-6, Rb-11, Rb-12, Rb-15, Rb-22 Rb-23, Rb-29, Rb-37 and Rb-38, and proving to be the fast growing, while isolates Rb-5 and Rb-26 showed the minimum radial growths and were rated as medium growing. **Lokesha and Benagi (2004)** reported that among four isolates of *M. phaseolina* in pigeon pea maximum growth was recorded in ICRISAT isolate which was significantly superior over other isolates (82.7 mm) followed by Bidar isolate (78.35 mm). **Iqbal and Mukhtar (2014)** noticed significant differences among 65 isolates of *R. bataticola* with sixteen isolate rated as fast growing, 11 as slow growing and the rest as medium growing. Maximum colony diameter of 87.17 and 86.67 mm was observed in isolate MP- 7 (Dera Ghazi Khan) and MP-26 (Layyah) proved to be the fast growing, while isolates MP8, MP-29 and MP-30 showed minimum radial growth and rated as slow growing.

On the basis of visual observation, colony colour were divided in four groups viz., black colour, black with grey aerial, dark brown colour and grey colour. Out of

sixteen isolates, black colour colony was observed in 4 isolates (Rb-1, Rb-7, Rb-8, Rb-11) while black with grey aerial mycelium was recorded in 7 isolates (Rb-2, Rb-3, Rb-5, Rb-9, Rb-14, Rb-15, Rb-16). Dark brown (Rb-6, Rb-12, Rb-13) and grey colour (Rb-4, Rb-10) was observed in 3 and 2 isolates respectively. Similarly, Sharma *et al.*, (2012) reported the variation among 94 isolates of *R. bataticola* collected from chickpea and concluded that the light black (52.12%) colony colour was more predominant. Srinivas (2016) reported that out of 68 isolates of *R. bataticola*, black colour colony was observed in 26 isolates while black with grey aerial mycelium was recorded in 20 isolates. Dark brown and grey colour was observed in 11 each respectively. Charul and Biswas (2009) reported that the mycelium of *R. bataticola* was varied from fluffy dark brown to partially fluffy mycelium colony, with smooth margin. Aboshosha *et al.* (2007) observed colonies of *Macrophomina* isolates of sunflower charcoal rot were mostly grey, dark grey, black and brown for fourteen stem isolates in frequencies of 4, 3, 5 and 1 on PDA and 4, 5, 2 and 2 on Czapek-Dox Agar while white colony was observed in only one isolate on both media.

On the basis of colony texture, Isolates were categorized into three groups viz., fluffy colony, appressed colony and velvety colony. Out of sixteen, maximum of 8 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-9, Rb-14 Rb-15, Rb-16) produced fluffy colony while appressed (Rb-1, Rb-11, Rb-12, Rb-13) and velvety growth (Rb-4, Rb-7, Rb-8, Rb-10) produced by 4 isolates each respectively. An experiment was conducted by Sharma *et al.* (2012) and reported that most of the isolates grew very fast and covered the plate within 96 h, while other isolates such as RB 21, RB 31, RB 7, RB 84 and RB 87 grew slowly and showed appressed growth. Subramanian *et al.* (2011) studied the pathogenic and genetic variability among the eleven isolates of *R. bataticola* (Taub.) Butler from different pulse crops. Based on morphological characters, 11 isolates were categorized into three groups viz., linear, fluffy, and linear at the end with fluffy growth at the center. Aghakhani and Dubey (2009) reported that nineteen isolates produced suppressed mycelium whereas 4 isolates in which three from Kanpur and one from Rohtak produced aerial mycelium. Edraki and Banishashemi (2010) studied sixty isolates of *M. phaseolina* collected from different parts of Iran on various crops like soyabean, cucumber and sesame. Isolates were grouped in to four phenotypes viz., fluffy with abundant sclerotia, fluffy with few sclerotia, partially fluffy and appressed growth.

Maximum number of 9 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-8, Rb-9, Rb-14, Rb-15, Rb-16) produced aerial mycelium and 7 isolates (Rb-1, Rb-4, Rb-7, Rb-10, Rb-11, Rb-12, Rb-13) did not develop any aerial mycelium. [Fernandez et al. \(2006\)](#) analyzed diversity patterns of 15 isolates from Mexico and 15 isolates from other countries of *M. phaseolina* on beans based on presence or absence of aerial mycelium. They observed 15 isolates had intermediate and 15 isolates had poor aerial mycelium. Eight from Mexican isolates and 7 isolates of non Mexican isolates had intermediate aerial growth.

Morphological variability among the *Rhizoctonia bataticola* isolates like sclerotial size, length and width ratio, shape, intensity, hyphal cell size were studied. The size of sclerotia varied from 101.15 x 98.85 μm (Rb-15) to 181.35 x 170.72 μm (Rb-6). Maximum sclerotial size was observed in case of isolate Rb-6 followed by Rb-7, Rb-12, Rb-5, Rb-1, Rb-11, while the isolate Rb-15 produced the smallest size sclerotia followed by Rb-9, Rb-8, Rb-2, Rb-14, Rb-3, Rb-13, Rb-4. Sclerotial diameter of *R. bataticola* in gram varied from 98.1-119.2 μm (Ghosh and Sen, 1973). Forty isolates of *R. bataticola* from chickpea were studied by Gupta et al. (2012) and observed that sclerotial sizes varied from 72.7- 117.2 x 57.1-106.5 μm . Based on sclerotial size, the isolates were divided into three groups viz., minute size (72.7 - 87.5 x 57.1-73.0 μm), medium size (87.6 - 103.2 x 73.5 - 90.0 μm) and bold (103.3 -117.2 x 90.1- 106.5 μm) size of sclerotia groups. Morphological variability among thirty six isolates of *M. phaseolina* from mung bean collected from different region of North, South, North East and Central India were studied by Devi and Singh (1998) and observed bigger sclerotia in isolate MP-2 (400 x 280 μm).

The length : width ratio varied from 1.01 (Rb-5) to 1.13 (Rb-14). Rb-5 found least length : width ratio followed by Rb-15, Rb-7, Rb-8, Rb-9, Rb-1, Rb-16, while maximum length : width ratio Rb-14, Rb-10, Rb-12, Rb-13, Rb-3. Sharma et al. (2012) reported the variation among 94 isolates of *R. bataticola* collected from chickpea fields and reported that the isolates varied for length and width of sclerotia. However, the length/width ratio ranged from 1.1 to 1.8. Prasad et al. (2014) studied the variability in shape, length, width and length/width morphological characters of microsclerotia and observed AP 21 isolate varied in length/width ratio (483.3/200.0 μm) while no significant differences were observed in others.

On the basis of shape of sclerotia, sixteen isolates were categorized into irregular, round and ovoid groups. Irregular shaped sclerotia were observed in 7 isolates (Rb-1, Rb-3, Rb-6, Rb-9, Rb-13, Rb-14, Rb-16) while round shaped sclerotia were observed in 5 isolates (Rb-4, Rb-5, Rb-7, Rb-10, Rb-11). Ovoid shape of sclerotia was observed in 4 isolates (Rb-2, Rb-8, Rb-12, Rb-15). Srinivas (2016) categorized isolates into irregular, round and ovoid groups based on shape of sclerotia. The Irregular shaped sclerotia were observed in 17 isolates while ovoid shaped sclerotia were observed in 23 isolates and round shape of sclerotia was observed in 28 isolates. Hildebrand *et al.* (1945) observed the sclerotia were black, smooth varying from spherical to oblong to irregular shaped. The sclerotial shape of any isolate varied with substrate on which fungus was grown. Based on sclerotial morphology, Gupta *et al.* (2012) and Shekhar *et al.* (2006) categorized the isolates into two groups viz., oblong shape with irregular edges and round with regular edges. Sharma *et al.* (2012) reported that the shape of the sclerotia varied from oblong (13.8%), ellipsoid (42.6%), irregular (36.2%) and round (7.4%)

The number of sclerotia per microscopic field when observed through 10x objective varied from 16.60 (Rb-12) to 44.60 (Rb-15). Maximum number of sclerotia was produced by Rb-15 followed by Rb-4, Rb-2, Rb-10, Rb-16, Rb-9, Rb-3, Rb-6, Rb-8, however minimum number of sclerotia was produced by Rb-12, Rb-13, Rb-7, Rb-1, Rb-5, Rb-11, Rb-14. Srinivas (2016) agreed with the finding, who reported that the number of sclerotia per microscopic field varied from 11.67 (Rb 63) to 70.67 (Rb 56). Gupta *et al.* (2012) observed that the isolate from Betul, Madhya Pradesh produced maximum number of sclerotia (22.3 sclerotia per microscopic field) and isolate from Dhule, Maharashtra exhibited minimum number of sclerotia (8.1 sclerotia per microscopic field).

The hyphal cell size varied from 10.35 x 5.10 μm (Rb-3) to 14.65 x 7.40 μm (Rb-4). Maximum hyphal cell size was recorded in Rb-4 followed by Rb-16, Rb-1, Rb-13, Rb-5, Rb-12, Rb-8, Rb-9, Rb-10, Rb-6, while minimum hyphal cell size was recorded in Rb-3 followed by Rb-15, Rb-14, Rb-2, Rb-11, Rb-7. Ratio between length and width of hyphal cell ranged from 1.55 (Rb-14) to 2.43 (Rb-10). Minimum ratio between length and width of hyphal cell was observed in Rb-14 followed by Rb-2, Rb-11, Rb-8, Rb-7, Rb-12, Rb-16, Rb-4, Rb-3, Rb-15, Rb-13, while maximum ratio between length and width of hyphal cell was observed in Rb-10 followed by Rb-5, Rb-1, Rb-6, Rb-9. Gupta *et al.* (2012) observed variation in size

of hyphal width of 40 *R. bataticola* isolates from chickpea. Eighteen highly virulent isolates were classified under very thin (5.2-6.5 μm), 17 highly virulent and 2 virulent isolates under thin (6.6 -7.9 μm) and only three highly virulent isolates from Chhattisgarh and Gujrat exhibited (8.0 - 9.3 μm) thick hyphal width. Sobti and Sharma (1992) reported that the width of hyphae varied from 4.16- 8.48 μm in *R. bataticola* isolates from groundnut. Srinivas (2016) reported that the ratio between length and width of hyphal cell varied from 1.35 (Rb 27) to 2.96 (Rb 65).

Pathogenicity of the sixteen isolates of *R. bataticola* was tested by artificial root dip inoculation using paper towels. Among sixteen isolates, four isolates viz., Rb-1, Rb-2, Rb-5 and Rb-12 showed highly virulent reaction, while seven isolates showed virulent reaction viz., Rb-4, Rb-6, Rb-8, Rb-10, Rb-11, Rb-15 and Rb16, however five isolates viz., Rb-3, Rb-7, Rb-9 Rb-13 and Rb-14 showed least virulent reaction. The incubation period ranged from 2.80 days (Rb-5) to 5.00 days (Rb-7). Shortest incubation period was recorded in Rb-5 while longest incubation period was recorded in Rb-7. Similarly, Pande *et al.* (2004) observed the maximum disease severity rating of 9 in the susceptible cultivar BG 212 when screened for source of resistance to dry root rot. Sharma *et al.* (2012) also proved the pathogenicity of 50 isolates of *R. bataticola* using chickpea cultivar BG 212 and reported the dry root rot disease severity ranging between 4 and 9 (on the basis of 1 to 9 severity rating scale). Their study also indicated that pathogenic and non pathogenic isolates were not concentrated in any one particular region /state. Srinivas (2016) reported that the incubation period ranged from 1.0 (Rb 21, Rb 35, Rb 60, Rb 61 and Rb 63) to 4.8 (Rb 4) days. Sharma *et al.* (2012) tested the pathogenicity of 94 isolates of *R. bataticola*. Among the 94 isolates, 8 isolates from Andhra Pradesh and Madhya Pradesh showed less aggressiveness on chickpea cultivar BG 212.

Chemical control of dry root rot is not effective as *R. bataticola* has a broad host range and survives in soil for longer periods in the form of sclerotia. The sclerotia can survive up to 10 months even in the absence of the host plants and under prevailing dry soil conditions. Use of host plant resistance is the most economical approach for management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot have been identified, but high levels of resistance are scarce in cultivated genotypes (Anon., 2010). In the experiment, total 126 chickpea entries were screened against dry root rot during rabi season 2014-15 and

2015-16. During 2014-15, twenty four entries viz, Phule G 0405, GL 10023, RSG 888, BAUG 15, DBGV 102, GJG 1209, GJG 1202, NBeG 453, JG 41, Phule G 08108, BGD 1082, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, GJG 1207, NBeG 439, HK 4, NBeG 460, HK 10-124, NBeG 399, BG 3059 showed resistance reaction. Forty nine entries found moderately resistant reaction. Thirty two entries have been found moderately susceptible Sixteen entries showed susceptible reaction and five entries have been found to show highly susceptible reaction, while during 2015-16, seventeen entries viz, GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059 were found resistant. Forty two entries found moderately resistant reaction. Forty two entries have been found moderately susceptible, seventeen entries showed susceptible reaction and eight entries have been found to show highly susceptible reaction.

Average of two year data was calculating for all the entries and according to disease incidence the entries were classified in five groups viz., Resistant, Moderately resistant, Moderately susceptible, Susceptible and Highly susceptible. On the basis on mean of two year data (2014-15 and 2015-16), of 126 entries, twenty two entries viz., GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059, NBeG 439 were found resistant, forty three entries found moderately resistant reaction, thirty seven entries have been found moderately susceptible, seventeen entries show susceptible reaction and seven entries have been found to show highly susceptible reaction.

Mishra *et al.* (2005) tested 470 germplasm lines and found KG-86 KWR-4, KWR-108 and KWR-277 as a resistant genotype. Chaturvedi and Dua (2009) have reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot. Khan *et al.* (2012) screened sixty germplasm lines of chickpea for their resistance against dry root rot disease in pot. Out of sixty germplasm lines, only nine lines namely, KGD-1189, KGD-1201, KGD-1209, KGD-1215, KGD-1217, KGD-1220, KGD-1221, KGD-1248 and KGD-1289, were found resistant. Ten lines namely KWR-7, KWR-26, KWR-28, KWR-50, KGR-18, KGR-48, KGR-159, KKG-

103, KKG-111 and KGD-1201 were found moderately resistant. Seven lines namely KWR-54, KWR-55, KWR-60, KWR-63, KWR-77, KWR-78, KLWR-79, were found moderately susceptible. 17 lines namely KGD-1238, KGD-1239, C-304, KWR-12, KWR-14, KWR-15, KWR-16, KWR-17, KWR-18, KWR-22, KWR-61, KWR-65, KWR-70, KWR-71 KWR-1111, KWR-1211 were found susceptible and rest 17 lines were found highly susceptible against the disease.

Botanical extracts are biodegradable (Devlin and Zettel, 1999) and their use in crop protection is a practical sustainable alternative. It reduces environmental contamination and health hazards (Grange and Ahmed, 1988). Botanical fungicides are unique because they can be produced easily by the farmers and small industries (Roy *et al.*, 2005). Different plant extracts also used separately or in combination to control some other fungi by the farmers. Very few studies have conducted using plant extracts and fungicides in the country to control *R. bataticola* causing dry root rot of chickpea. *In the present investigation, efficacy of nine botanicals was evaluated against growth of R. bataticola. All the botanicals significantly inhibited the growth of R. bataticola. Significantly maximum mycelium inhibition was recorded in Zingiber officinale rhizome followed by Datura stramonium leaf, Allium sativum clove and Eucalyptus globulus leaf, however minimum mycelium inhibition was recorded in Calotropis spp. leaf followed by Allium cepa bulb and Parthenium hysterophorus leaf. Savaliya et al. (2015) reported that the extract of garlic cloves (A. sativum L.) was proved excellent with maximum inhibiting (77.65 %) mycelial growth and scanty sclerotial formation followed by onion bulb extract (A. cepa L.) (63.98 %). Mandhare and Suryawanshi (2009) screened the antifungal properties of the extracts (at 10% each) of Azadirachta indica (leaves), Ocimum sanctum [O. tenuiflorum] (leaves), Eucalyptus sp. (leaves), Nerium indicum (leaves), A. sativum (cloves) and Z. officinale (rhizomes) against R. bataticola by poisoned food technique. The growth of R. bataticola was inhibited by 77.77 and 64.44% by the extracts of A. sativum and A. indica, respectively. The other extracts were not effective. Sharma et al. (2014) reported that the ethanoilic extract of D. stramonium leaf exhibit maximum inhibition against R. solani (88%) followed by Fusarium solani (76%). Tiwari and Das (2011), who reported that plant extracts of Psoralea corylifolia, Asparagus racemosus and Coleus forskohlii were highly efficacious against the mycelial growth as well as in checking the disease severity of R. solani. Hussain et al. (2014), Wadikar and*

Nimbalkar (2015) and Naik *et al.* (2016) evaluated the efficacy of four plants extracts viz., Neem, Pongamia, Subabul Gliricidia and observed the radial growth of the mycelium. Neem leaf extract found maximum inhibition (90 mm) of the fungus at 5, 10 and 15 percent concentration followed by Pongamia leaf extract at seven days after incubation. Among the leaf extract, neem and Pongamia leaf extracts were found to be effective against root rot pathogens, *R. solani*.

Efficacy of 11 systemic and five non systemic fungicides were tested at 0.1 % and 0.3 % concentrations respectively in the laboratory against *Rhizoctonia bataticola*. The results indicated that there was significant difference among the fungicides in inhibiting the mycelial growth of *R. bataticola*. Among the fungicides evaluated, 100 per cent mycelium inhibition was recorded in Carbendazim (0.1 %), Tricyclazole (0.1 %), Difenconazole (0.1 %), Hexaconazole (0.1 %), Propiconazole (0.1 %), Tebiconazole (0.1 %), Carboxin (0.1 %), Mancozeb (0.3 %), Copper oxychloride (0.3 %), Mancozeb + Copper oxychloride (0.3 %) and Mancozeb + Chlorothalonil (0.3 %), while maximum mycelium growth in Chlorothalonil (0.3%) followed by Pyroclostrobin (0.1 %), Azoxystrobin (0.1 %) and Thiophanate methyl (0.1 %). After the experiment all the selected fungicides were evaluated again at lower concentration. Among the fungicides, 100 per cent mycelium inhibition was recorded in Carbendazim (0.05 %), Difenconazole (0.05 %), Propiconazole (0.05 %), Tebiconazole (0.05 %), Carboxin (0.05 %) and Mancozeb (0.2%). Least mycelium inhibition was recorded in Tricyclazole (0.05 %) followed by Mancozeb + Chlorothalonil (0.2 %) and Mancozeb + Copper oxychloride (0.2%).

Ammajamma and Hegde (2009) evaluated the efficacy of systemic and non-systemic fungicides against *R. bataticola* at different concentrations. Among systemic fungicides, Carboxin (0.05 and 0.1), Hexaconazole, Metalaxyl and Traidemifon at 0.1 per cent completely (100%) inhibited the growth of *R. bataticola* and among the non-systemic fungicides tested, Thiram at 0.1, 0.2 and 0.3 per cent concentrations was found effective against *R. bataticola*. Prashanthi *et al.*, (2000) reported that Carbendazim (Bavistin) and Propiconazole were highly effective with the mean mycelial growth inhibition of 100 and 93.7%, respectively. Carbendazim showed 100% inhibition at all concentrations. Among the non-systemic fungicides, Chlorothalonil SC at 3000 ppm was the most effective (82.45% inhibition of mycelial growth). Similarly Prajapati (2002) conducted laboratory experiment to determine the efficacy of 11 fungicides against the causative agent (*R. bataticola* [M.

phaseolina) of dry root rot in chickpeas. Carbendazim, Carbendazim + Thiram, Carboxin, and Thiophanate-methyl completely inhibited the growth of *R. bataticola*.

Fifty four soil samples were collected from rhizosphere of five different crops and from seven districts of west M.P. Seven strains of *Trichoderma* recovered from fifty four soil sample. T-1, T-2, T-3, T-4, T-5, T-6 and T-7 strains were isolated from rhizospheres of cotton, chilli, chickpea, greengram, chickpea, pigeonpea and chickpea respectively. The result indicated that *Trichoderma* spp. could grow and survive in various kinds of soil conditions. This evidence agrees with the report of Harman *et al.* (2004) who had described the members of genus *Trichoderma* are free-living fungi that are common in soil and root ecosystems. For *Trichoderma* isolation, the dilution plating technique remains adequate, agreeing with the findings of Sivakumar *et al.* (2000). Kader *et al.* (1999) mixed the soil samples with sterile distilled water and made a series of dilutions. From the dilutions, 0.5ml of the soli solution was placed on potato dextrose agar (PDA) and incubated at 30°C for three days. Fungi were isolated from the mixed isolates from each plate and sub cultured on PDA, until a pure isolate was obtained.

Effective and practical chemical control is not feasible. Biological control appears to be the only solution for long-term sustainability and effective management of soil borne diseases. *Trichoderma* spp. are among the most studied fungal bio-control agents and commercially marketed as biopesticides, biofertilizers and soil amendments (Harman *et al.*, 2004; Lorito *et al.*, 2004; Baruah and Dutta, 1978). In the present investigation, seven native strains of *Trichoderma* spp. were evaluated against *R. bataticola*. **All the native antagonists i.e *Trichoderma* isolates showed significant reduction in mycelial growth of *R. bataticola* when compared to control. Among seven *Trichoderma* isolates, T-6 showed maximum inhibition of growth of *R. bataticola* followed by T-7, T-3 and T-5. The least inhibition was shown by T-1 followed by T-4 and T-2. Statically, T-6 and T-7 were found at par with each other, while T-1 followed by T-4 showed similar effect on mycelium inhibition of *R. bataticola*.** Veena, *et al.*, (2014) tested potential of ten strains of *Trichoderma* spp. against *R. bataticola*. Out of the ten isolates tested against *Rhizoctonia bataticola*, *Trichoderma* isolate-7 showed highest inhibition percentage. These results are in agreement with the findings reported by Bandyopadhyay *et al.* (2003) who reported that strain of *Trichoderma* inhibited the growth of *R. bataticola* by 51.1 per cent under *in vitro* conditions. Paul

et al. (2008) reported that among 11 *Trichoderma* isolates tested, *T. harzianum* showed higher inhibition of the growth of *R. solani* by 77 per cent under *in vitro* conditions. Kaushal (2008) reported that *T. harzianum* was effective in inhibiting the mycelial growth of *R. bataticola* the causal organism of chickpea dry root rot. Pan (2009) reported the antagonistic potential of *Trichoderma* isolates through production of volatile and non-volatile substances against *Macrophomina phaseolina*. Choudhary et al. (2010) tested four bioagents viz, *T. viride*, *T. harzianum*, *Aspergillus versicolor* and *Bacillus firmus* for the control of dry root rot in mungbean. In dual cultures, *T. viride*, *T. harzianum* and *A. versicolor* were effective in inhibiting the growth of *M. phaseolina* to an extent of 61 to 65%.

Rhizoctonia bataticola is a polyphagous soil borne pathogen infecting over 500 plant species worldwide causing huge losses. Though, the fungus is seed and soil borne (Dingra and Sinclair 1994), soil borne inoculum is more developed. For soil borne pathogen, use of fungicide is not practical due to exorbitant cost and environmental hazards involved. Hence, integrated management of the disease using bio-control agents and chemicals is the best alternative (Ramarethinum et al., 2001). Investigations were made in the present study to develop an effective management strategy for dry root rot of chick pea. Six fungicides, four botanicals and four strains of *Trichoderma* were selected for pot experiment after *In vitro* screening and evaluated against dry root rot caused by *R. bataticola*. Among the treatments, minimum disease incidence was recorded in Carbendazim which was found at par with Difenconazole, Carboxin, *Trichoderma* Strain T7, Propiconazole and Mancozeb. Maximum disease incidence was recorded in untreated control which was found at par with *Eucalyptus globulus* leaf and *Trichoderma* Strain T5. The results are in accordance with Surianchandraselvan, 1997; Manoranjitham et al. 2000; Tewari and Mukhopadhyay, 2003; Poddar et al., 2004; Gaur et al., 2005). Khan et al. (2012) tested eight fungicides against dry root rot fungus, among them Mancozeb, Carbendazim, Copper-oxy-chloride and Benomyl completely inhibited the growth of the fungus compared to control. Further, in fungicidal trails on management of dry root rot of chickpea caused by *R. bataticola*, Carbendizim (0.2 per cent) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 per cent and highest grain yield of 192 q/ha respectively (Mohan et al., 2006). Nagamani et al (2011) reported that the lowest

per cent disease incidence (4.33) was recorded in seed treatment with carbendazim@ 2 g/kg of seed + seed treatment with *T.viride* @ 4 g/kg of seed + soil application of FYM fortified with *T. viride* (T7) and highest per cent disease incidence was recorded in control (35.03), the next best treatment was recorded in seed treatment with *T.viride* @ 4 g/kg of seed +soil application FYM fortified with *T. viride* (T6) (8.21) followed by seed treatment with *T.viride* @ 4 g/kg of seed (12.45) (T3).

Chapter -VI

SUMMARY, CONCLUSION AND SUGGESTIONS FOR FURTHER WORK

Chickpea (*Cicer arietinum* L.) is one of the most important pulse crops of India. It occupies very important position in semi-arid farming system both for human nutrition and restoring the soil fertility. The average production of chickpea is 15-20 quintal per hectare which is low in spite of high yielding varieties and new agronomic practices. The reasons of low yield are so many part from other reasons. The main cause of low yield of this crop is due to incidence of diseases. The chickpea crop is attacked by 172 pathogens (67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma) from all over the world. The soil borne diseases, which severely damage the chickpea under favourable conditions are dry root rot caused by *Rhizoctonia bataticola*, wilt caused by *Fusarium oxysporum* f. sp. *ciceri* and collar rot caused by *Sclerotium rolfsii*. Among them, occurrence of dry root rot caused by *Rhizoctonia bataticola* (Taub.) Butler is emerging as a serious biotic constraint in recent years. Dry root rot generally appears during late flowering, podding stages and the infected plants appear completely dried.

6.1: SUMMARY

- The incidence of *Rhizoctonia* root rot recorded at 40 places of 15 districts surveyed which range from 2.54 to 20.89 %. Dry root rot disease of chickpea crop was found in all the sites which were surveyed during 2014-2015 and 2015-16. The mean maximum dry root rot incidence was noticed in Lukwasa followed by Shivpuri and Kolarash while the least dry root rot incidence recorded in Khargone followed by Rajpur and Barwani.
- During the survey, irrigation condition of all the places were recorded. Per cent disease incidence in the irrigated locations varied from 2.54 (Khargone) to 7.67 (Biaora) whereas, in the locations where no irrigation facility was available there, it varied from 5.17 (Sendwa) to 20.89 (Lukwasa). Comparatively, the disease incidence was less in irrigated fields over un-irrigated ones.
- Three types of soil were recorded in which chickpea crop is cultivated. The per cent disease incidence varied from 4.58 (Guna) to 20.89 (Lukwasa) in locations where the chickpea was crop grown in alluvial soil while it was from 3.20 (Rajpur) to 12.07 (Panchor) in locations where the chickpea crop grown in

medium black soil. The locations where the chickpea crop grown in deep black soil, it varied from 2.54 (Khargone) to 15.34 (Mau).

- Potato dextrose agar medium was found significantly superior as compared to the other tested culture media in respect of growth and sclerotial development.
- Four screening techniques were evaluated in which maximum seedling mortality was recorded in soil inoculation method. This method is significantly superior over other techniques.
- 16 isolates of *R. bataticola* collected from 15 districts showed difference in colony diameter, colour, texture, aerial mycelium, sclerotial size, length and width ratio, shape, intensity and hyphal cell size. Maximum colony diameters was recorded in Isolate Rb-5 (Makshi)
- Colony colour were divided in four groups viz., black colour, black with grey aerial, dark brown colour and grey colour. Out of sixteen isolates, black colour colony was observed in 4 isolates (Rb-1, Rb-7, Rb-8, Rb-11) while black with grey aerial mycelium was recorded in 7 isolates (Rb-2, Rb-3, Rb-5, Rb-9, Rb-14, Rb-15, Rb-16). Dark brown (Rb-6, Rb-12, Rb-13) and grey colour (Rb-4, Rb-10) was observed in 3 and 2 isolates respectively.
- Isolates were categorized into three groups based on colony texture viz., fluffy colony, appressed colony and velvety colony. Out of sixteen, maximum 8 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-9, Rb-14, Rb-15, Rb-16) produced fluffy colony while appressed (Rb-1, Rb-11, Rb-12, Rb-13) and velvety growth (Rb-4, Rb-7, Rb-8, Rb-10) produced by 4 isolates.
- 9 isolates (Rb-2, Rb-3, Rb-5, Rb-6, Rb-8, Rb-9, Rb-14, Rb-15 and Rb-16) produced aerial mycelium while, remaining 7 isolates (Rb-1, Rb-4, Rb-7, Rb-10, Rb-11, Rb-12, Rb-13) did not develop any aerial mycelium.
- Maximum sclerotial size was observed in case of isolate Rb-6 isolated from Panchor (181.35 x 170.72 μm)
- Maximum length : width ratio was found in Rb-14 (Kukshi).
- Irregular shaped sclerotia were observed in 7 isolates while round shaped sclerotia were observed in 5 isolates and ovoid shape of sclerotia was observed in 4 isolates.
- Maximum number of sclerotia was produced by Rb-15 (Dhar).
- Maximum hyphal cell size was recorded in Rb-4 (Biaora)

- Maximum ratio between length and width of hyphal cell was observed in Rb-10 (Segaon)
- Among sixteen isolates, four isolates viz., Rb-1, Rb-2, Rb-5 and Rb-12 showed highly virulent reaction, while seven isolates showed virulent reaction, however, five isolates showed least virulent reaction.
- The incubation period ranged from 2.80 days to 5.00 days. Shortest incubation period was recorded in Rb-5 (Makshi) while, longest incubation period was recorded in Rb-7 (Dewas).
- The screening of chickpea cultivars under root rot sick microfield for two continuous years indicated that cultivar viz., GL 10023, RSG 888, BAUG 15, GJG 1209, GJG 1202, JG 41, GNG 2215, BG 372, GNG 2261, BG 3054, NDG 13-21, GNG 2263, GJG 1205, NBeG 460, HK 10-124, NBeG 399, BG 3059, NBeG 439 were found resistant, other cultivars exhibited moderately resistant to highly susceptible reaction.
- *All the tested nine botanicals significantly inhibited the growth of R. bataticola. Significantly however, maximum mycelium inhibition was recorded in Zingiber officinale rhizome.*
- *In vitro* bioassay study, 100 per cent mycelium inhibition was recorded in Carbendazim @ 0.05 %, Difenconazole @ 0.05 %, Propiconazole @ 0.05 %, Tebiconazole @ 0.05 %, Carboxin @ 0.05 % and Mancozeb @ 0.2%
- Seven isolates of *Trichoderma* recovered from fifty four soil sample isolated from rhizospheres of Cotton, Chilli, Chickpea, Greengram, Chickpea, Pigeonpea and Chickpea Respectively.
- **Among seven *Trichoderma* isolates, T-6 showed maximum inhibition of growth of *Rhizoctonia bataticola***
- Minimum disease incidence was recorded in Carbendazim which was found at par with Difenconazole, Carboxin, *Trichoderma* isolate T7, Propiconazole and Mancozeb

6.2 : CONCLUSION

- During the survey, maximum dry root rot incidence was noticed in Lukwasa (20.89 %), while, the least dry root rot incidence recorded in Khargone (2.54 %).

- Less dry root rot incidence was recorded in irrigated field (4.59 %) as compared to un-irrigated fields (10.48 %).
- Comparatively the disease incidence was less in deep black soil (6.75 %) over medium black soil (7.47 %) and alluvial soil (12.24 %).
- Among the eight culture media tested, Potato dextrose agar was found best for the growth of *R. bataticola*.
- Among the inoculation techniques, maximum seedling mortality (87.50 %) was recorded in soil inoculation method.
- A maximum colony diameter (85.67 mm) was recorded in Isolate Rb-5 (Makshi).
- Out of sixteen isolates, black colour colony was observed in 4 isolates while black with grey aerial mycelium was recorded in 7 isolates. Dark brown and grey colour was observed in 3 and 2 isolates respectively.
- Out of sixteen isolates, maximum of 8 isolates produced fluffy colony while appressed and velvety growth produced by 4 isolates each respectively.
- 9 isolates produced aerial mycelium while remaining 7 isolates did not develop any aerial mycelium.
- Maximum sclerotial size ($180 \times 170.72 \mu\text{m}$) was observed in case of isolate Rb-6 (Panchor).
- Maximum length : width ratio (1.13) recorded in isolate Rb-14 (Kukshi).
- Irregular shaped sclerotia were observed in 7 isolates while round shaped sclerotia were observed in 5 isolates. Ovoid shape of sclerotia was observed in 4 isolates.
- Maximum number of sclerotia (44.60) was produced by Rb-15 (Dhar).
- Maximum hyphal cell size ($14.65 \times 7.40 \mu\text{m}$) was recorded in Rb-4, while minimum hyphal cell size ($10.35 \times 5.10 \mu\text{m}$) was recorded in Rb-3
- Maximum ratio between length and width of hyphal cell (2.43) was observed in Rb-10 (Segaon).
- Among sixteen isolates, four isolates viz., Rb-1, Rb-2, Rb-5 and Rb-12 showed highly virulent reaction, while seven isolates showed virulent reaction, however, five isolates showed least virulent reaction.
- Longest incubation period (5.00 days) was recorded in Rb-7 (Dewas).
- On the basis on mean of two year data (2014-15 and 2015-16), out of 126 entries, twenty two entries were found resistant, while forty three, thirty

seven, seventeen and seven entries showed moderately resistant, moderately susceptible, susceptible reaction and highly susceptible reaction.

- Under *in-vitro* study, *Zingiber officinale* rhizome extract was found most effective for reducing the mycelial growth of *R. bataticola*.
- Among the fungicides evaluated, under *in-vitro* condition Carbendazim @ 0.05 %, Difenconazole @ 0.05 %, Propiconazole @ 0.05 %, Tebiconazole @ 0.05 %, Carboxin @ 0.05 % and Mancozeb @ 0.2% were found significantly effective against *R. bataticola*.
- Among seven *Trichoderma* isolates, T-6 showed maximum inhibition of growth of *R. bataticola* (67.22 %) followed by T-7, T-3 and T-5.
- In pot experiment, Carbendazim found most effective which was found at par with Difenconazole, and Propiconazole.

6.3: SUGGESTIONS FOR FURTHER WORK

- Survey and surveillance of disease should be undertaken every year to observe rhythmic changes in the incidence of the disease in relation to changes in weather.
- Collection and isolation of *Rhizoctonia bataticola* should be from more number of chickpea growing areas of Madhya Pradesh for intensive studies on variability in *R. bataticola*.
- There is need to verify the resistance of genotypes under multi locations to know the wide base of resistance.
- Morphological, physiological and biochemical factors responsible for disease resistance needs investigation.
- More work needed with respect to validation of integrated approaches like use of bio-agents and botanicals for eco-friendly disease management.
- Research should further be intensified to reduce the incidence of dry root rot of chickpea with parameter viz., sowing dates, sowing depth, soil characteristics, soil moisture and temperature.
- Analysis of genetic diversity in populations of host and pathogen.
- Induced resistance and systemically acquired resistance.
- The advanced studies on biotechnological approaches gene transfer, *Agrobacterium tumefecience* mediated transformation and somatic hybridization,

exploitation for developing transgenic plants of crop with superior resistance to *R. bataticola* are needed.

APPENDICES

Anova table for table-15 (3 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Medium	7	147.96	21.14	13.01	2.66
Error	16	26.00	1.63		
TOTAL	23	173.96	22.76		

Anova table for table-15 (5 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Medium	7	865.83	123.69	70.68	2.66
Error	16	28.00	1.75		
TOTAL	23	893.83	125.44		

Anova table for table-15 (7 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Medium	7	1126.29	160.90	77.23	2.66
Error	16	33.33	2.08		
TOTAL	23	1159.63	162.98		

Anova table for table-16 (1%)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Inoculation technique	4	3607.64	901.91	38.46	3.06
Error	15	351.74	23.45		
TOTAL	19	3959.39	925.36		

Anova table for table-16 (2%)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Inoculation technique	4	5570.54	1392.64	34.20	3.06
Error	15	610.75	40.72		
TOTAL	19	6181.29	1433.35		

Anova table for table-16 (3%)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Inoculation technique	4	9412.07	2353.02	54.31	3.06
Error	15	649.89	43.33		
TOTAL	19	10061.97	2396.34		

Anova table for table-16 (4%)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Inoculation technique	4	12402.14	3100.53	85.70	3.06
Error	15	542.70	36.18		
TOTAL	19	12944.84	3136.71		

Anova table for table-17 (3 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Isolates	15	88.31	5.89	6.57	1.99
Error	32	28.67	0.90		
TOTAL	47	116.98	6.78		

Anova table for table-17 (5 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Isolates	15	350.65	23.38	9.20	1.99
Error	32	81.33	2.54		
TOTAL	47	431.98	25.92		

Anova table for table-17 (7 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Isolates	15	804.58	53.64	13.70	1.99
Error	32	125.33	3.92		
TOTAL	47	929.92	57.56		

Anova table for table-26 (3 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Botanicals	9	902.53	100.28	59.87	2.21
Error	30	50.25	1.68		
TOTAL	39	952.78	101.96		

Anova table for table-26 (5 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Botanicals	9	4510.60	501.18	68.81	2.21
Error	30	218.50	7.28		
TOTAL	39	4729.10	508.46		

Anova table for table-26 (7 DAI)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Botanicals	9	5836.13	648.46	37.11	2.21
Error	30	524.25	17.48		
TOTAL	39	6360.38	665.93		

Anova table for table-27					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Fungicides	16	28239.92	1765.00	849.20	1.95
Error	34	70.67	2.08		
TOTAL	50	28310.59	1767.07		

Anova table for table-28					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Fungicides	11	23679.18	2152.65	668.06	2.22
Error	24	77.33	3.22		
TOTAL	35	23756.51	2155.87		

Anova table for table-29					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Trichoderma	7	10232.72	1461.82	193.03	2.42
Error	24	181.75	7.57		
TOTAL	31	10414.47	1469.39		

Anova table for table-30 (2014-15)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Treatment	14	1928.64	137.76	16.27	2.04
Error	30	262.16	8.74		
TOTAL	44	2190.80	146.50		

Anova table for table-30 (2015-16)					
SV	DF	SS	MSS	F _{cal}	F _{tab}
Treatment	14	2275.45	162.53	12.46	2.04
Error	30	375.41	12.51		
TOTAL	44	2650.86	175.05		

Anova table for table-30 (Mean)					
SV	DF	SS	MSS	F_{cal}	F_{tab}
Year (Y)	1	38.94	38.94	3.67	4.00
Treatment	14	4079.61	291.40	27.44	1.86
Interaction (Y × T)	14	103.23	7.37	0.69	1.86
Error	60	637.09	10.62		
TOTAL	89	4858.87	348.33		

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